

Rational Voter Learning, Issue Alignment, and Polarization*

Martin Vaeth[†]

[Latest version here](#)

August 31, 2026

Abstract

We model electoral competition between two parties when voters rationally learn about their own political positions through flexible information acquisition. Rational learning polarizes voter preferences and aligns them across policy issues, even when true positions are unimodally distributed and independent. When parties strategically choose their positions to influence voter learning, party and voter polarization increase as information costs decline, and parties may adopt positions more extreme than their ideal policies. Endogenous learning introduces two forces on party positions: parties gain from moderating by biasing learning in their favor, and the more extreme party gains from polarizing by triggering more learning.

Keywords: rational inattention, voter ideology, electoral competition, polarization

JEL Classifications: D72, D83

*I am indebted to Roland Bénabou and Alessandro Lizzeri for their guidance throughout the project. For helpful discussions and feedback, I would like to thank Alexander Bloedel, Mira Frick, Germán Gieczewski, Francesco Fabbri, Jiadong Gu, Faruk Gul, En Hua Hu, Matias Iaryczower, Ryota Iijima, Gleason Judd, Andreas Kleiner, Botond Kőszegi, Stephan Lauermann, Elliot Lipnowski, John Londregan, Filip Matějka, Nolan McCarty, Dan McGee, Jessica Min, Xiaosheng Mu, Pietro Ortoleva, Jacopo Perego, Wolfgang Pesendorfer, Doron Ravid, Joseph Ruggiero, Fedor Sandomirskiy, Christopher Sims, Guido Tabellini, João Thereze, Leeat Yariv, and participants at various seminars and the 2025 Conference on Rational Inattention. Financial support was provided by The William S. Dietrich II Economic Theory Center. All errors are my own.

[†]Paris School of Economics. martin.vaeth@psemail.eu

1 Introduction

Voter positions in the United States display two puzzling features. First, one can predict a voter’s position on most policy issues remarkably well by knowing just how left or right they are (Jessee 2012; Hare, Highton, and Jones 2024; among others). Voter positions on different issues are therefore strongly aligned, which is surprising given the wide variety of issues, such as taxation, immigration, and the environment. Second, evidence suggests that voter positions are increasingly polarized, in the sense of being clustered around two poles on the left-right axis (Pew Research Center 2014).¹ A concern in the polarization literature is that such a distribution creates two homogeneous groups with few individuals bridging the divide between them, increasing the risk of conflict (Esteban and Ray 1994).

This paper provides a joint explanation of issue alignment and polarization based on rational voter learning. Previous research and public discourse attribute issue alignment and polarization to voter biases like confirmation bias, herding due to echo chambers, or partisan news media. By contrast, we show issue alignment and polarized ideology emerge naturally from rational (i.e., no biases) and individual (i.e., no herding) voter learning, driven by voters’ own information choices (i.e., no media effects). Central to the mechanism is that voters learn about their political position through flexible information acquisition in order to decide between two parties. Concretely, learning about one’s political position can take the form of learning about the effects of policies, for example, learning about the effects of tariffs to inform one’s position on trade policy. We assume that learning is costly, whether in time, effort, or money. This paper shows that the resulting cost-minimization motive structures rational voter learning in a way that generates issue alignment and polarization.

Whether voter ideology is driven by biases or by rational learning is important for assessing the functioning of democratic elections (Achen and Bartels 2016). If biases dominate, candidates may cater to these biases rather than advancing policies that address societal needs. If instead voter positions reflect rational learning, we may be more optimistic that elections produce policies aligned with voters’ interests. The second part of this paper asks whether such optimism is warranted, studying the effects of voter learning on party platforms.

We analyze rational voter learning in an otherwise standard political-economy model. To study issue alignment, we incorporate a multidimensional policy space, where each dimension represents a different policy issue. Voters face two parties, each adopting a policy platform in this space, which is the policy the party implements if elected. Party platforms are exogenous in the first part of the paper. Following the literature, a voter’s utility decreases quadratically in the distance between a policy and her *ideal point*, which captures her political position. In our model, voters do not know their ideal points but can form political opinions through learning. We assume that voters start with a homogeneous prior that coincides with the true distribution of ideal points, which in the simplest case is normal and independent across issues.² Voters then update their beliefs by acquiring information flexibly, following the rational-inattention framework (Sims 2003).

¹Section 1.1 examines the evidence for issue alignment and ideological polarization in detail.

²The paper allows for any log-concave spherical distribution. Section 3.1 discusses heterogeneous priors.

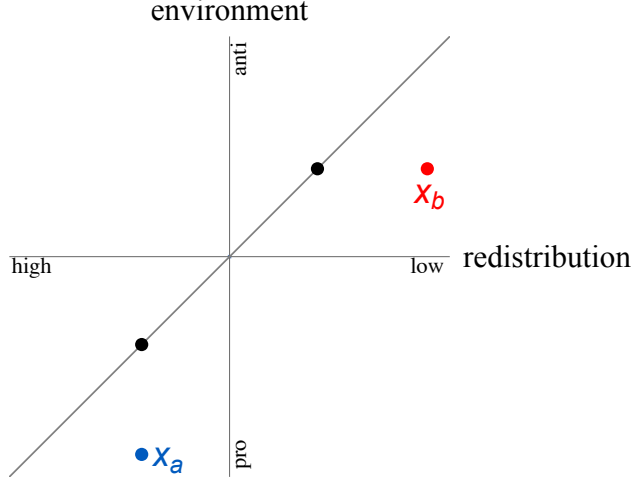


Figure 1: Example policy platforms x_a and x_b of parties a and b , respectively, under two policy issues. Voter learning induces revealed ideal points on the diagonal line.

This framework allows voters to choose both how much and what type of information to acquire, subject to a cost increasing in information. Because utility is quadratic, after acquiring information a voter votes for the party whose platform is closer to her expected ideal point. We refer to this expected ideal point as her *revealed ideal point*, distinguishing revealed ideology—the distribution of revealed ideal points induced by learning—from the distribution of true ideal points.

Our first result is that rational voter learning generates issue alignment and polarization: revealed ideal points align along a left-right axis and cluster around two poles, even if true ideal points are independent across issues and centrally clustered. At the core of this mechanism is that voters trade off making an informed choice and minimizing the cost of information. By acquiring only the information necessary to determine which party platform is closer, voters align their preferences along a single axis and polarize.

To understand the high-level intuition for why cost-effective learning generates issue alignment, consider a two-dimensional policy space with two issues, redistribution and environmental protection. Suppose the two parties propose policy platforms, x_a and x_b , as shown in Figure 1, where one platform favors more redistribution and stronger environmental protection than the other. The two platforms determine a unique line, whose direction we refer to as the *axis of disagreement*. The axis of disagreement is the direction along which parties differentiate, combining the two issues in proportion to how much the parties disagree on each. Crucially, voters learn only where their ideal point lies along this axis, because only the position along the axis determines which platform is closer. Where a voter’s ideal point lies in the orthogonal direction does not affect the *relative* proximity of the two platforms and is therefore irrelevant to her vote. Consequently, a voter learns only whether she leans toward more redistribution and stronger environmental protection or toward less of each. Learning of this kind aligns revealed ideal points across issues, placing them on a line that can be interpreted as an endogenous left-right axis. The left-right axis inherits the direction of party disagreement and passes through voters’ prior expectation of their ideal points.

This result holds for any number of policy issues, assuming a reflection-invariant information cost and a spherical prior.³

The mechanism by which cost-effective learning generates ideological polarization is as follows. Voters only need to determine which party they are closer to, not by how much. Thus, they optimally acquire a *binary* signal about their ideal points. This signal sorts voters into two groups: those who believe they are closer to one party and those who believe they are closer to the other. Even centrist voters categorize themselves based on this binary information, causing them to perceive their preferences as more extreme than they truly are.

The second part of this paper examines whether elections produce policies that serve voters' interests when voter ideology results from rational learning. To address this question, we endogenize party platforms and study their interaction with voter learning. Parties strategically position themselves in the policy space, balancing electoral prospects against their own policy preferences (Wittman 1973; Calvert 1985).

Characterizing equilibrium is difficult because of the multidimensional policy space and endogenous voter learning. Policy-motivated parties are usually analyzed in one-dimensional policy spaces in which a median voter is pivotal (Duggan 2017). Endogenous voter learning further complicates the analysis, because vote shares follow from voters' optimal learning strategies rather than from the share of voters whose ideal point is closer to one platform. Despite these obstacles, the model admits a unique pure-strategy equilibrium in closed form over an explicitly characterized region of the parameter space. Tractability is achieved through a combination of modeling choices, in particular the posterior-variance information cost. Under these choices voter learning has a closed-form solution, which in turn makes it possible to characterize parties' best responses.

In our model, parties move first and voter learning responds to party platforms. Moving first gives parties an agenda-setting role: through their platforms, parties influence which issues voters pay attention to. A party that polarizes on a policy issue draws attention to that issue, because a voter's position on the issue now does more to determine which platform is closer. Endogenous voter learning introduces two novel strategic motives in party positioning, a *moderation motive* and a *polarization motive*. First, parties may moderate their platforms to bias voter learning in their favor. As information costs decline, the moderation motive weakens, leading to more platform polarization, which we show reduces aggregate voter welfare. The model therefore offers a mechanism by which technologies that make information cheaper, such as the internet, can raise polarization, even in the absence of echo-chamber effects. Second, the more extreme party may find it optimal to polarize, that is, to move away from the moderate party, in order to trigger more voter learning, thereby reducing the bias toward its opponent. The polarization motive implies that parties may adopt platforms more extreme than their own ideal policies. In models where voter ideology is exogenous, by contrast, platforms necessarily lie between the parties' ideal policies (Roemer 1997).

³The proof is less immediate than this sketch suggests and rests on a reflection argument that uses the stated properties of the prior and the information cost, illustrated in Figure 3.

Our results caution against interpreting party influence on voter ideology necessarily as a sign of voter irrationality. Political scientists have long argued political elites have a large influence on the ideology of voters (Campbell, Converse, Miller, and Stokes 1960; Zaller 1992; Lenz 2012). Such influence has traditionally been taken as evidence of voter irrationality, a view shared by Achen and Bartels (2016), who argue such influence presents a threat to democracy. If parties can shape voter ideology instead of merely responding to it, elections may fail to produce governments responsive to the preferences of voters. We contribute to this debate by showing that some forms of party influence on voter ideology are consistent with voter rationality and leave room for policy to respond to voters’ true preferences. In our model, both the issue alignment and the polarization of revealed ideal points depend on party platforms. First, the alignment of revealed ideal points across policy issues is determined by the relative positions of the platforms. As illustrated in Figure 1, because party *a* favors more redistribution and stronger environmental protection than party *b*, support for redistribution and support for environmental protection align in the resulting voter ideology. Second, more polarized platforms can result in more polarized revealed ideology (Proposition 1). Although voters appear in both cases to simply follow party platforms, our model shows that such effects result from rational voter learning. At the same time, party platforms in our equilibrium do respond to voters’ true ideal points, namely to the center of their distribution, the policy that maximizes voter welfare (Theorems 2 and 3). More broadly, our model illustrates how revealed preferences may differ systematically from true preferences in a context-dependent way.

Our model has policy implications for addressing issue alignment and polarization. Traditional approaches—such as improving political knowledge (Delli Carpini and Keeter 1996) or breaking up echo chambers (Sunstein 2018)—may be ineffective if issue alignment and polarization stem from rational voter learning. Instead, more fundamental changes to voters’ choice sets would be necessary to incentivize multidimensional and non-binary learning. Electoral reforms that increase the number of parties, such as replacing plurality voting with proportional representation, can induce multidimensional learning (Corollary 1). Participation mechanisms that elicit the intensity of voter preferences, such as storable votes (Casella 2005) or quadratic voting (Lalley and Weyl 2018), can induce non-binary learning.

The paper is organized as follows. The rest of this section discusses related literature and evidence on voter ideology. Section 2 introduces the model. Section 3 analyzes voter learning. Section 4 studies electoral competition when voter ideology results from optimal learning. Section 5 concludes.

Related Literature This paper contributes to the growing literature on rationally inattentive voters. What sets our paper apart is the object of learning: voters learn about their own *ideal points* rather than about the parties. This type of learning allows us to explain properties of voter ideology, which is exogenous in other papers. Matějka and Tabellini (2021) study electoral competition in which voters pay attention to *party platforms*. Voters who pay more attention respond more strongly to platforms, so parties cater to them more. In our model, by contrast, more attention weakens the bias in voter learning toward the moderate party, so parties cater less to voters. Li

and Hu (2023) study attention to *implemented policies* in an electoral-accountability setting. They show that the welfare effects of increased attention and voter polarization are ambiguous. Hu, Li, and Segal (2023) analyze learning about *valence* through an attention-maximizing intermediary. They show that such learning generates policy divergence even with office-motivated candidates. We instead derive divergence from policy-motivated parties and study its interaction with voter learning. Closer to our work in its object of learning is Yuksel (2022), where voters also learn about their ideal points. She analyzes learning under partitional signals and finds that more specialized learning increases platform polarization. We allow for flexible learning and show cheaper information increases polarization through a different mechanism. More broadly, we contribute to the rational inattention literature by studying two players who compete to influence an agent’s learning about her own type. Learning is biased toward the player whose offer is ex-ante more attractive, so players cater to the agent’s prior. The player with the ex-ante less attractive offer benefits from differentiation, which triggers more learning and weakens the bias.

The literature has proposed several explanations for issue alignment and polarized ideology. Converse (1964) introduced the concept of issue alignment, which he termed ideological constraint, attributing it to logical, psychological, and social sources. Enke, Rodríguez-Padilla, and Zimmermann (2023) propose that moral universalism, the degree of altruism toward strangers versus in-group members, explains correlations in policy views across domains. Spector (2000) derives one-dimensional ideology from cheap talk between two groups with differing priors, while DeMarzo, Vayanos, and Zwiebel (2003) derive it from networks and persuasion bias. Ideological polarization has been linked to biased news (Martin and Yurukoglu 2017) and to cognitive limitations, such as overconfidence (Ortoleva and Snowberg 2015) and bounded rationality (Eguia and Hu 2022). Other papers have also proposed rational inattention as a mechanism for belief polarization, albeit through ex-ante heterogeneity in preferences (Novák, Matveenko, and Ravaioli 2024) or path-dependent sequential information acquisition (Nimark and Sundaresan 2019). Callander and Carbajal (2022) explain dynamic polarization through voters adjusting their ideal points toward the party they voted for to rationalize their choice. Our model complements these works by explaining issue alignment and polarization jointly, and by generating comparative statics for both. Revealed ideology is more aligned and more polarized when parties are further apart and when information is cheaper. These predictions match the rise in alignment and polarization alongside party polarization in the US, and the fact that both are stronger among better-informed voters. Section 3.1 presents these comparative statics and the related evidence.

This paper contributes to the study of electoral competition in multidimensional policy spaces with divergent platforms, as is typical of actual elections. Both features are essential to our analysis of voter learning, since issue alignment requires at least two issues and information has no value when the platforms coincide. The literature has provided many mechanisms for platform divergence, breaking the median voter result of Downs (1957). We do not propose a new mechanism but show how endogenous ideology formation through voter learning interacts with the most prominent one, policy-motivated parties facing electoral uncertainty, known as the Calvert-Wittman model.

Policy-motivated parties still converge to the median voter unless the electoral outcome is uncertain (Calvert 1985; Wittman 1983). In a one-dimensional policy space, pure-strategy equilibria exist under appropriate conditions and parties diverge (Roemer 1997). Multidimensional policy spaces are far less well understood. Calvert (1985) shows only that convergent platforms are not an equilibrium. Existing extensions to multiple dimensions drop either policy motivation (Lindbeck and Weibull 1987) or electoral uncertainty (Duggan and Fey 2005), and platforms converge again. To our knowledge, the literature has provided no existence or characterization result for equilibrium under both policy motivation and electoral uncertainty in more than one dimension.⁴ We provide a version of the Calvert-Wittman model that, for a range of parameters, admits a unique pure-strategy equilibrium in closed form even in a multidimensional policy space.

1.1 Evidence on Voter Ideology

Our model predicts that revealed ideology displays issue alignment and ideological polarization. We discuss the evidence on both features for US voters.

1.1.1 Issue Alignment

Recent evidence shows that the ideology of US voters is approximately one-dimensional (Jessee 2009; Jessee 2012; Tausanovitch and Warshaw 2012; Shor and Rogowski 2018; Fowler, Hill, Lewis, Tausanovitch, Vavreck, and Warshaw 2023). These studies use voter surveys, such as the American National Election Studies, to estimate voter ideal points, similar to the ideal point estimation of legislators from roll-call data (Poole and Rosenthal 1985). Each study fits a one-dimensional spatial model with quadratic utility to the binary responses that individuals give to policy questions, such as whether the minimum wage should be raised. These studies find that a one-dimensional model already explains voter responses well, accounting for about 80% of binary responses, and that adding dimensions raises explanatory power only marginally. They conclude that voter ideology is well described by a one-dimensional ideological spectrum.

In contrast, an older literature starting with Converse (1964) studies the correlation between voter responses to different policy questions rather than estimating ideal points. These papers typically find low correlation and conclude that ideology is not well represented by a one-dimensional spectrum, or that there is little *constraint* on voter ideology in the terminology of Converse. Later research has shown that the low correlation is partly driven by response mistakes, such as those arising from inattentive respondents (Ansolabehere, Rodden and Snyder 2008). There is a further reason why the correlation between responses is a poor measure of how one-dimensional voter ideology is. Even if voters answer according to a one-dimensional spatial model, they do not give left or right answers consistently but pick whichever of a question’s two options (e.g., raising the minimum wage or not, expanding military presence abroad or not) lies closer to their ideal point.

⁴Roemer (2001) instead obtains divergent equilibria in multiple dimensions by modeling each party as a coalition of factions with conflicting objectives. A deviation requires the assent of all, which is restrictive enough to establish existence and delivers a continuum of equilibria.

Each question therefore splits the electorate at a cutoff on the left-right axis, with voters to the right of the cutoff giving the right response and voters to the left giving the left response. Only voters near the extremes fall on the same side of every cutoff and answer consistently, so moderates give a mix of left and right responses even under a fully one-dimensional model.

1.1.2 Polarized Ideology

The literature on polarization measures defines a distribution as more polarized the more it is concentrated on a small number of clusters and the farther apart the clusters lie (Esteban and Ray 2012). Whether US voters are ideologically polarized, in the sense of their ideal points having a bimodal distribution, is the subject of a long debate, with Abramowitz and Saunders (2008) on the affirmative side and Fiorina and Abrams (2008) on the other, and Lelkes (2016) providing a critical overview. Neither side estimates voter ideal points and both rely on raw survey responses, which makes their findings hard to interpret. Much of the evidence against polarization rests on ideological self-placements on 7-point scales (Fiorina and Abrams 2008). Such self-placements have been criticized on several grounds, but a more basic problem has received less attention. Bimodality is not preserved under monotone transformations of the scale, so assessing polarization requires a cardinal measure of ideology, and a 7-point scale is merely ordinal. Reading bimodality off such a scale presumes that the seven categories correspond to intervals of equal length on the underlying cardinal scale of ideology, an assumption that is typically left implicit. Estimating ideal points from survey responses to multiple questions avoids the problem, as it yields a meaningful cardinal scale.

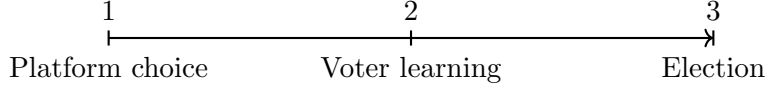
Several newer papers do estimate voter ideal points from survey responses. Bafumi and Herron (2010) find a bimodal distribution, while most find a unimodal one (Hill and Tausanovitch 2015; Jessee 2012). However, Fowler, Hill, Lewis, Tausanovitch, Vavreck, and Warshaw (2023) argue that the estimation procedures understate polarization, because respondents who pay little attention to the survey answer close to randomly and are artificially placed near the center. Screening for such respondents, they find a bimodal distribution of ideal points in some survey years.

Our model is also consistent with voters being polarized while the population as a whole, including those who abstain, is not. As discussed in Section 3, when voting is costly and an individual turns out only if the expected utility difference between the parties is large enough, the model predicts a third centrist group emerging alongside the two polarized ones. The centrist group abstains, so revealed ideal points are bimodal among those who turn out while a mass of moderates remains in the population. Abramowitz (2010) indeed finds that voters are more polarized than the electorate as a whole.

The evidence on the evolution of polarization is more uniform and suggests a rise in the last two decades. Several studies aggregate responses to multiple policy questions into an ideology score and find that scores become more extreme starting in the 2000s (Pew Research Center 2014; Martin and Yurukoglu 2017, Online Appendix; Eguia and Hu 2022). Our model predicts that voter polarization rises as information costs decline and as party polarization increases.

2 Model

Game The policy space is $\mathbb{R}^n$ with $n \in \mathbb{N}$, where each dimension corresponds to a policy issue. Voter ideal points as well as party platforms live in this space. There is a continuum of voters indexed by $i \in [0, 1]$ and two parties, a and b . The timing is as follows.



- (1) Parties a and b simultaneously and publicly commit to policy platforms $x_a \in \mathbb{R}^n$ and $x_b \in \mathbb{R}^n$, respectively.
- (2) Each voter $i \in [0, 1]$ learns about her ideal point $\theta_i \in \mathbb{R}^n$.
- (3) Each voter votes for one of the two parties, and the election determines which platform is implemented.

Parties choose platforms $x_a, x_b \in \mathbb{R}^n$ before voters learn about their ideal points. Thus, parties act as agenda-setters: through their platforms, parties affect what issues voters pay attention to. For example, if a party polarizes on migration, then voters will pay more attention to this issue because it becomes more relevant to their voting decision. If, alternatively, voters learned before parties moved, the main result on voter learning (Theorem 1) would remain unchanged in a pure-strategy equilibrium. The reason is that the qualitative predictions on voter ideology do not depend on whether voters learn given observed or given anticipated platforms.

Voters The voter utility includes policy utility u and information cost c ,

$$U_i = u(x, \theta_i) - \kappa \cdot c. \quad (1)$$

We vary the information cost parameter $\kappa > 0$ for comparative statics. We specify the information cost in the paragraph on learning below. Following the workhorse model of spatial voting, we assume voters' policy utility is quadratic in the difference between their ideal point and the policy. That is,

$$u(x, \theta) = -(x - \theta)^\top A(x - \theta), \quad (2)$$

where $A \in \mathbb{R}^{n \times n}$ is an arbitrary symmetric, positive definite matrix representing voters' *priorities*. The matrix A induces the A -distance $((x - y)^\top A(x - y))^{1/2}$ between points x, y in the policy space, so the policy utility is minus the squared A -distance between the policy and the ideal point. The empirical literature on voter ideal points assumes quadratic utility with homogeneous priorities A and finds that this specification explains survey responses well (see Section 1.1). Quadratic utility allows us to speak about "revealed ideal points" of voters who hold a non-degenerate belief π over their true ideal point θ , as the following remark shows.

Remark 1. *A voter with belief π over her ideal point θ votes as if she had the known ideal point $\mathbb{E}_\pi[\theta]$.*

By a bias-variance decomposition of the expected utility from policy x under belief π over θ ,

$$\mathbb{E}_\pi[u(x, \theta)] = u(x, \mathbb{E}_\pi[\theta]) - \mathbb{E}_\pi[(\theta - \mathbb{E}_\pi[\theta])^\top A(\theta - \mathbb{E}_\pi[\theta])]. \quad (3)$$

The latter term does not depend on x , so when a voter compares two platforms (or survey response items), she chooses the one that is closer to her posterior mean in the A -distance. Hence, a voter with belief π acts like a voter with a known ideal point of $\mathbb{E}_\pi[\theta]$. Accordingly, we call the posterior mean of a voter's belief π over θ her *revealed ideal point*. This ideal point is the one that is estimated from survey responses, which is important when we interpret empirical findings about voter ideology. We refer to the distribution $\rho \in \Delta(\mathbb{R}^n)$ over posterior means induced by learning as the *revealed ideology* in the population, to distinguish revealed ideology from the true distribution of ideal points, which we specify next.⁵

Voter ideal points $\theta_i \in \mathbb{R}^n$ are distributed according to a distribution $\mu \in \Delta(\mathbb{R}^n)$ with a log-concave spherical density f_μ and mean normalized to 0. The density f_μ is spherical if it is of the form $f_\mu(\theta) = g(\theta^\top \theta)$ with $g: \mathbb{R}_+ \rightarrow \mathbb{R}_+$. A spherical density is rotationally symmetric and implies that the ideal points are uncorrelated across dimensions, so there is no issue alignment at the outset. Log-concavity implies that the density is unimodal, so there is no polarization at the outset either. These assumptions allow for a non-parametric class of distributions, including the normal distribution $\mathcal{N}(0, k I_n)$ and a uniform distribution on a ball.

Learning Our departure from the standard spatial model is that voters do not know their ideal points. Instead, they share the prior μ , coinciding with the true distribution of ideal points, and can learn at a cost. Each voter can acquire *any* signal structure (Blackwell experiment) about her ideal point subject to an information cost, as in the rational-inattention literature following Sims (2003), surveyed by Maćkowiak, Matějka, and Wiederholt (2023). The information cost is the expected reduction in the variance of the voter's belief about her ideal point, rather than the expected reduction in entropy that is standard in this literature. The entropy-based cost possesses the three properties of the information cost on which the proof of Theorem 1 rests, so the main result on voter learning holds under the standard cost as well. We adopt the variance-based cost for tractability in the analysis of endogenous platforms.

To define the information cost, recall that a signal structure specifies a conditional distribution over signal realizations given any ideal point. Upon a signal realization, the voter forms a posterior via Bayesian updating. Thus, a signal structure induces a distribution $\tau \in \Delta(\Delta(\mathbb{R}^n))$ over posterior beliefs $\pi \in \Delta(\mathbb{R}^n)$, with which we identify the signal structure. The information cost of a signal structure is the expected reduction in the variance of the voter's belief,

$$c(\tau) = \text{Var}_\mu(\theta) - \mathbb{E}_\tau[\text{Var}_\pi(\theta)],$$

where $\text{Var}_\nu(X) := \mathbb{E}_\nu[\|X - \mathbb{E}_\nu[X]\|^2]$ is the multidimensional analogue of variance, defined for any distribution ν of an $\mathbb{R}^n$ -valued random vector X . The cost is non-negative and increases in the

⁵For any topological space S , we denote by $\Delta(S)$ the set of Borel probability measures on S .

Blackwell informativeness of the signal structure, capturing that learning takes time and effort. Like the information costs in recent contributions (e.g., Ravid, Roesler, and Szentes 2022; Thereze 2022), our information cost belongs to the class of posterior-mean separable information costs axiomatized in Mensch and Malik (2024). The information cost is particularly tractable in our setting because it equals the variance of posterior means by the law of total variance,

$$c(\tau) = \text{Var}_\tau(\mathbb{E}_\pi[\theta]),$$

which enables the posterior-mean formulation (4) of the voter learning problem below.

For voters to acquire costly information despite never being pivotal among the continuum of voters, we make the behavioral assumption that voters engage in *expressive voting and learning*, as is standard in the literature on rationally inattentive voters (Matějka and Tabellini 2021; Hu, Li, and Segal 2023; Li and Hu 2023). That is, voters care about voting for the preferred party given their true preferences and are willing to incur an information cost to do so. Possible reasons are that voters derive a psychological benefit from voting correctly or that they regard correct voting as a civic duty (see also Feddersen and Sandroni 2006). Thus, κ measures the cost of information relative to the psychological benefit of voting for the preferred party.

Expressive voting, the quadratic policy utility, and the information cost together allow us to state the voter’s learning problem as a choice of the distribution $\rho \in \Delta(\mathbb{R}^n)$ over posterior means induced by τ , as we derive in Appendix A.1:

$$\begin{aligned} \max_{\rho \in \Delta(\mathbb{R}^n)} \mathbb{E}_{\theta \sim \rho} \left[\left| (x_b - x_a)^\top A \left(\theta - \frac{x_a + x_b}{2} \right) \right| - \kappa \theta^\top \theta \right] \\ \text{s.t. } \rho \preceq_{\text{MPS}} \mu. \end{aligned} \quad (4)$$

The first term in the objective is the instrumental value of information, the payoff from the voting decision. A voter with revealed ideal point θ votes for the platform that is closer in the A -distance to θ , and the term measures by how much. Only the projection of θ onto the direction $A(x_b - x_a)$ relative to the party midpoint enters and information is valuable insofar as it moves the revealed ideal point away from the midpoint along that direction. The second term is the information cost. Because ρ preserves the prior mean of zero, the variance of the posterior means equals $\mathbb{E}_\rho[\theta^\top \theta]$. The constraint is a feasibility condition. A distribution $\rho \in \Delta(\mathbb{R}^n)$ over posterior means is induced by some signal structure if and only if ρ is a mean-preserving contraction of the prior μ , $\rho \preceq_{\text{MPS}} \mu$.⁶ The solution is unique (Lemma 1), so all voters acquire the same distribution over revealed ideal points, and we identify ρ with the revealed ideology in the population.⁷

Parties We assume that parties are policy-motivated (Wittman 1973; Calvert 1985). That is, party $j \in \{a, b\}$ has an ideal point $x_j^* \in \mathbb{R}^n$, known to both parties, and evaluates policies according

⁶ A distribution $\rho \in \Delta(\mathbb{R}^n)$ is a mean-preserving contraction of another distribution $\mu \in \Delta(\mathbb{R}^n)$ (or, equivalently, μ is a mean-preserving spread of ρ) if there exist $\mathbb{R}^n$ -valued random variables $X \sim \rho, Y \sim \mu$ with $\mathbb{E}[Y|X] = X$.

⁷ We follow the common practice of identifying a population distribution with the distribution of an individual agent. Deriving the identity from independent signal realizations across a continuum of voters raises measurability problems (Judd 1985), for which Uhlig (1996) and Sun (2006) provide solutions.

to the policy utility $u(\cdot, x_j^*)$. The utilities U_a and U_b of parties a and b , respectively, are

$$\begin{aligned} U_a(x_a, x_b) &= P_a(x_a, x_b)u(x_a, x_a^*) + (1 - P_a(x_a, x_b))u(x_b, x_a^*) \\ U_b(x_a, x_b) &= P_a(x_a, x_b)u(x_a, x_b^*) + (1 - P_a(x_a, x_b))u(x_b, x_b^*), \end{aligned}$$

where $P_a(x_a, x_b)$ is the probability that party a 's policy x_a gets implemented. We make the common assumption that this implementation probability is the vote share (Wittman 1983; Wittman 1990; Callander and Carbajal 2022; Yuksel 2022). A continuous mapping from vote share to implementation probability can result from noise voters or random turnout of partisans (e.g., Feddersen and Pesendorfer 1996), or from non-majoritarian institutions in parliament.⁸ Continuity of this mapping introduces electoral uncertainty, which is what allows for platform divergence in equilibrium. The identity mapping we adopt for tractability, but the forces we identify do not depend on it. The moderation and polarization motives operate through the response of the vote share to a party's platform, so both are present for any implementation probability that is strictly increasing in the vote share.

Equilibrium We study subgame perfect pure-strategy equilibria in which both parties receive positive vote shares. In such an equilibrium, each voter votes for her preferred party at the election stage, voters learn optimally given party platforms, and parties choose platforms optimally anticipating voter behavior.⁹ By Remark 1, a voter with revealed ideal point θ votes at the election stage for the platform with smaller A -distance to θ , and we assume that indifferent voters vote for either party with equal probability.¹⁰ Given the voting behavior at the election stage, a strategy profile reduces to a triple $(x_a, x_b, \rho(\cdot, \cdot))$ consisting of two platforms and a learning strategy $\rho: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \Delta(\mathbb{R}^n)$ that assigns a revealed ideology to every pair of platforms. The vote share of party a is then

$$\begin{aligned} P_a(x_a, x_b) &= \rho(x_a, x_b)(\{\theta \in \mathbb{R}^n \mid (x_a - \theta)^\top A(x_a - \theta) < (x_b - \theta)^\top A(x_b - \theta)\}) \\ &\quad + \frac{1}{2} \rho(x_a, x_b)(\{\theta \in \mathbb{R}^n \mid (x_a - \theta)^\top A(x_a - \theta) = (x_b - \theta)^\top A(x_b - \theta)\}). \end{aligned} \quad (5)$$

Equilibrium amounts to three conditions on $(x_a, x_b, \rho(\cdot, \cdot))$. First, for all $(x'_a, x'_b) \in \mathbb{R}^n \times \mathbb{R}^n$, the revealed ideology $\rho(x'_a, x'_b)$ solves the voter's learning problem (4) given (x'_a, x'_b) . Second, x_a maximizes $U_a(\cdot, x_b)$ and x_b maximizes $U_b(x_a, \cdot)$, where P_a is given by (5). Third, both parties receive a positive vote share, $P_a(x_a, x_b) \in (0, 1)$.

⁸As Yuksel (2022) notes, if half the population consists of noise voters and their vote share for party a is uniformly distributed, then the implementation probability is exactly the vote share from non-noise voters.

⁹If one party received no votes, that party's payoff would be independent of its own platform, so the party would be indifferent across all platforms that leave its vote share at zero. Platform pairs of this kind do not survive an arbitrarily small office benefit, because the party without votes can adopt the platform of its opponent, which leaves the implemented policy unchanged and raises its vote share and thereby office benefit.

¹⁰The exact tie-breaking rule is inconsequential. If the two platforms differ, then no voter is indifferent at the election stage (Lemma 5). If the two platforms coincide, then both parties obtain the same payoff irrespective of the vote shares.

2.1 Discussion of Assumptions

Our model of voter learning is intentionally abstract and admits several concrete interpretations, which we now spell out.

Learning about Ideal Points A voter’s ideal point is the policy she would most like to see implemented, so learning about her ideal point means learning which policies are good for her. Such learning can be interpreted as (i) introspecting on how to *value* the consequences of policies or as (ii) learning about the private *consequences* of policies. As an example, consider a voter’s position on income taxation. To determine her optimal tax policy, the voter may want to (i) introspect on her values for equity versus efficiency, and (ii) learn which tax bracket she is in and what other economic consequences the policy has. Learning about the consequences of policies can proceed through the news media, an interpretation we return to below. Voters form political opinions in many ways, and the force we identify can be at work under either interpretation. We therefore remain agnostic as to which interpretation should be adopted.

Voter learning is most readily interpreted as the formation of a political opinion early in life. However, our model of voter learning also applies to older voters who formed political beliefs ahead of earlier elections and now revisit these beliefs because platforms or their ideal points have changed. For such voters, a common prior expectation is implausible, but Section 3.1 shows that our results extend in a meaningful way to heterogeneous prior expectations.

We assume voters can acquire costly information about their ideal points, but they costlessly observe party platforms. This approach allows us to make clear what mechanisms result from endogenous voter learning about ideal points, as opposed to learning about platforms (as in Matějka and Tabellini 2021). However, as we illustrate in Online Appendix D.1, our results on voter ideology also hold when voters do not know platforms but observe a noisy signal about platforms, based on which they choose how to learn about ideal points. The substantive restriction is that voters do not learn *jointly* about platforms and ideal points, which is an interesting avenue for future research.

Flexible Information Acquisition We place no exogenous restriction on the information a voter can acquire, so her incentives alone determine what she learns. In particular, we do not require signals about different policy issues to be independent. Voters can therefore acquire a signal that tells them whether they are left or right, aggregating their positions on several issues into a single dimension.

Our results require only the optimal signal structure to be available to voters, so the plausibility of flexible learning can be judged from that signal structure alone. By Theorem 1, the optimal signal is binary and aggregates the policy issues into a single dimension. We discuss the plausibility of such learning under each of the two interpretations of voter learning.

When voter learning is interpreted as introspection on values, we can think of the voter as imagining two policy bundles that differ across several issues and introspecting on their relative desirability. Comparisons of that kind are the object of drift-diffusion models in psychology and neuroscience, in which an agent accumulates noisy evidence on the relative value of two alternatives

until she is confident enough to decide (Ratcliff 1978). Static information costs of the posterior-separable form we use arise as reduced forms of such sequential sampling (Hébert and Woodford 2023; Morris and Strack 2019). By not restricting information, our approach also stays true to the original motivation of rational inattention as modeling the brain as an efficient information processor subject only to information capacity constraints (Sims 2003).

When voter learning is interpreted as learning about the private consequences of policies, aggregation across issues can be performed by information intermediaries such as the news media, as in Hu, Li, and Segal (2023). Voters may obtain such information from sufficiently personalized media outlets, such as news feeds or newspapers catering to specific demographic groups or values. Such outlets may aggregate information about different policy issues into a one-dimensional signal, as other models of media assume (Duggan and Martinelli 2011; Perego and Yuksel 2022; Yuksel 2022). The signal tells readers whether the left or the right policy bundle serves them better. Editorial endorsements of candidates, as well as opinion programs and columns that argue for a left or a right worldview across issues, can be interpreted in this way.

3 Voter Learning

This section characterizes optimal voter learning given party platforms x_a and x_b , assuming $x_a \neq x_b$ (otherwise, voters learn nothing). The next section endogenizes the platforms. This section can be read independently of the political-economy application: the results apply analogously to an industrial-organization setting with horizontally differentiated goods in a product-attribute space with consumers learning about their ideal points.

The following lemma shows by standard arguments that a solution to the voter learning problem exists and is unique.

Lemma 1. *The voter learning problem (4) has a unique solution.*

The main result of this section describes the revealed ideal points (posterior means) that are induced by the optimal voter learning strategy.

Theorem 1. *The solution to the voter learning problem (4) is supported on at most two revealed ideal points on the line through the prior mean with direction $A(x_b - x_a)$.*

The left panel of Figure 2 visualizes the result. Even though the true distribution of ideal points is multidimensional and unimodal, revealed ideology is one-dimensional and supported on at most two points. By implication, revealed ideal points are perfectly aligned across issues, even when the true ideal points are independent across them. Because the line in the theorem does not depend on κ , the revealed ideal points of all voters lie on the same line even when voters differ in their information cost. Heterogeneity in κ spreads the revealed ideal points along the line, and revealed ideology remains bimodal, that is, polarized, when the distribution of κ is sufficiently concentrated (right panel of Figure 2). We interpret the line as an *endogenous left-right axis* of voter ideology, the one-dimensional spectrum that much of formal political economy assumes. In Online Appendix 1.1,

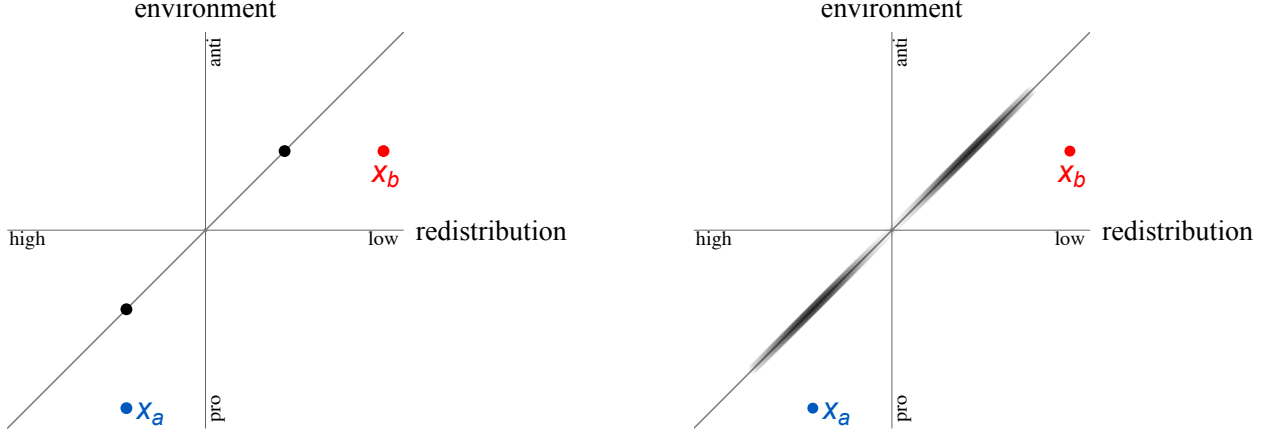


Figure 2: Revealed ideal points ($\bullet$) under party platforms x_a, x_b and priorities $A = I_n$. Left panel: common information cost parameter κ . Right panel: κ heterogeneous across voters.

we show that revealed ideal points lying on a line is exactly the property identified by the survey evidence on issue alignment (Proposition 6).

On a high level, the intuition for the result is that voters learn only about decision-relevant information, which leads them to ignore all but one dimension of the policy space and to learn in a binary fashion. This is familiar from the rational-inattention literature, where agents ignore payoff-irrelevant dimensions and acquire no more signals than there are actions. Surprisingly, the connection to issue alignment and polarization has gone unnoticed, and the literature has pursued more complicated explanations for these phenomena instead. Having said that, the alignment result does not follow mechanically from a standard rational-inattention argument. Instead, the proof rests on a novel reflection argument that exploits symmetries of the decision problem. Importantly, the argument does not rely on the controversial invariance property of mutual information (Dean and Neligh 2023).

Suppose by way of contradiction that voters acquired information such that the induced distribution ρ over posterior means was *not* supported on at most two points on the line L through the prior mean with direction $A(x_b - x_a)$ (Figure 3(a)). We construct in four steps a distribution with the same instrumental value but a lower cost. First, consider the distribution obtained by reflecting ρ across L (Figure 3(b)). The reflected distribution is feasible because the prior is spherical. The instrumental value in the voter objective (4) depends on a posterior mean θ only through its projection onto L , whereas the cost depends only on the squared distance $\theta^\top \theta$ to the prior mean. Reflection across L preserves the projection onto L as well as the distance to the prior mean, so the voter is indifferent between ρ and its reflection. Second, because the information cost is posterior separable, the voter is indifferent to mixing the two distributions, and hence to acquiring the mixture $\frac{1}{2}\rho + \frac{1}{2}\text{ref}(\rho)$ (Figure 3(c)) instead. This mixture is symmetric about L . Third, this symmetry implies that projecting the mixture orthogonally onto L is a mean-preserving contraction (Figure 3(d)). This mean-preserving contraction corresponds to a garbling that reduces the information cost while preserving the instrumental value, since it maintains the projection on

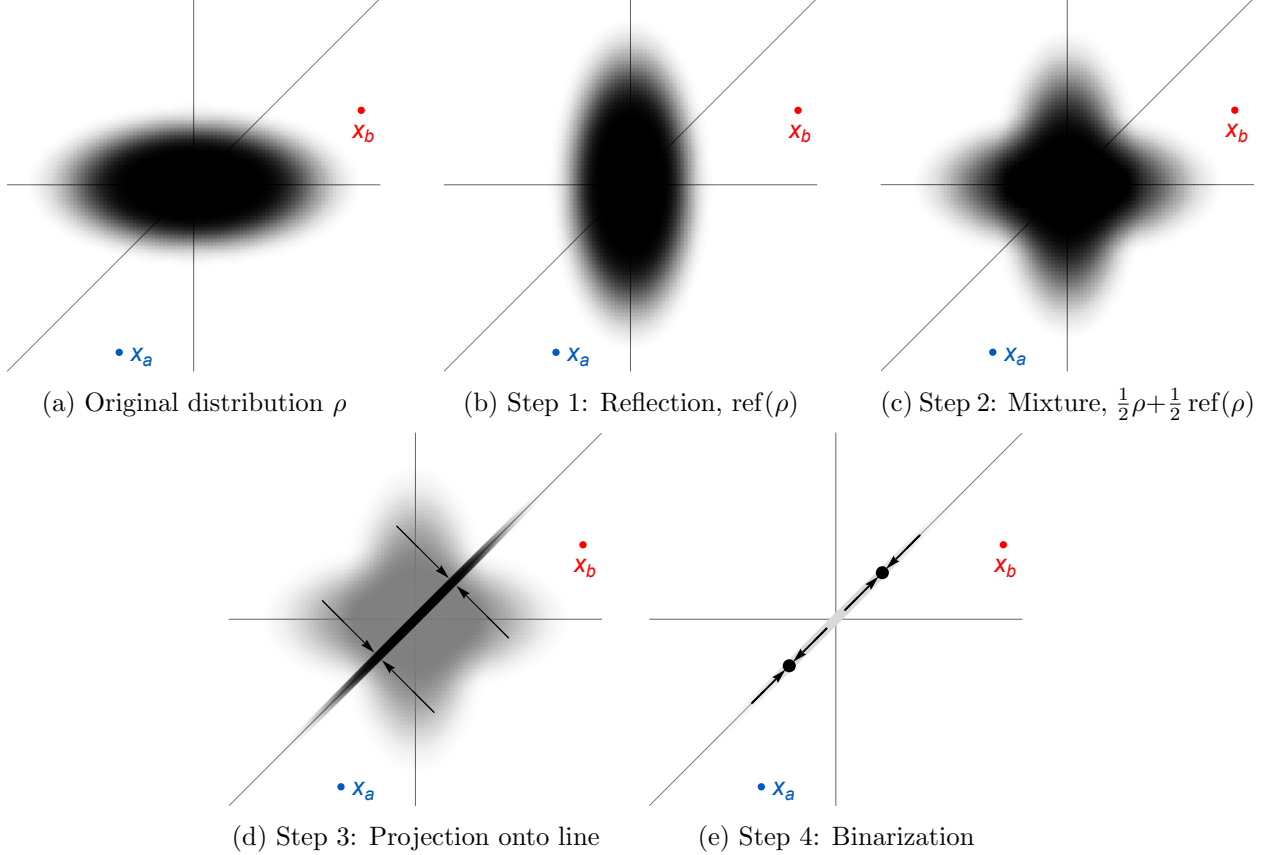


Figure 3: Reflection argument underlying the proof of Theorem 1, for $A = I_n$.

L. Finally, garbling once more according to the action recommendation yields a distribution supported on at most two points, with the same instrumental value and a further reduced information cost (Figure 3(e)). The argument uses only reflection invariance, posterior separability, and strict Blackwell monotonicity of the information cost.

Issue alignment survives when voters also want to learn how strongly they prefer one party over the other. Suppose voters care about the intensity of their preference, for instance because they vote only when the expected utility difference between the parties is large enough and abstain otherwise. The first three steps of the proof go through unchanged, so revealed ideal points still lie on the line with direction $A(x_b - x_a)$. The binarization step no longer applies. Alongside the two groups, a third emerges, which can be interpreted as disengaged voters who do not feel strongly enough about either party to vote.

The sharp form of issue alignment in Theorem 1 rests on quadratic utility, the spherical prior, and reflection invariance of the information cost. Without those assumptions the revealed ideal points need not lie exactly on a fixed line. However, the force behind the result is more general. Information is valuable to a voter only insofar as it helps her compare the two parties, and that comparison aggregates all issues on which the parties differ. She therefore would like to learn about the issues jointly rather than one at a time, which correlates her positions across issues. Our assumptions deliver this mechanism in its clearest form. The resulting characterization yields

comparative statics of revealed ideology and makes the analysis of endogenous party platforms in Section 4 tractable.

3.1 Comparative Statics and Extensions

The model delivers comparative statics of voter ideology that behavioral accounts of polarization and issue alignment do not readily generate. Polarization and issue alignment are both stronger among better-informed voters, and both rise with party polarization. In the model, low information costs and distant party platforms induce more voter learning, and voter learning is what produces polarization and issue alignment. Comparative statics of flexible information acquisition are notoriously difficult, but our setting is tractable enough to deliver them.

Cheaper information and more distant platforms both polarize revealed ideology, in the sense that the two ideological poles of revealed ideology move further apart. Lower information costs induce more voter learning, and voter learning is what spreads revealed ideal points apart.

Proposition 1. *Suppose $x_a^\top Ax_a = x_b^\top Ax_b$. Then revealed ideology is symmetric around the prior mean, and the revealed ideal points are weakly further apart when the information cost parameter κ is smaller or when $\|A(x_b - x_a)\|$ is larger.*

The proof adapts the reflection argument behind Theorem 1, reflecting around the hyperplane on which the voter is indifferent between the two platforms rather than around the line L . The hypothesis $x_a^\top Ax_a = x_b^\top Ax_b$ says the two platforms are equally far from voters' prior mean according to voter priorities A , which places that hyperplane through the prior mean. The reflection then preserves both feasibility and the voter's payoff, and uniqueness implies the solution is invariant under the reflection. Together with Theorem 1, only a one-dimensional family of candidate solutions survives, and the comparative statics follow from increasing differences along the family.

Under symmetric party platforms, the comparative statics follow from reflection invariance, posterior separability, and strict Blackwell monotonicity of the information cost alone. For the posterior-variance cost, the closed form solutions in (7) below extend the comparative statics to asymmetric party platforms, as long as the pair $\{\theta_a, \theta_b\}$ defined in (7) remains feasible. Once feasibility binds, the distance between the ideological poles increases in κ , so the comparative statics reverse for small enough information costs. Equally distant party platforms are a reasonable first approximation for US politics.

The prediction that better-informed voters, those with a low κ , are more polarized is consistent with the evidence from the US. Palfrey and Poole (1987) construct an index of voter information and find that better-informed voters take more extreme positions, and Abramowitz and Saunders (2008) and Abramowitz (2010) find that more educated and more politically engaged voters are more ideologically extreme. Causal evidence points the same way. Baysan (2022) finds that randomized information campaigns before a Turkish referendum persistently polarized voters, and Schneider-Strawczynski and Valette (2025) find that heavier television coverage of immigration in France moves initially moderate viewers toward both extremes.

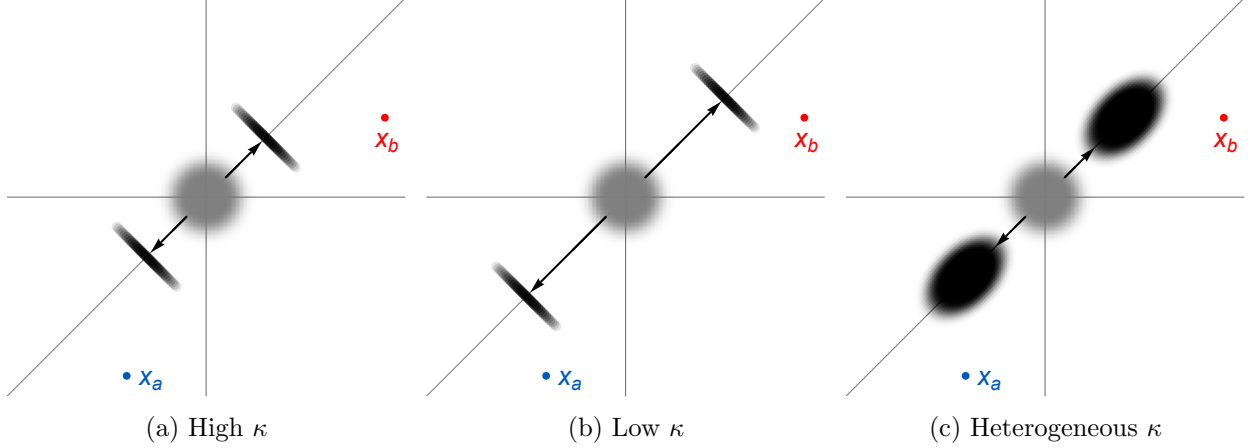


Figure 4: Heterogeneous prior means: the gray cloud represents prior expectations and the black clouds revealed ideal points.

Issue alignment does not rely on voters sharing the same prior expectation about their ideal point. Suppose voters differ in what they expect their ideal point to be before learning, and that each voter's prior is spherical. By Theorem 1, each voter still learns along the direction $A(x_b - x_a)$, and the two revealed ideal points lie on the line through her own prior expectation with direction $A(x_b - x_a)$. The marginal distribution of revealed ideal points orthogonal to $A(x_b - x_a)$ is therefore unchanged by learning, while the marginal along $A(x_b - x_a)$ undergoes a mean-preserving spread. Revealed ideology is no longer exactly one-dimensional, but positions across issues still become correlated through learning. We take the share of the variance of revealed ideal points that is explained by the component along $A(x_b - x_a)$ as a measure of issue alignment. Cheaper information and more polarized parties raise the spread along $A(x_b - x_a)$ and leave the orthogonal dispersion unchanged, so both raise issue alignment.¹¹ Better-informed voters should therefore display more issue alignment, in line with evidence relating issue alignment to political interest and knowledge (Converse 1964; Hare 2022). Issue alignment should also rise with party polarization, in line with the rise of issue alignment over the decades in which US parties polarized (Freeze and Montgomery 2016; Hare 2022).

Figure 4 (a)-(c) illustrates. The gray cloud collects prior expectations and the black shapes collect the revealed ideal points they give rise to. The position along $A(x_b - x_a)$ of the two revealed ideal points is common to all voters, so the cloud of prior expectations is mapped onto two segments orthogonal to $A(x_b - x_a)$, at a distance from each other that is governed by the cost of information. Panels (a) and (b) differ only in that cost, and cheaper information moves the two segments further apart while leaving their length unchanged. Panel (c) lets the cost of information differ across voters as well, which replaces the two segments by two clouds.

Theorem 1 relies on there being only two parties, but an appropriate extension survives with more parties. Under plurality rule, in a Duvergerian equilibrium where voters decide between the two front-runners, the mechanism still applies.¹² Under proportional representation, expressive

¹¹See Online Appendix D.3 for a formal analysis.

¹²Although we assume expressive voting, strategic voting does not undermine our mechanism under private values

voting may remain a reasonable assumption. With $k > 2$ parties an analogous reflection argument shows that revealed ideal points lie on a fixed affine subspace of dimension at most $k - 1$. If the k platforms happen to lie on a line, voter ideology is still one-dimensional. If party platforms lie in general position, the dimensionality of voter ideology rises with the number of relevant parties.

Corollary 1. *Suppose voters face k party platforms $x_1, \dots, x_k$ and choose a Bayes-consistent $\tau \in \Delta(\Delta(\mathbb{R}^n))$ to maximize $\mathbb{E}_\tau[\max_{j=1, \dots, k} \mathbb{E}_\pi[u(x_j, \theta)]] - \kappa c(\tau)$. Then any solution is supported on at most k revealed ideal points, contained in the subspace through the prior mean spanned by $\{A(x_j - x_1)\}_{j=2, \dots, k}$ with dimension at most $k - 1$.*

This result aligns with evidence from Western European countries with more than two relevant parties. For these countries, political scientists argue a two-dimensional space is needed to describe voter and party platforms (Kriesi, Grande, Lachat, Dolezal, Bornschier, and Frey 2006; Kriesi, Grande, Lachat, Dolezal, Bornschier, and Frey 2008; Bornschier 2010a; Bornschier 2010b; however, see Van Der Brug and Van Spanje 2009).

3.2 Vote Shares

The vote shares induced by voter learning, and the bias established below, are the central inputs to the analysis of electoral competition in the next section. By Theorem 1, revealed ideology is supported on at most two points, so the vote share $P_a(x_a, x_b)$ in (5) is the weight that the unique optimal revealed ideology places on the revealed ideal point that is A -closer to x_a than to x_b . The vote shares differ in general from the shares of voters whose true ideal point is A -closer to one platform than to the other, given by the same masses under the prior μ .

The next result shows that costly learning biases the vote share toward the moderate party, the party whose position is A -closer to the prior mean 0. The moderate party receives a larger vote share than the share of voters whose true ideal point is closer to its position, even though voters update beliefs in an unbiased, Bayesian manner.

Proposition 2. *Suppose $x_a^\top A x_a < x_b^\top A x_b$. Then party a 's vote share $P_a(x_a, x_b)$ weakly exceeds the share of voters whose true ideal point is A -closer to x_a than to x_b . The vote share $P_a(x_a, x_b)$ weakly increases if we raise the information cost parameter κ , and weakly decreases if we scale up $x_b - x_a$ while holding the midpoint $\frac{x_a + x_b}{2}$ fixed.*

The high-level reason for the bias is that asymmetric signals are less informative and hence cheaper. Voters acquire signals biased toward the moderate party to economize on their information cost. The extreme case is immediate: when information is prohibitively costly, voters learn nothing and all vote for the moderate party. The proposition shows that the bias persists at every finite cost and identifies comparative statics. The bias is smaller when information is cheaper or when parties are more polarized, because in those cases the motive to economize on information is weaker.

as long as voters perceive a positive probability of being pivotal.

Under rational inattention, behavior can be biased toward ex-ante more attractive options (Matějka and McKay 2015). Precise statements outside the two-state, two-action case are not available to our knowledge, because the unconditional choice probabilities are hard to characterize under a mutual-information cost. Our setting, with quadratic utility and a posterior-variance cost, is more tractable. The key is that the voter learning problem reduces to a one-dimensional mean-measurable information-design problem, which is well understood.

To unpack the intuition in our setting, consider the signal structures that implement the distribution of revealed ideal points as κ rises. By Theorem 1 this distribution is one-dimensional and it can be implemented by a signal that informs voters only about the scalar projection of their ideal point onto the axis $A(x_b - x_a)$. The following discussion therefore concerns signal structures that are functions of this scalar projection. Voters are indifferent between the two parties at the scalar projection of $\frac{x_a + x_b}{2}$, which we call the party midpoint. When κ is small, voters acquire a threshold signal with threshold near the party midpoint. The signal almost perfectly separates voters according to which party their true ideal point is closer to, so vote shares approximate the true shares. As κ increases, the optimal threshold becomes more extreme than the party midpoint and moves toward the extreme party. This happens because, under the log-concave prior, a more extreme threshold reduces the posterior-variance information cost. A threshold more extreme than the party midpoint assigns more voters to the moderate party, which is the source of the bias. As κ rises further, voters shift from adjusting the threshold to garbling the binary signal. For these values of κ , the two revealed ideal points are symmetric about the party midpoint, which equalizes the cost of a mistaken vote for each party from garbling. Higher κ contracts these two revealed ideal points toward the party midpoint while preserving this symmetry. The vote shares are exactly the weights that make the two revealed points average to the prior mean. Since the prior mean lies on the moderate party’s side of the party midpoint, contracting the points toward the party midpoint raises the vote share for the moderate party. Finally, when the cost is high enough, voters acquire no information and all vote for the moderate party.

Our notion of bias toward the moderate party matches the definition of *media bias* in Gentzkow, Shapiro, and Stone (2015), so the model illustrates a rational-inattention-based rationale for demand-driven media bias. Recall that one interpretation of voter learning is that information is supplied by an intermediary such as the news media. Because the moderate party is defined as the one closer to the voter’s prior, the bias favors prior-aligned news. Most demand-driven explanations of media bias rest on a psychological utility from news. We instead show that media bias can be a rational way for voters to economize on their information cost.¹³ The two channels behind this bias map onto the two forms of media bias in Gentzkow, Shapiro, and Stone (2015): a more extreme threshold is a filtering bias and asymmetric garbling adds a distortion bias.

Proposition 2 is central for electoral competition. Endogenous information creates two new

¹³Our result complements Che and Mierendorff (2019). In their dynamic model voters choose between left- and right-biased news under a time cost, and voters with extreme beliefs acquire news biased toward their current belief. Our model allows flexible information acquisition, so voters may acquire unbiased news, yet the information they choose is always biased toward the party closer to their prior expectation of the optimal policy.

motives in party positioning, a **moderation motive** and a **polarization motive**. First, the bias toward the moderate party gives parties an additional reason to moderate, and this incentive strengthens as information becomes more costly. Second, the extreme party has an incentive to polarize, since greater polarization reduces the bias toward the moderate party. The same logic gives the moderate party an incentive to move toward the extreme party, since proximity raises the bias in its own favor. The next section shows that the moderation motive drives symmetric equilibria and the polarization motive is central for understanding asymmetric equilibria.

4 Electoral Competition

Solving for the equilibrium platforms is challenging because of the multidimensional policy space and endogenous voter learning. Throughout this section we set uniform priorities, $A = I_n$, for tractability. The revealed ideal points and the vote shares then admit particularly simple closed-form solutions. To state them, let $\text{Proj}_{y,z}$ denote the orthogonal projection onto the line through $y, z \in \mathbb{R}^n$,

$$\begin{aligned} \text{Proj}_{y,z} : \mathbb{R}^n &\rightarrow \mathbb{R}^n \\ x &\mapsto y + \frac{(z - y)^\top (x - y)}{\|z - y\|^2} (z - y). \end{aligned} \quad (6)$$

When the prior is dispersed enough, in a sense made precise below, optimal learning at platforms $x_a \neq x_b$ induces the two revealed ideal points

$$\begin{aligned} \theta_a(x_a, x_b) &:= \text{Proj}_{0, x_b - x_a} \left(\frac{x_a + x_b}{2} \right) - \frac{x_b - x_a}{2\kappa}, \\ \theta_b(x_a, x_b) &:= \text{Proj}_{0, x_b - x_a} \left(\frac{x_a + x_b}{2} \right) + \frac{x_b - x_a}{2\kappa}. \end{aligned} \quad (7)$$

Both revealed ideal points lie on the line through the origin with direction $x_b - x_a$. The first term of (7) is the orthogonal projection of the party midpoint onto that line, and the two revealed ideal points lie at equal distance on either side of that projection. These solutions imply a simple expression for the vote share,

$$P_a(x_a, x_b) = \frac{1}{2} + \frac{\kappa}{2} \frac{\|x_b\|^2 - \|x_a\|^2}{\|x_b - x_a\|^2}.$$

The closed-form solutions hold only when the revealed ideal points are feasible, that is, when some distribution on $\{\theta_a, \theta_b\}$ is a mean-preserving contraction of the voter prior μ . We return to the consequences of binding feasibility below.

To characterize equilibrium, we prove a best response lemma for the multidimensional case that helps reduce the electoral competition game to one dimension. The best response of a party splits into two choices: the direction in which the party differentiates from its opponent, and, given that direction, the extent of differentiation. Fixing the direction restricts the two platforms to a line. We show that the multidimensional party game restricted to such a line is equivalent

to a one-dimensional party game in which the policy space is collapsed to that line. This game is tractable because the voter learning problem reduces to a one-dimensional mean-measurable information design problem, which is well understood. The resulting solution to the voter problem makes a party's best response affine in the opponent's projected platform wherever feasibility does not bind, which gives the optimal extent of differentiation along a given direction. Substituting the optimal extent of differentiation back into the party objective leaves a maximization over the direction of differentiation alone, which we solve in closed form. Party j 's best-response direction passes through $(1 - \kappa)x_a^*$. In equilibrium the line spanned by the two platforms therefore passes through $(1 - \kappa)x_a^*$ and through $(1 - \kappa)x_b^*$, and the platforms along that line coincide with those of the corresponding one-dimensional model.

The argument is considerably complicated by the possibility that feasibility binds, in which case the closed-form solutions fail. Even if feasibility does not bind at a given pair of platforms, establishing that a platform is a best response requires ruling out deviations into the region where feasibility binds. Much of the work in the proof goes into these deviations and into verifying that feasibility does not bind at the platforms we characterize.

The remainder of this section characterizes equilibrium, first under a symmetric setup in which parties are equally ideologically extreme, then under an asymmetric setup in which one party is more ideologically extreme. Two motives shape the equilibrium. A moderation motive pulls parties toward the center of the voter distribution, and a polarization motive pushes them apart. The symmetric setup isolates the moderation motive, and the asymmetric setup brings in the polarization motive.

4.1 Symmetric Setup

We first characterize the equilibrium under a symmetric setup, that is, when parties are equally ideologically extreme from the viewpoint of voters, $\|x_a^*\| = \|x_b^*\|$. For the statement of the theorem, define $\underline{\kappa}$ via

$$\frac{1 - \underline{\kappa}}{\underline{\kappa}} \|x_a^*\| = \mathbb{E}_\mu[|\theta_1|].$$

where θ_1 denotes the first component of θ . The parameter $\underline{\kappa}$ is the smallest information cost parameter for which the closed-form solution (7) for the voter learning problem holds at the reflected platform pairs $(-x_b, x_b)$ and $(x_a, -x_a)$, and hence also at the equilibrium platforms. Throughout, δ_x denotes the Dirac measure placing unit mass at x .

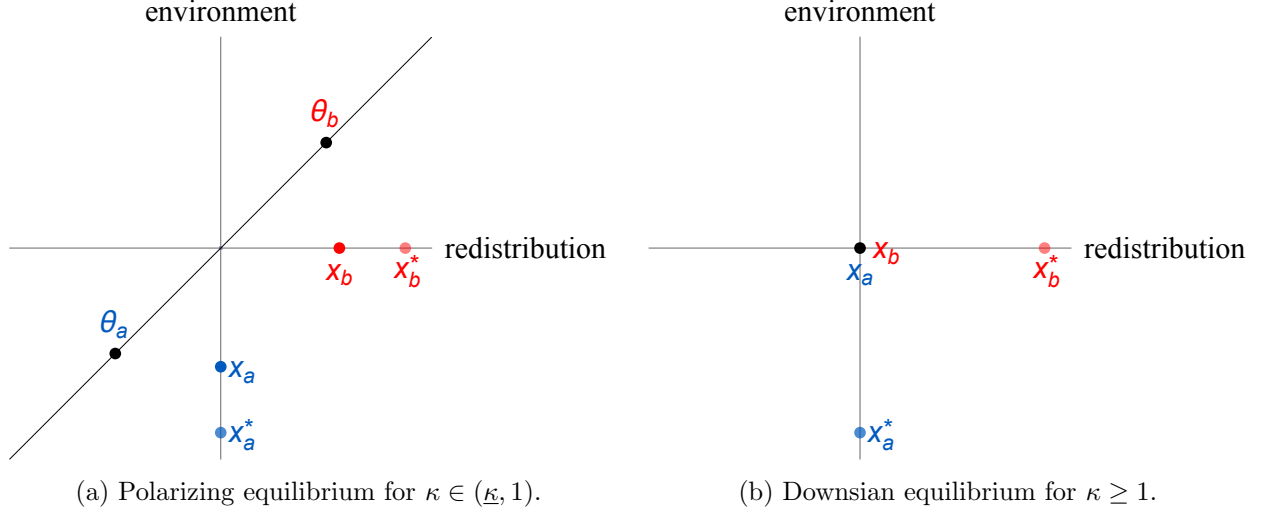


Figure 5: Equilibria under symmetric setup.

Theorem 2. Let $\|x_a^*\| = \|x_b^*\|$ and $x_a^* \neq x_b^*$. Define $\theta_j(\kappa) := \text{Proj}_{0, x_b^* - x_a^*}((1 - \kappa)x_j^*/\kappa)$ for $j \in \{a, b\}$. The following describes an equilibrium in two parameter regimes.

	Party positions	Revealed voter ideology
$\kappa \in (\underline{\kappa}, 1)$: Polarizing equilibrium	$(x_a, x_b) = (1 - \kappa)(x_a^*, x_b^*)$	$\rho = \frac{1}{2}\delta_{\theta_a(\kappa)} + \frac{1}{2}\delta_{\theta_b(\kappa)}$
$\kappa \geq 1$: Downsian equilibrium	$(x_a, x_b) = (0, 0)$	$\rho = \delta_0$

The equilibrium is unique for $\kappa \geq 1$. For $\kappa \in (\underline{\kappa}, 1)$ with $\kappa \neq 1/2$, the equilibrium is unique if $\|x_b^* - x_a^*\| < \kappa \mathbb{E}_\mu[|\theta_1|]$.

The theorem describes two types of equilibria, depending on the information cost parameter, displayed in Figure 5.

For large information costs, there is a *Downsian equilibrium*, reminiscent of the median-voter theorem (Downs 1957). In this equilibrium, parties fully converge at the voter's prior expectation of the optimal policy. Voters acquire no information and split their votes arbitrarily between the two parties.

For smaller information costs, there is what we call a *polarizing equilibrium*. In this equilibrium, parties diverge and voters acquire information, which polarizes the population. The platforms and the revealed ideal points are available in closed form.

The smaller the information cost, the more parties and voters polarize. The prediction on party polarization may be surprising, since one would expect parties to cater more to better informed voters. That is what happens when voter learning is about party platforms rather than about their own ideal points (Matějka and Tabellini 2021). Our comparative statics are driven by the moderation motive of parties. For a high-level intuition, recall Proposition 2: because voters economize on the cost of information, their learning is biased toward the moderate party. This bias encourages parties to moderate. As information becomes cheaper, voters have less reason to economize, the bias weakens, and parties polarize more.

To explain the intuition in more detail, let us first recall parties' trade-off when choosing positions. Policy-motivated parties trade off votes (winning often) and ideology (implementing a favorable policy when winning). Parties gain votes by moving closer to voters' prior expectation. Parties gain ideological utility by moving closer to their own ideal policies, polarizing. The optimal party platforms equate the loss of votes with the gain in ideological utility from polarizing further. As a consequence, the smaller the loss of votes, the larger is the equilibrium level of polarization. The bias toward the moderate party creates an additional loss of votes from polarizing further than the other party, dampening polarization. However, the bias toward the moderate party recedes as information becomes cheaper, creating more party polarization.

A natural conjecture is that cheaper information benefits voters, both directly through a lower cost of learning and indirectly through better-informed voting. The next result shows the opposite in the polarizing regime: cheaper information reduces aggregate voter welfare. We define aggregate voter welfare W under party platforms (x_a, x_b) and revealed voter ideology ρ as the total policy utility minus the total information cost:

$$W(x_a, x_b, \rho) = P_a(x_a, x_b)\mathbb{E}_\mu[u(x_a, \theta)] + (1 - P_a(x_a, x_b))\mathbb{E}_\mu[u(x_b, \theta)] - \kappa c(\rho).$$

We comment below on how the welfare comparative static would change if we included expressive utility from voting for one's preferred party in voter welfare.

Corollary 2. *Let $\|x_a^*\| = \|x_b^*\|$ and $x_a^* \neq x_b^*$. On $\kappa \in (\underline{\kappa}, 1]$, equilibrium aggregate voter welfare W is strictly increasing in κ .*

The intuition has two parts. First, lower κ increases party polarization, moving implemented policies further from the welfare-maximizing policy at the center of the voter distribution, the origin. Second, one might expect voters to reduce their information cost as the information becomes cheaper. However, cheaper information and greater platform polarization make voters acquire more information, and the increase in information acquisition overturns the direct benefit of a smaller information cost parameter.

Under expressive utility from voting for one's preferred party, platform polarization can be beneficial by giving voters a choice to vote for the closer party. A similar benefit arises under aggregate uncertainty about voter ideal points, where platform polarization makes policy responsive to the realized preference distribution and can raise welfare (Bernhardt, Duggan and Squintani 2009). Polarization in our model, however, is not a pure spread of the platforms. Lower information costs also move the midpoint of the platforms away from the origin and toward the midpoint $\frac{x_a^* + x_b^*}{2}$ of the parties' ideal points, which biases policy and lowers welfare. The negative effects on policy utility and on the information cost discussed above remain.

4.2 Asymmetric Setup

We now turn to the equilibrium when one party is more ideologically extreme, which without loss of generality we take to be party b , so that $\|x_a^*\| < \|x_b^*\|$.

Even under asymmetry, the equilibrium is in closed form on an explicitly characterized region of the parameter space. Define the party platforms

$$\begin{aligned} x_a(\kappa) &:= \text{Proj}_{(1-\kappa)x_a^*, (1-\kappa)x_b^*} \left(\frac{1-\kappa}{1-2\kappa} ((1-\kappa)x_a^* + \kappa x_b^*) \right) \\ x_b(\kappa) &:= \text{Proj}_{(1-\kappa)x_a^*, (1-\kappa)x_b^*} \left(\frac{1-\kappa}{1-2\kappa} ((1-\kappa)x_b^* + \kappa x_a^*) \right), \end{aligned} \quad (8)$$

the revealed ideal points

$$\begin{aligned} \theta_a(\kappa) &:= \text{Proj}_{0, x_b^* - x_a^*} \left(\frac{1-\kappa}{1-2\kappa} \frac{x_a^* + x_b^*}{2} \right) - \frac{1-\kappa}{2\kappa} (x_b^* - x_a^*) \\ \theta_b(\kappa) &:= \text{Proj}_{0, x_b^* - x_a^*} \left(\frac{1-\kappa}{1-2\kappa} \frac{x_a^* + x_b^*}{2} \right) + \frac{1-\kappa}{2\kappa} (x_b^* - x_a^*), \end{aligned} \quad (9)$$

and party a 's vote share

$$P_a(\kappa) := \frac{1}{2} + \frac{\kappa}{2(1-2\kappa)} \frac{\|x_b^*\|^2 - \|x_a^*\|^2}{\|x_b^* - x_a^*\|^2}. \quad (10)$$

The characterization covers the case in which the two parties are ideologically on opposite sides of the center of the voter distribution. Let $\text{proj}_v(x) := \frac{v^\top x}{\|v\|}$ denote the scalar projection onto $v \in \mathbb{R}^n$, and set $\check{x}_a^* := \text{proj}_{x_b^* - x_a^*}(x_a^*)$ and $\check{x}_b^* := \text{proj}_{x_b^* - x_a^*}(x_b^*)$. The relevant case is $\check{x}_a^* < 0 < \check{x}_b^*$, in which the ideal policies project onto opposite sides of the origin along the ideological difference $x_b^* - x_a^*$.

Define the set of information cost parameters

$$\mathcal{K} = \left\{ \kappa \in \left(0, \frac{\check{x}_b^* - \check{x}_a^*}{3\check{x}_b^* - \check{x}_a^*} \right) \cup \left(\frac{\check{x}_b^* - \check{x}_a^*}{\check{x}_b^* - 3\check{x}_a^*}, 1 \right) \mid \max\{\|x_a(\kappa)\|, \|x_b(\kappa)\|\} < \kappa \mathbb{E}_\mu[|\theta_1|] \right\}.$$

Under $\|x_a^*\| < \|x_b^*\|$, the value $1/2$ falls strictly between the two intervals in the definition of $\mathcal{K}$.

Theorem 3. *Suppose $\|x_a^*\| < \|x_b^*\|$ and $\check{x}_a^* < 0 < \check{x}_b^*$. The following describes an equilibrium in two parameter regimes.*

	Party positions	Revealed voter ideology
$\kappa \in \mathcal{K}$:	Polarizing equilibrium $(x_a, x_b) = (x_a(\kappa), x_b(\kappa))$	$\rho = P_a(\kappa) \delta_{\theta_a(\kappa)} + (1 - P_a(\kappa)) \delta_{\theta_b(\kappa)}$
$\kappa \geq 1$:	Downsian equilibrium $(x_a, x_b) = (0, 0)$	$\rho = \delta_0$

The equilibrium is unique for $\kappa \geq 1$. For $\kappa \in \mathcal{K}$, the equilibrium is unique if $\|x_b^ - x_a^*\| \leq \kappa \mathbb{E}_\mu[|\theta_1|]$.*

Theorem 3 states that in equilibrium, party platforms x_a and x_b are on the line through $(1-\kappa)x_a^*$ and $(1-\kappa)x_b^*$, for all $\kappa \in \mathcal{K}$. The positions of parties on said line are given by (8). As in the equilibrium under a symmetric setup, party polarization is given by the simple formula

$$x_b(\kappa) - x_a(\kappa) = (1-\kappa)(x_b^* - x_a^*).$$

In particular, party polarization increases as the information cost κ decreases.

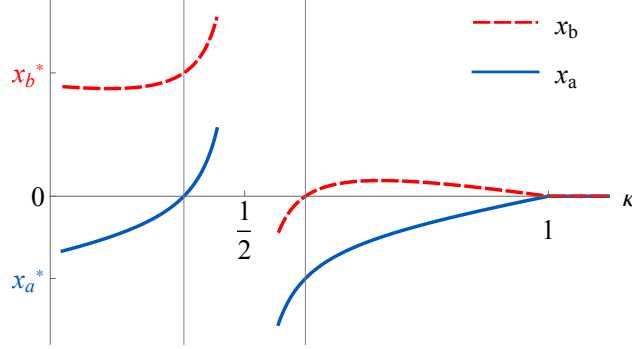


Figure 6: Equilibrium platforms (vertical axis) as a function of the information cost parameter κ , in a one-dimensional policy space with $x_a^* = -1$ and $x_b^* = 1.5$. Platforms are plotted for $\kappa \in \mathcal{K}$ and $\kappa \geq 1$.

Figure 6 plots the equilibrium platforms (x_a, x_b) as a function of κ under a one-dimensional policy space for $\kappa \in \mathcal{K}$ and for $\kappa \geq 1$. For large information costs, $\kappa \geq 1$, parties converge fully as in the symmetric setup. For smaller information costs, $\kappa < 1$, parties polarize as κ decreases. For an intermediate range of κ , there is no equilibrium in pure strategies where both parties obtain positive vote shares. We comment on this below.

A notable feature of the equilibrium is that for intermediate information costs, both parties are on the same side of zero and one party is more extreme than its ideal policy. This contrasts with the Calvert-Wittman model under exogenous voter positions, where party platforms necessarily lie between their ideal policies (Roemer 1997).

This feature of the equilibrium is driven by the **polarization motive**, which creates *chase-and-evade incentives*, similar to those in the literature on valence advantage (Groseclose 2001; Aragones and Palfrey 2002; Bernhardt, Buisseret, and Hidir 2020). By the logic of the polarization motive, the party with the more extreme platform has an incentive to differentiate from its opponent (“evade”) and the party with the more moderate platform has an incentive to move closer (“chase”). When the information cost parameter κ is close to $1/2$, these incentives are strong enough to push the more extreme platform past that party’s ideal policy and to pull the more moderate platform across the center of the voter distribution. Endogenous voter learning thus gives a party an incentive to adopt a platform more extreme than its ideal policy. If κ is very close to $1/2$, the chase-and-evade incentives become so strong that a pure-strategy equilibrium fails to exist, as is common in the literature on valence advantage (Aragones and Palfrey 2002; Hummel 2010).

5 Conclusion

This paper relaxes the assumption that voters perfectly know their political preferences. Instead, voters can flexibly learn about their ideal points at a cost and do so for the purpose of expressing their political opinion in two-party elections. The choice set voters face shapes revealed ideology in the population by determining the incentives to learn. Because a two-party election offers only two options, voters learn about their ideal points in a one-dimensional and binary way. Revealed ideol-

ogy therefore displays issue alignment and polarization. A reduction in the information cost raises party and voter polarization and can harm voters, and parties can strategically adopt platforms more extreme than their own ideal policies. These results highlight the importance of understanding political behavior not only as a function of underlying preferences, but also of the institutional and informational environment in which voter decisions are made.

This paper opens several avenues for future research. One is to extend the model to repeated elections. The present model is stylized in treating the election as a one-shot event, whereas elections in reality reoccur. The model can also be used as a building block to study other questions in political economy, especially in multidimensional policy spaces. Standard models with divergent parties are restricted to one-dimensional policy spaces (Duggan 2017). Our model allows for a multidimensional policy space and, on an explicitly characterized region of the parameter space, gives closed-form solutions for party platforms and voter ideology.

A Appendix: Voter Learning

A.1 Voter Learning Problem

A voter who acquires the distribution τ over posteriors π , votes, given each posterior, for the platform with the higher expected policy utility. She chooses τ subject to Bayes plausibility, $\mathbb{E}_\tau[\pi] = \mu$, which is the only restriction that Bayesian updating imposes by the disintegration theorem, to maximize

$$\mathbb{E}_\tau \left[\max \left\{ \mathbb{E}_\pi \left[-(\theta - x_a)^\top A(\theta - x_a) \right], \mathbb{E}_\pi \left[-(\theta - x_b)^\top A(\theta - x_b) \right] \right\} \right] - \kappa \text{Var}_\tau (\mathbb{E}_\pi[\theta]).$$

The first term is the instrumental value of information for the voting decision, and the second term is the information cost. We rewrite the two terms in turn. The instrumental utility can be rewritten as follows:

$$\begin{aligned} & \mathbb{E}_\tau \left[\max \left\{ \mathbb{E}_\pi [-(\theta - x_a)^\top A(\theta - x_a)], \mathbb{E}_\pi [-(\theta - x_b)^\top A(\theta - x_b)] \right\} \right] \\ &= \mathbb{E}_\tau \left[\max \left\{ -x_a^\top A x_a + 2x_a^\top A \mathbb{E}_\pi[\theta], -x_b^\top A x_b + 2x_b^\top A \mathbb{E}_\pi[\theta] \right\} - \mathbb{E}_\pi[\theta^\top A \theta] \right] \\ &= \mathbb{E}_\tau \left[\max \left\{ -x_a^\top A x_a + 2x_a^\top A \mathbb{E}_\pi[\theta], -x_b^\top A x_b + 2x_b^\top A \mathbb{E}_\pi[\theta] \right\} \right] - \mathbb{E}_\mu[\theta^\top A \theta] \\ &= \mathbb{E}_\tau \left[\left| (x_b - x_a)^\top A \left(\mathbb{E}_\pi[\theta] - \frac{x_a + x_b}{2} \right) \right| \right] + C, \end{aligned} \tag{11}$$

where

$$C = \mathbb{E}_\mu \left[(x_a + x_b)^\top A \theta - \theta^\top A \theta \right] - \frac{1}{2} \left(x_a^\top A x_a + x_b^\top A x_b \right)$$

does not depend on τ and can hence be ignored.

By the law of total variance, the information cost equals the variance of the posterior means. The average posterior mean equals the prior mean, which is normalized to zero, so

$$c(\tau) = \text{Var}_\tau (\mathbb{E}_\pi[\theta]) = \mathbb{E}_\tau [\|\mathbb{E}_\pi[\theta]\|^2].$$

Together, the voter learning problem reads

$$\max_{\substack{\tau \in \Delta(\Delta(\mathbb{R}^n)) \\ \mathbb{E}_\tau[\pi] = \mu}} \mathbb{E}_\tau \left[\left| (x_b - x_a)^\top A \left(\mathbb{E}_\pi[\theta] - \frac{x_a + x_b}{2} \right) \right| - \kappa \mathbb{E}_\pi[\theta]^\top \mathbb{E}_\pi[\theta] \right]. \tag{12}$$

The objective depends on τ only through the induced distribution of the posterior mean $\mathbb{E}_\pi[\theta]$. This distribution is well defined for every feasible τ . Applying the constraint $\mathbb{E}_\tau[\pi] = \mu$ to the nonnegative function $\theta \mapsto \|\theta\|$ (extending the equality from bounded to nonnegative functions by monotone convergence) yields $\mathbb{E}_\tau[\mathbb{E}_\pi[\|\theta\|]] = \mathbb{E}_\mu[\|\theta\|]$, which is finite by log-concavity of μ , so τ -almost every posterior has a finite mean. As is known, a distribution $\rho \in \Delta(\mathbb{R}^n)$ over posterior means is induced by some feasible τ if and only if $\rho \preceq_{\text{MPS}} \mu$. The voter learning problem (12) is therefore equivalent to (4).

A.2 Theorem 1

Proof. We first introduce some terminology. For $\rho \in \Delta(\mathbb{R}^n)$, write

$$b(\rho) = \mathbb{E}_{\theta \sim \rho} \left[\left| (A(x_b - x_a))^\top \left(\theta - \frac{x_a + x_b}{2} \right) \right| \right] \quad (13)$$

for the instrumental value and

$$c(\rho) = \mathbb{E}_{\theta \sim \rho} [\theta^\top \theta]$$

for the cost of ρ , so that the voter's objective (4) equals $b(\rho) - \kappa c(\rho)$. We say ρ is feasible if $\rho \preceq_{\text{MPS}} \mu$. Define

$$v := A(x_b - x_a), \quad L := \text{span}(\{v\}).$$

L is the line through the prior mean with direction $A(x_b - x_a)$.

Fix a feasible ρ . We construct a feasible ρ^* that is supported on L , induces at most two posterior means, and satisfies $b(\rho^*) = b(\rho)$ and $c(\rho^*) \leq c(\rho)$, where the inequality is strict whenever ρ does not place mass 1 on L or induces more than two posterior means. This proves the theorem. We construct ρ^* in four steps.

Step 1: Reflection. Let ref denote the Euclidean reflection around L ,

$$\text{ref}(\theta) := 2 \frac{v^\top \theta}{v^\top v} v - \theta,$$

and denote the induced pushforward on $\Delta(\mathbb{R}^n)$ also by ref . The distribution $\text{ref}(\rho)$ is feasible: since ref is linear, $\rho \preceq_{\text{MPS}} \mu$ implies $\text{ref}(\rho) \preceq_{\text{MPS}} \text{ref}(\mu)$, and $\text{ref}(\mu) = \mu$ because ref is orthogonal and μ is spherical. Moreover, $b(\text{ref}(\rho)) = b(\rho)$, because the instrumental value (13) depends on θ only through $v^\top \theta$ and $v^\top \text{ref}(\theta) = v^\top \theta$. Finally, $c(\text{ref}(\rho)) = c(\rho)$, because ref preserves $\theta^\top \theta$. Together, the voter is indifferent between ρ and $\text{ref}(\rho)$.

Step 2: Randomization. Consider the mixture

$$\rho^m := \frac{1}{2} \rho + \frac{1}{2} \text{ref}(\rho).$$

The mixture is feasible because the set of mean-preserving contractions of μ is convex. Equivalently, ρ^m is induced by the voter randomizing between the two distributions over posteriors that induce ρ and $\text{ref}(\rho)$. Since b and c are linear in ρ , Step 1 implies $b(\rho^m) = b(\rho)$ and $c(\rho^m) = c(\rho)$. Thus the voter is indifferent between ρ and ρ^m . By construction, ρ^m is symmetric around L in the sense that $\text{ref}(\rho^m) = \rho^m$, which we use in the next step.

Step 3: Projection. Let

$$\text{Proj}_L(\theta) := \frac{v^\top \theta}{v^\top v} v$$

denote the orthogonal projection onto L , extended to distributions via the pushforward. We claim

$$\rho^p := \text{Proj}_L(\rho^m)$$

is a mean-preserving contraction of ρ^m . Let $Y \sim \rho^m$ and $X := \text{Proj}_L(Y)$. Because $\text{ref}(\rho^m) = \rho^m$

and $\text{Proj}_L \circ \text{ref} = \text{Proj}_L$, the conditional distribution of Y given X is invariant under ref , so $\mathbb{E}[Y | X]$ is a fixed point of ref and hence lies on L . Since $\text{Proj}_L(\mathbb{E}[Y | X]) = \mathbb{E}[\text{Proj}_L(Y) | X] = X$, it follows that $\mathbb{E}[Y | X] = X$. Thus ρ^p is a mean-preserving contraction of ρ^m and hence feasible.

The instrumental value is unchanged, $b(\rho^p) = b(\rho^m)$, because $v^\top \text{Proj}_L(\theta) = v^\top \theta$. The cost weakly decreases: pointwise, $\text{Proj}_L(\theta)^\top \text{Proj}_L(\theta) \leq \theta^\top \theta$, with strict inequality for $\theta \notin L$. Hence, if ρ (and therefore ρ^m) does not place mass 1 on L , then $c(\rho^p) < c(\rho^m) = c(\rho)$.

Step 4: Binarization. Finally, we replace ρ^p by the at most binary distribution given by its conditional means on the two half-spaces in which voting for a and b , respectively, is optimal. This corresponds to a garbling based on the action recommendation, which preserves the instrumental value and weakly reduces the cost. Partition the state space into the half-spaces

$$R_a := \{\theta \in \mathbb{R}^n \mid v^\top (\theta - \frac{x_a + x_b}{2}) \leq 0\}, \quad R_b := \{\theta \in \mathbb{R}^n \mid v^\top (\theta - \frac{x_a + x_b}{2}) > 0\}.$$

Let ρ^* denote the (at most binary) distribution placing mass $\rho^p(R_j)$ on the conditional mean $\mathbb{E}_{\rho^p}[\theta | R_j]$ for each half-space with positive probability. The distribution ρ^* is a mean-preserving contraction of ρ^p and hence feasible, ρ^* induces at most two posterior means, and ρ^* places mass 1 on L because ρ^p does. The instrumental value is preserved, $b(\rho^*) = b(\rho^p)$, because on each half-space, $b(\theta)$ is affine. The cost weakly decreases, $c(\rho^*) \leq c(\rho^p)$, because ρ^* is a mean-preserving contraction of ρ^p and $\theta^\top \theta$ is convex in θ . If ρ^p has support greater than two, then the conditional distribution on at least one cell is non-degenerate, so strict Jensen's inequality for the strictly convex map $\theta \mapsto \theta^\top \theta$ yields $c(\rho^*) < c(\rho^p)$.

Conclusion. The distribution ρ^* is feasible, supported on L with at most two support points, and satisfies $b(\rho^*) = b(\rho)$ and $c(\rho^*) \leq c(\rho)$. If ρ does not place mass 1 on L , the inequality $c(\rho^*) \leq c(\rho)$ is strict by Step 3. If ρ places mass 1 on L but induces more than two posterior means, then $\rho^p = \rho^m = \rho$, and the inequality is strict by Step 4. Hence any solution to the voter problem induces at most two posterior means and places mass 1 on L . $\square$

Comment. The proof extends to information costs that are not posterior-mean separable, provided the cost satisfies three properties—a version of strict Blackwell monotonicity, mixing, and reflection invariance—stated and proved in Online Appendix D.2. The argument reflects, randomizes, and garbles the distribution over posteriors, rather than over posterior means.

A.3 Lemma 1

Proof. Let $M := \{\rho \in \Delta(\mathbb{R}^n) \mid \rho \preceq_{\text{MPS}} \mu\}$ denote the feasible set of (4). By Strassen's theorem, the MPS order coincides with the convex order: $\rho \preceq_{\text{MPS}} \mu$ if and only if $\int \phi d\rho \leq \int \phi d\mu$ for every convex $\phi: \mathbb{R}^n \rightarrow \mathbb{R}$. In particular, M is convex, every $\rho \in M$ has mean $\mathbb{E}_\rho[\theta] = \mathbb{E}_\mu[\theta] = 0$, and $\mathbb{E}_\rho[\|\theta\|^k] \leq \mathbb{E}_\mu[\|\theta\|^k] < \infty$ for every $k \geq 1$, where all moments of μ are finite by log-concavity.

M is compact: Because $\mathbb{R}^n$ is Polish, $\Delta(\mathbb{R}^n)$ equipped with the weak topology is Polish as well, so Prokhorov's theorem applies: tightness is equivalent to relative compactness. M is tight because Markov's inequality and convexity of $\phi(\theta) = \|\theta\|$ give $\rho(\{\|\theta\| > R\}) \leq \mathbb{E}_\rho[\|\theta\|]/R \leq \mathbb{E}_\mu[\|\theta\|]/R$

uniformly over $\rho \in M$. M is closed: let $\rho_k \in M$ converge weakly to ρ . The uniform second-moment bound makes first moments uniformly integrable, so $\mathbb{E}_\rho[\theta] = \lim_k \mathbb{E}_{\rho_k}[\theta] = 0$. Now let ϕ be convex and let a be an affine function with $a \leq \phi$. The integral of a under any mean-zero distribution equals its intercept, so the Portmanteau theorem applied to the non-negative continuous function $\phi - a$ yields

$$\int \phi d\rho \leq \liminf_k \int \phi d\rho_k \leq \int \phi d\mu.$$

Hence $\rho \in M$, and M is compact.

Existence: The integrand of (4) is continuous and bounded above by some $C > 0$, because its quadratic term dominates its piecewise linear term for large $\|\theta\|$. The objective is therefore weakly upper semicontinuous by the Portmanteau theorem. An upper semicontinuous function on the compact set M attains its maximum.

Uniqueness: As in the proof of Theorem 1, partition the state space into the half-spaces

$$R_a = \{\theta \in \mathbb{R}^n \mid v^\top(\theta - \frac{x_a + x_b}{2}) \leq 0\}, \quad R_b = \{\theta \in \mathbb{R}^n \mid v^\top(\theta - \frac{x_a + x_b}{2}) > 0\}.$$

The binarization argument in Step 4 of the proof of Theorem 1 applies verbatim to any feasible ρ : unless ρ equals its binarization, the binarization is feasible, attains the same instrumental value, and has strictly smaller cost. Hence any optimal ρ is supported on at most one point in each half-space. Now let ρ_1 and ρ_2 be optimal. Because M is convex and the objective is linear in ρ , the mixture $\rho^m := \frac{1}{2}\rho_1 + \frac{1}{2}\rho_2$ is optimal, and its support is $\text{supp } \rho_1 \cup \text{supp } \rho_2$. Since ρ^m is supported on at most one point per half-space, ρ_1 and ρ_2 are supported on a common set $\{\theta_a, \theta_b\}$, where θ_j denotes the point in R_j , if any. If this set is a singleton $\{p\}$, then $\rho_1 = \rho_2 = \delta_p$. Otherwise, each ρ_i places some weight w_i on θ_a with $w_i\theta_a + (1 - w_i)\theta_b = 0$, because every $\rho \in M$ has mean zero; since $\theta_a \neq \theta_b$, this equation has at most one solution, so $w_1 = w_2$ and $\rho_1 = \rho_2$. $\square$

A.4 Proposition 1

Proof. The argument is similar to the reflection argument in the proof of Theorem 1, with the reflection around the line L replaced by the reflection around the hyperplane on which the voter is indifferent between the two platforms. Invariance of the solution under that reflection, together with Theorem 1, leaves a one-dimensional family of candidate solutions, indexed by the distance between the two revealed ideal points. Because the prior is spherical, the family and its feasible set are the same in every direction, so the platforms and the information cost enter the reduced problem only through $\|A(x_b - x_a)\|/\kappa$, and both comparative statics follow from increasing differences.

We keep the notation of the proof of Theorem 1 and write $v := A(x_b - x_a)$, $\hat{v} := v/\|v\|$, and $L := \text{span}(\{v\})$. The hypothesis $x_a^\top A x_a = x_b^\top A x_b$ is equivalent to $v^\top \frac{x_a + x_b}{2} = 0$, so the instrumental value (13) is $b(\rho) = \mathbb{E}_{\theta \sim \rho}[\|v^\top \theta\|]$, and the hyperplane $H := \{\theta \in \mathbb{R}^n \mid v^\top \theta = 0\}$, on which the voter is indifferent between the two platforms, contains the prior mean.

Let

$$\text{ref}_H(\theta) := \theta - 2 \frac{v^\top \theta}{v^\top v} v$$

denote the reflection around H , with the induced pushforward on $\Delta(\mathbb{R}^n)$ also denoted by ref_H . Step 1 of the proof of Theorem 1 applies with ref_H in place of the reflection around L . The map ref_H is orthogonal and μ is spherical, so $\text{ref}_H(\rho)$ is feasible whenever ρ is, and $c(\text{ref}_H(\rho)) = c(\rho)$ because ref_H preserves $\theta^\top \theta$. The instrumental value is also unchanged, $b(\text{ref}_H(\rho)) = b(\rho)$, because $v^\top \text{ref}_H(\theta) = -v^\top \theta$.

Let ρ solve (4). Then $\text{ref}_H(\rho)$ solves (4) as well, so uniqueness of the solution (Lemma 1) gives that ρ is invariant under ref_H . By Theorem 1, ρ is supported on at most two points of L . The reflection ref_H maps $t\hat{v}$ to $-t\hat{v}$, so invariance leaves only the one-dimensional family

$$\rho_t := \frac{1}{2}\delta_{-t\hat{v}} + \frac{1}{2}\delta_{t\hat{v}}, \quad t \geq 0,$$

and the two revealed ideal points lie at distance $2t$. Revealed ideology is thus symmetric around the prior mean, which is the first claim.

Let $T := \{t \geq 0 \mid \rho_t \preceq_{\text{MPS}} \mu\}$ collect the feasible members of the family. Both T and the cost $c(\rho_t)$ are the same for every direction $\hat{v}$. Indeed, any two unit vectors are related by a rotation, a rotation is a composition of two reflections, and each reflection leaves μ invariant because μ is spherical and leaves the information cost invariant by reflection invariance. Since $b(\rho_t) = \|v\|t$, dividing the voter's objective by κ shows that the solution is the distribution ρ_t whose index maximizes

$$\frac{\|v\|}{\kappa} t - c(\rho_t)$$

over T . Neither T nor $c(\rho_t)$ depends on the platforms or on κ , so the objective has increasing differences in t and $\|v\|/\kappa$. The optimal t therefore weakly increases in $\|v\|/\kappa$, which gives the second claim. $\square$

References

- Abramowitz, A. (2010). *The disappearing center: Engaged citizens, polarization, and American democracy*. Yale University Press.
- Abramowitz, A., & Saunders, K. L. (2008). Is Polarization a Myth? *The Journal of Politics*, 70(2), 542–555.
- Achen, C., & Bartels, L. (2016). *Democracy for Realists: Why Elections Do Not Produce Responsive Government*. Princeton University Press.
- Ansolabehere, S., Rodden, J., & Snyder, J. M. (2008). The strength of issues: Using multiple measures to gauge preference stability, ideological constraint, and issue voting. *American Political Science Review*, 102(2), 215–232.
- Aragones, E., & Palfrey, T. R. (2002). Mixed Equilibrium in a Downsian Model with a Favored Candidate. *Journal of Economic Theory*, 103(1), 131–161.
- Aumann, R. J., & Maschler, M. (1995). *Repeated games with incomplete information*. MIT press.
- Bafumi, J., & Herron, M. C. (2010). Leapfrog Representation and Extremism: A Study of American Voters and Their Members in Congress. *American Political Science Review*, 104(3), 519–542.
- Baysan, C. (2022). Persistent Polarizing Effects of Persuasion: Experimental Evidence from Turkey. *American Economic Review*, 112(11), 3528–3546.

- Bernhardt, D., Buisseret, P., & Hidir, S. (2020). The Race to the Base. *American Economic Review*, 110(3), 922–942.
- Bernhardt, D., Duggan, J., & Squintani, F. (2009). The case for responsible parties. *American Political science review*, 103(4), 570–587.
- Bornschier, S. (2010a). Cleavage Politics and the Populist Right. The New Cultural Conflict in Western Europe. *The Social Logic of Politics*.
- Bornschier, S. (2010b). The New Cultural Divide and the Two-Dimensional Political Space in Western Europe. *West European Politics*, 33(3), 419–444.
- Callander, S., & Carbajal, J. C. (2022). Cause and Effect in Political Polarization: A Dynamic Analysis. *Journal of Political Economy*, 130(4), 825–880.
- Calvert, R. L. (1985). Robustness of the multidimensional voting model: Candidate motivations, uncertainty, and convergence. *American Journal of Political Science*, 69–95.
- Campbell, A., Converse, P. E., Miller, W. E., & Stokes, D. E. (1960). *The American Voter*. University of Chicago Press.
- Casella, A. (2005). Storable votes. *Games and Economic Behavior*, 51(2), 391–419.
- Che, Y.-K., & Mierendorff, K. (2019). Optimal dynamic allocation of attention. *American Economic Review*, 109(8), 2993–3029.
- Converse, P. (1964). The nature of belief systems in mass publics. *Ideology and Discontent*.
- Dean, M., & Neligh, N. (2023). Experimental Tests of Rational Inattention. *Journal of Political Economy*, 131(12), 3415–3461.
- Delli Carpini, M. X., & Keeter, S. (1996). *What Americans know about politics and why it matters*. Yale University Press.
- DeMarzo, P. M., Vayanos, D., & Zwiebel, J. (2003). Persuasion bias, social influence, and unidimensional opinions. *The Quarterly journal of economics*, 118(3), 909–968.
- Downs, A. (1957). An Economic Theory of Political Action in a Democracy. *Journal of Political Economy*, 65(2), 135–150.
- Duggan, J., & Martinelli, C. (2011). A Spatial Theory of Media Slant and Voter Choice. *The Review of Economic Studies*, 78(2), 640–666.
- Duggan, J. (2017). A survey of equilibrium analysis in spatial models of elections. *Unpublished manuscript*.
- Duggan, J., & Fey, M. (2005). Electoral competition with policy-motivated candidates. *Games and Economic Behavior*, 51(2), 490–522.
- Eguia, J., & Hu, T.-W. (2022). *Voter polarization and extremism* (tech. rep.).
- Enke, B., Rodríguez-Padilla, R., & Zimmermann, F. (2023). Moral Universalism and the Structure of Ideology. *The Review of Economic Studies*, 90(4), 1934–1962.
- Esteban, J., & Ray, D. (1994). On the measurement of polarization. *Econometrica: Journal of the Econometric Society*, 819–851.
- Esteban, J., & Ray, D. (2012). Comparing Polarization Measures. In M. R. Garfinkel & S. Skaperdas (Eds.), *The Oxford Handbook of the Economics of Peace and Conflict*. Oxford University Press.
- Feddersen, T. J., & Pesendorfer, W. (1996). The swing voter’s curse. *The American economic review*, 408–424.
- Feddersen, T. J., & Sandroni, A. (2006). Ethical voters and costly information acquisition. *Quarterly Journal of Political Science*, 1(3), 287–312.
- Fiorina, M. P., & Abrams, S. J. (2008). Political polarization in the American public. *Annual Review of Political Science*, 11, 563.
- Fowler, A., Hill, S. J., Lewis, J. B., Tausanovitch, C., Vavreck, L., & Warshaw, C. (2023). Moderates. *American Political Science Review*, 1–18.

- Freeze, M., & Montgomery, J. M. (2016). Static Stability and Evolving Constraint: Preference Stability and Ideological Structure in the Mass Public. *American Politics Research*, 44(3), 415–447.
- Gentzkow, M., & Kamenica, E. (2016). A Rothschild-Stiglitz approach to Bayesian persuasion. *American Economic Review*, 106(5), 597–601.
- Gentzkow, M., Shapiro, J. M., & Stone, D. F. (2015). Chapter 14 - Media Bias in the Marketplace: Theory. In S. P. Anderson, J. Waldfogel, & D. Strömberg (Eds.), *Handbook of Media Economics* (pp. 623–645). North-Holland.
- Groseclose, T. (2001). A Model of Candidate Location When One Candidate Has a Valence Advantage. *American Journal of Political Science*, 45(4), 862–886.
- Hare, C. (2022). Constrained Citizens? Ideological Structure and Conflict Extension in the US Electorate, 1980–2016. *British Journal of Political Science*, 52(4), 1602–1621.
- Hare, C., Highton, B., & Jones, B. (2024). Assessing the Structure of Policy Preferences: A Hard Test of the Low Dimensionality Hypothesis. *The Journal of Politics*.
- Hébert, B., & Woodford, M. (2023). Rational inattention when decisions take time. *Journal of Economic Theory*, 208, 105612.
- Hill, S. J., & Tausanovitch, C. (2015). A Disconnect in Representation? Comparison of Trends in Congressional and Public Polarization. *The Journal of Politics*, 77(4), 1058–1075.
- Hu, L., Li, A., & Segal, I. (2023). The Politics of Personalized News Aggregation. *Journal of Political Economy Microeconomics*, 1(3), 463–505.
- Hummel, P. (2010). On the nature of equilibria in a Downsian model with candidate valence. *Games and Economic Behavior*, 70(2), 425–445.
- Jessee, S. A. (2009). Spatial voting in the 2004 presidential election. *American Political Science Review*, 103(1), 59–81.
- Jessee, S. A. (2012). *Ideology and spatial voting in American elections*. Cambridge University Press.
- Jewitt, I. (2004). Notes on the ‘Shape’ of Distributions.
- Judd, K. L. (1985). The law of large numbers with a continuum of IID random variables. *Journal of Economic Theory*, 35(1), 19–25.
- Kamenica, E., & Gentzkow, M. (2011). Bayesian persuasion. *American Economic Review*, 101(6), 2590–2615.
- Kriesi, H., Grande, E., Lachat, R., Dolezal, M., Bornschier, S., & Frey, T. (2006). Globalization and the transformation of the national political space: Six European countries compared. *European Journal of Political Research*, 45(6), 921–956.
- Kriesi, H., Grande, E., Lachat, R., Dolezal, M., Bornschier, S., & Frey, T. (2008). *West European Politics in the Age of Globalization*. Cambridge University Press.
- Ladha, K. (1991). A spatial model of legislative voting with perceptual error. *Public Choice*, 68(1-3).
- Lalley, S. P., & Weyl, E. G. (2018). Quadratic voting: How mechanism design can radicalize democracy. *AEA Papers and Proceedings*, 108, 33–37.
- Lelkes, Y. (2016). Mass polarization: Manifestations and measurements. *Public Opinion Quarterly*, 80(S1), 392–410.
- Lenz, G. S. (2012). *Follow the Leader?: How Voters Respond to Politicians’ Policies and Performance*. University of Chicago Press.
- Li, A., & Hu, L. (2023). Electoral accountability and selection with personalized information aggregation. *Games and Economic Behavior*, 140, 296–315.
- Lindbeck, A., & Weibull, J. W. (1987). Balanced-budget redistribution as the outcome of political competition. *Public choice*, 52(3), 273–297.
- Maćkowiak, B., Matějka, F., & Wiederholt, M. (2023). Rational inattention: A review. *Journal of Economic Literature*, 61(1), 226–273.

- Martin, G. J., & Yurukoglu, A. (2017). Bias in cable news: Persuasion and polarization. *American Economic Review*, 107(9), 2565–2599.
- Matějka, F., & McKay, A. (2015). Rational Inattention to Discrete Choices: A New Foundation for the Multinomial Logit Model. *American Economic Review*, 105(1), 272–298.
- Matějka, F., & Tabellini, G. (2021). Electoral Competition with Rationally Inattentive Voters. *Journal of the European Economic Association*, 19(3), 1899–1935.
- Mensch, J., & Malik, K. (2024). Posterior-Mean Separable Costs of Information Acquisition.
- Morris, S., & Strack, P. (2019). The wald problem and the relation of sequential sampling and ex-ante information costs. *Available at SSRN 2991567*.
- Nimark, K. P., & Sundaresan, S. (2019). Inattention and belief polarization. *Journal of Economic Theory*, 180, 203–228.
- Novák, V., Matveenko, A., & Ravaioli, S. (2024). The Status Quo and Belief Polarization of Inattentive Agents: Theory and Experiment. *American Economic Journal: Microeconomics*.
- Ortoleva, P., & Snowberg, E. (2015). Overconfidence in Political Behavior. *American Economic Review*, 105(2), 504–535.
- Palfrey, T. R., & Poole, K. T. (1987). The relationship between information, ideology, and voting behavior. *American journal of political science*, 511–530.
- Perego, J., & Yuksel, S. (2022). Media Competition and Social Disagreement. *Econometrica*, 90(1), 223–265.
- Pew Research Center. (2014). Political polarization in the american public.
- Poole, K. T., & Rosenthal, H. (1985). A Spatial Model for Legislative Roll Call Analysis. *American Journal of Political Science*, 29(2), 357–384.
- Ratcliff, R. (1978). A theory of memory retrieval. *Psychological review*, 85(2), 59.
- Ravid, D., Roesler, A.-K., & Szentes, B. (2022). Learning before trading: On the inefficiency of ignoring free information. *Journal of Political Economy*, 130(2), 346–387.
- Roemer, J. E. (1997). Political-economic equilibrium when parties represent constituents: The unidimensional case. *Social Choice and Welfare*, 14(4), 479–502.
- Roemer, J. E. (2001). *Political competition*. Harvard University Press Cambridge.
- Schneider-Strawczynski, S., & Valette, J. (2025). Media Coverage of Immigration and the Polarization of Attitudes. *American Economic Journal: Applied Economics*, 17(1), 337–368.
- Shaked, M., & Shanthikumar, J. G. (2007). *Stochastic Orders*. Springer New York.
- Shor, B., & Rogowski, J. C. (2018). Ideology and the US Congressional Vote. *Political Science Research and Methods*, 6(2), 323–341.
- Sims, C. A. (2003). Implications of rational inattention. *Journal of Monetary Economics*, 50(3), 665–690.
- Spector, D. (2000). Rational debate and one-dimensional conflict. *The Quarterly Journal of Economics*, 115(1), 181–200.
- Sun, Y. (2006). The exact law of large numbers via Fubini extension and characterization of insurable risks. *Journal of Economic Theory*, 126(1), 31–69.
- Sunstein, C. (2018). *#Republic: Divided Democracy in the Age of Social Media*. Princeton University Press.
- Tausanovitch, C., & Warshaw, C. (2012). *How Should We Choose Survey Questions to Measure Citizens’ Policy Preferences?* (Tech. rep.). Citeseer.
- Thereze, J. (2022). Adverse Selection and Endogenous Information.
- Uhlig, H. (1996). A law of large numbers for large economies. *Economic Theory*, 8, 41–50.
- Van Der Brug, W., & Van Spanje, J. (2009). Immigration, Europe and the ‘new’ cultural dimension. *European Journal of Political Research*, 48(3), 309–334.

- Wittman, D. (1973). Parties as utility maximizers. *American Political Science Review*, 67(2), 490–498.
- Wittman, D. (1983). Candidate Motivation: A Synthesis of Alternative Theories. *The American Political Science Review*, 77(1), 142–157.
- Wittman, D. (1990). Spatial strategies when candidates have policy preferences. *Advances in the spatial theory of voting*, 66–98.
- Yuksel, S. (2022). Specialized Learning and Political Polarization. *International Economic Review*, 63(1), 457–474.
- Zaller, J. (1992). *The nature and origins of mass opinion*. Cambridge University Press.

B Online Appendix: Additional Results on Voter Learning

The previous results use only certain properties of the information cost. This section uses the posterior-variance information cost to characterize the solution to the voter learning problem and the resulting vote shares, which will be used in the analysis of endogenous party platforms. We begin with the one-dimensional case $n = 1$, in which party platforms are scalars $x_a, x_b \in \mathbb{R}$ and the voter prior $\mu \in \Delta(\mathbb{R})$ has a log-concave density f_μ that is symmetric around 0. At the end of the section we turn to the general n -dimensional case and show how it reduces to the one-dimensional one with transformed parameters.

In one dimension, the posterior-mean formulation of the voter learning problem is

$$\begin{aligned} \max_{\rho \in \Delta(\mathbb{R})} \mathbb{E}_{\theta \sim \rho} \left[\left| (x_b - x_a) \left(\theta - \frac{x_a + x_b}{2} \right) \right| - \kappa \theta^2 \right] \\ \text{s.t. } \rho \preceq_{\text{MPS}} \mu. \end{aligned} \quad (14)$$

The following definition will be useful below.

Definition 1. For $\theta_a, \theta_b \in \mathbb{R}$, we say $\{\theta_a, \theta_b\}$ is **feasible** if there exists a distribution ρ on $\{\theta_a, \theta_b\}$ that is a mean-preserving contraction of the prior μ .

If ρ is a mean-preserving contraction of μ , then ρ has the same expectation as μ , which is 0. So if $0 \notin [\theta_a, \theta_b]$, then $\{\theta_a, \theta_b\}$ is not feasible. If instead $0 \in [\theta_a, \theta_b]$, then there is a unique distribution ρ over $\{\theta_a, \theta_b\}$ with expectation 0. Feasibility therefore reduces to whether this distribution ρ is a mean-preserving contraction of the prior μ .

We fix $x_a < x_b$ in the following. The case $x_a > x_b$ is analogous and the case $x_a = x_b$ is trivial since it induces no voter learning. The definition (7) of the posterior means from the multidimensional case simplifies in one dimension to

$$\begin{aligned} \theta_a &:= \frac{x_a + x_b}{2} - \frac{x_b - x_a}{2\kappa}, \\ \theta_b &:= \frac{x_a + x_b}{2} + \frac{x_b - x_a}{2\kappa}. \end{aligned}$$

By Lemma 1 a solution to the voter learning problem (14) exists and is unique. The following proposition characterizes it. The proofs of the results in this section follow below.

Proposition 3 (Revealed Ideal Points). *The solution ρ to (14) is characterized by:*

- (1) *If $\{\theta_a, \theta_b\}$ is feasible, ρ is the mean-zero distribution supported on $\{\theta_a, \theta_b\}$.*
- (2) *If $0 \notin [\theta_a, \theta_b]$, $\rho = \delta_0$.*
- (3) *If $\{\theta_a, \theta_b\}$ is not feasible and $0 \in [\theta_a, \theta_b]$, the optimal ρ is induced by a threshold signal structure $s(\theta) = \mathbb{1}(\theta > t)$, where $t \in \mathbb{R} \cup \{-\infty, \infty\}$ is strictly more extreme than the party midpoint, that is, either $0 < \frac{x_a + x_b}{2} < t$, $0 = \frac{x_a + x_b}{2} = t$, or $t < \frac{x_a + x_b}{2} < 0$. The induced posterior means are in $[\theta_a, \theta_b]$. As κ decreases, the threshold t moves weakly closer to the party midpoint $\frac{x_a + x_b}{2}$.*

The following result summarizes the implications for parties' vote shares.

Corollary 3 (Vote Shares). *The vote shares induced by the solution ρ to (14) are characterized by:*

(1) *If $\{\theta_a, \theta_b\}$ is feasible,*

$$P_a(x_a, x_b) = \frac{1}{2} + \frac{\kappa}{2} \frac{x_b + x_a}{x_b - x_a}.$$

(2) *If $0 \notin (\theta_a, \theta_b)$, the party whose position is closer to 0 obtains all votes.*

(3) *If $\{\theta_a, \theta_b\}$ is not feasible and $0 \in (\theta_a, \theta_b)$, $P_a(x_a, x_b) = F_\mu(t)$, where t is as in Proposition 3.*

If party a is more extreme, that is, $|x_a| > |x_b|$, then $P_a(x_a, x_b) < \frac{1}{2} + \frac{\kappa}{2} \frac{x_b + x_a}{x_b - x_a}$.

As a further corollary, we obtain the bias toward the moderate party in one dimension.

Corollary 4 (Bias toward Moderate Party). *If $|x_a| < |x_b|$, then party a 's vote share $P_a(x_a, x_b)$ weakly exceeds the share of voters whose true ideal point is closer to x_a than to x_b . The vote share $P_a(x_a, x_b)$ weakly increases if we raise the information cost parameter κ , and weakly decreases if we raise $x_b - x_a$ while holding the party midpoint $\frac{x_a + x_b}{2}$ fixed.*

Feasibility: We now give a sufficient condition for feasibility of $\{\theta_a, \theta_b\}$ based on the following lemma due to Jewitt (2004). Define $\Delta(t) := \mathbb{E}_\mu[\theta | \theta > t] - \mathbb{E}_\mu[\theta | \theta < t]$.

Lemma 2 (Jewitt's Lemma). *The map $\Delta(t)$ is weakly decreasing in t for $t < 0$ and weakly increasing in t for $t > 0$.*

Proof. Jewitt (2004) shows that $\Delta(t)$ is quasiconvex when the density of θ is quasiconcave. A log-concave density is quasiconcave, so $\Delta(t)$ is quasiconvex. Because the density of θ is symmetric around 0, $\Delta(t)$ is symmetric around 0, so $\Delta(t)$ is weakly decreasing for $t < 0$ and weakly increasing for $t > 0$. $\square$

Define $\Delta := \Delta(0) = \mathbb{E}_\mu[\theta | \theta > 0] - \mathbb{E}_\mu[\theta | \theta < 0]$. By symmetry of μ , $\Delta = 2\mathbb{E}_\mu[\theta | \theta > 0] = 2\mathbb{E}_\mu[|\theta|]$.

Lemma 3. *If $|x_b - x_a| \leq \kappa\Delta$ and $\theta_a \leq 0 \leq \theta_b$, then $\{\theta_a, \theta_b\}$ is feasible.*

Proof. The posterior means θ_a, θ_b satisfy $|\theta_b - \theta_a| = |x_b - x_a|/\kappa \leq \Delta$. Since $\theta_a \leq 0 \leq \theta_b$ and $\theta_b - \theta_a \leq \Delta$, at least one of θ_a, θ_b has absolute value at most $\Delta/2$. Suppose $|\theta_a| \leq \Delta/2$, the case $|\theta_b| \leq \Delta/2$ being analogous.

A threshold signal with threshold $t = 0$ induces the posterior means $\mathbb{E}_\mu[\theta | \theta < t] = -\Delta/2$ and $\mathbb{E}_\mu[\theta | \theta > t] = \Delta/2$. As t increases to ∞ , $\mathbb{E}_\mu[\theta | \theta < t]$ converges to $\mathbb{E}_\mu[\theta] = 0$. The function $\mathbb{E}_\mu[\theta | \theta < t]$ is continuous in t because θ has a density, so there is a $t' \geq 0$ with $\mathbb{E}_\mu[\theta | \theta < t'] = \theta_a$. By Jewitt's Lemma and $\theta_b - \theta_a \leq \Delta$, $\mathbb{E}_\mu[\theta | \theta > t'] \geq \theta_b$. The mean-zero distribution on $\{\theta_a, \theta_b\}$ is a mean-preserving contraction of the mean-zero distribution on $\{\mathbb{E}_\mu[\theta | \theta < t'], \mathbb{E}_\mu[\theta | \theta > t']\}$, since $\theta_a \leq 0 \leq \theta_b \leq \mathbb{E}_\mu[\theta | \theta > t']$. Hence $\{\theta_a, \theta_b\}$ is feasible. $\square$

Dimension Reduction: We now turn to the general n -dimensional case and show that it reduces to the one-dimensional problem solved above. From here on x_a , x_b , and μ are again the general primitives, $x_a, x_b \in \mathbb{R}^n$ and $\mu \in \Delta(\mathbb{R}^n)$. Assume $x_a \neq x_b$, so that the direction $A(x_b - x_a)$ is well defined. The voter problem is

$$\begin{aligned} \max_{\rho \in \Delta(\mathbb{R}^n)} \mathbb{E}_{\theta \sim \rho} \left[\left| (x_b - x_a)^\top A\left(\theta - \frac{x_a + x_b}{2}\right) \right| - \kappa \theta^\top \theta \right] \\ \text{s.t. } \rho \preceq_{\text{MPS}} \mu. \end{aligned} \quad (\text{Pn})$$

Let proj_v denote the scalar projection on $v \in \mathbb{R}^n$,

$$\begin{aligned} \text{proj}_v : \mathbb{R}^n &\rightarrow \mathbb{R} \\ x &\mapsto \frac{v^\top x}{\|v\|}, \end{aligned} \quad (15)$$

so $\text{proj}_{A(x_b - x_a)}$ is the scalar projection onto the line through the origin with direction $A(x_b - x_a)$, extended to measures by the pushforward. Define

$$\hat{x}_a = \text{proj}_{A(x_b - x_a)}(x_a), \quad \hat{x}_b = \text{proj}_{A(x_b - x_a)}(x_b) \quad (16)$$

$$\hat{\kappa} = \frac{(x_b - x_a)^\top A(x_b - x_a)}{\|A(x_b - x_a)\|^2} \kappa, \quad (17)$$

$$\hat{\mu} = \text{proj}_{A(x_b - x_a)}(\mu). \quad (18)$$

The one-dimensional problem we associate with (Pn) is

$$\begin{aligned} \max_{\hat{\rho} \in \Delta(\mathbb{R})} \mathbb{E}_{\hat{\theta} \sim \hat{\rho}} \left[\left| (\hat{x}_b - \hat{x}_a)(\hat{\theta} - \frac{\hat{x}_a + \hat{x}_b}{2}) \right| - \hat{\kappa} \hat{\theta}^2 \right] \\ \text{s.t. } \hat{\rho} \preceq_{\text{MPS}} \hat{\mu}. \end{aligned} \quad (\text{P1})$$

By Theorem 1, the solution to (Pn) is supported on the line through the origin with direction $A(x_b - x_a)$. Along that line the problems (Pn) and (P1) coincide up to a positive scaling of the objective, as the following lemma uses.

Lemma 4. *A distribution $\rho \in \Delta(\mathbb{R}^n)$ solves (Pn) if and only if $\hat{\rho} := \text{proj}_{A(x_b - x_a)}(\rho)$ solves (P1) and ρ is supported on the line through the origin with direction $A(x_b - x_a)$.*

If μ is spherical and log-concave, then $\hat{\mu}$ is symmetric around 0 and log-concave, the latter because log-concavity is preserved by marginalization. Thus, problem (P1) has exactly the form solved above, with cost $\hat{\kappa}$, platforms $\hat{x}_a, \hat{x}_b$, and prior $\hat{\mu}$. Specializing Proposition 3 and Corollaries 3 and 4 to (P1) and projecting back therefore characterizes revealed ideal points and vote shares in the general case, and gives the bias result of Proposition 2.

Finally, we observe that voter learning never leaves a voter indifferent between the two platforms, see also Hu, Li, and Segal (2023).

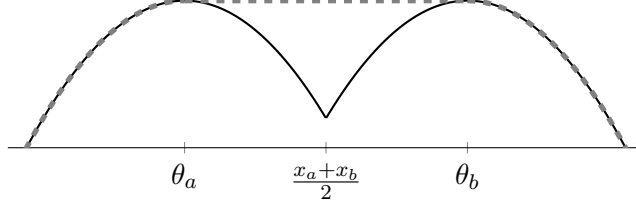


Figure 7: The value function $v(\theta)$ and its concave envelope (dashed).

Lemma 5. *Let $x_a \neq x_b$. The solution ρ of (4) places no mass on the indifference set*

$$\mathcal{I} := \{\theta \in \mathbb{R}^n \mid (x_b - \theta)^\top A(x_b - \theta) = (x_a - \theta)^\top A(x_a - \theta)\}.$$

There are two ways in which a voter could end up indifferent. She could acquire information and hold a revealed ideal point in $\mathcal{I}$, or she could acquire no information and already be indifferent at the prior mean, which is the case when $x_a^\top A x_a = x_b^\top A x_b$. The proof rules out the first because such learning delivers the same expected voting payoff as no learning at a strictly positive cost, and it rules out the second because a small symmetric signal along $A(x_b - x_a)$ raises the voting payoff at a first-order rate and the information cost only at a second-order rate.

B.0.1 Proposition 3

Proof. Recall that we assume $x_a < x_b$. We rewrite the voter learning problem using the Rothschild-Stiglitz approach (Gentzkow and Kamenica 2016):

$$\max_{\rho \in \Delta(\mathbb{R})} \mathbb{E}_{\theta \sim \rho} \left[\overbrace{\left| (x_b - x_a) \left(\theta - \frac{x_a + x_b}{2} \right) \right| - \kappa \left(\theta - \frac{x_a + x_b}{2} \right)^2}^{v(\theta)} \right] \quad (19)$$

$$\text{s.t.} \quad \mathbb{E}_{\theta \sim \rho}[\theta] = 0, \quad (20)$$

$$\int_{-\infty}^t F_\rho(\theta) d\theta \leq \int_{-\infty}^t F_\mu(\theta) d\theta \quad \forall t \in \mathbb{R}. \quad (21)$$

This reformulation replaces the mean-preserving contraction constraint $\rho \preceq_{\text{MPS}} \mu$ by the mean constraint (20) and the majorization constraints (21). By (20), adding an affine function to the integrand of (14) amounts to adding a constant, so the optimum is unchanged if we replace θ by $\theta - \frac{x_a + x_b}{2}$ in the objective.

We first solve the relaxed problem that drops the majorization constraints (21). This is a one-dimensional persuasion problem, solved by concavification of the value function v in posterior-mean space (Aumann and Maschler 1995; Kamenica and Gentzkow 2011). The value function v , shown with its concave envelope in Figure 7, has maxima at $\theta_a = \frac{x_a + x_b}{2} - \frac{x_b - x_a}{2\kappa}$ and $\theta_b = \frac{x_a + x_b}{2} + \frac{x_b - x_a}{2\kappa}$. If $0 \notin [\theta_a, \theta_b]$, the solution is δ_0 . If $0 \in [\theta_a, \theta_b]$, the solution is the mean-zero distribution ρ^* supported on $\{\theta_a, \theta_b\}$.

By Lemma 1, the original problem (19) has a unique solution. If the solution to the relaxed problem satisfies the majorization constraints, then it solves the original problem. The point mass

δ_0 is always feasible, and ρ^* satisfies $\rho^* \preceq_{\text{MPS}} \mu$ exactly when $\{\theta_a, \theta_b\}$ is feasible. This covers the first two cases of the proposition.

It remains to treat the case $0 \in [\theta_a, \theta_b]$ when $\{\theta_a, \theta_b\}$ is not feasible, for which we show that the optimal ρ is induced by a threshold signal $s(\theta) = \mathbf{1}(\theta > t)$ with $t \in \mathbb{R} \cup \{-\infty, \infty\}$ with posterior means $\mathbb{E}_\mu[\theta | \theta < t]$ and $\mathbb{E}_\mu[\theta | \theta > t]$. By Theorem 1, the solution ρ to (19) is supported on at most two points. If the support is a single point, then $\rho = \delta_0$, which is induced by the threshold signal with threshold $t = \pm\infty$.

Suppose instead ρ is supported on two points $\{y, z\}$ with $y < 0 < z$, implying $\rho(\{y\}) = \frac{z}{z-y}$. The resulting value of the voter problem (14) is $V(y, z) = \frac{z}{z-y}v(y) + \frac{-y}{z-y}v(z)$.

First, $\theta_a \leq y < z \leq \theta_b$. If a support point lay strictly outside $[\theta_a, \theta_b]$, say $z > \theta_b$, then by $0 < \theta_b < z$ there is a mean-zero distribution ρ' on $\{y, \theta_b\}$ that is a mean-preserving contraction of ρ , hence feasible. By $\rho'(\{\theta_b\}) > \rho(\{z\})$ and $v(\theta_b) \geq \max\{v(y), v(z)\}$, we obtain $V(y, \theta_b) > V(y, z)$, contradicting optimality.

Second, the majorization constraint binds at some $t \in [y, z]$. Write

$$G_{y,z}(t) := \int_{-\infty}^t (F_\mu - F_\rho) d\theta = \int_{-\infty}^t F_\mu(\theta) d\theta - \left(\frac{z}{z-y}(t-y)^+ + \frac{-y}{z-y}(t-z)^+ \right),$$

so the majorization constraint (21) reads $G_{y,z}(t) \geq 0$ for all $t \in \mathbb{R}$. Toward a contradiction, suppose $G_{y,z}(t) > 0$ for all $t \in [y, z]$. We show the mean-zero distribution ρ' on $\{ky, kz\}$ with $k > 1$ close enough to 1 is an improvement. Scaling preserves the probabilities, so $\rho'(\{ky\}) = \rho(\{y\})$ and $\rho'(\{kz\}) = \rho(\{z\})$. Without loss suppose $y \geq \theta_a$ and $z < \theta_b$. Optimality requires $y < \frac{x_a+x_b}{2} < z$. Then $v'(y) \leq 0$ and $v'(z) > 0$, so $\frac{d}{dk}V(ky, kz) = y\rho(\{y\})v'(y) + z\rho(\{z\})v'(z) > 0$ at $k = 1$, so for k close enough to 1, $V(ky, kz) > V(y, z)$. Finally, ρ' is feasible for k close enough to 1. The map $(y, z, t) \mapsto G_{y,z}(t)$ is jointly continuous, so by the maximum theorem $\min_{t \in [y,z]} G_{y,z}(t) > 0$ and $\min_{t \in [y',z']} G_{y',z'}(t) > 0$ for (y', z') close enough to (y, z) . Outside $[y', z']$ the majorization constraint holds automatically: since $F_{\rho'} = 0$ below y' , the integrand $F_\mu - F_{\rho'}$ is nonnegative there, so $G_{y',z'}$ increases from $G_{y',z'}(-\infty) = 0$ and stays nonnegative; since $F_{\rho'} = 1$ above z' , the integrand is nonpositive there, so $G_{y',z'}$ decreases to $G_{y',z'}(+\infty) = 0$, the latter by the mean-zero property, and again stays nonnegative. Hence ρ' satisfies (21) for all t , contradicting optimality. Therefore $G_{y,z}(t) = 0$ for some $t \in [y, z]$.

A binary distribution with support $\{y, z\}$ whose majorization constraint binds at $t \in [y, z]$ is induced by the threshold signal with threshold t and posterior means $y = \mathbb{E}_\mu[\theta | \theta < t]$ and $z = \mathbb{E}_\mu[\theta | \theta > t]$.

The optimal threshold t is strictly more extreme than $\frac{x_a+x_b}{2}$ (that is, t is strictly further from 0 than is $\frac{x_a+x_b}{2}$), as we now show. One can easily verify that the instrumental value of information $b(t)$ increases as t moves toward $\frac{x_a+x_b}{2}$:

$$\begin{aligned} b(t) &:= \int_{-\infty}^t -(x_a - \theta)^2 f_\mu(\theta) d\theta + \int_t^\infty -(x_b - \theta)^2 f_\mu(\theta) d\theta \\ \Rightarrow b'(t) &= -(x_a - t)^2 + (x_b - t)^2 f_\mu(t) = -2(t - \frac{x_a+x_b}{2})(x_b - x_a)f_\mu(t). \end{aligned}$$

In particular, $b'(\frac{x_a+x_b}{2}) = 0$. The information cost decreases as t becomes more extreme, provided the prior is log-concave. The information cost is

$$\begin{aligned} c(t) &:= \text{Var}[\theta] - (F_\mu(t) \text{Var}[\theta | \theta < t] + (1 - F_\mu(t)) \text{Var}[\theta | \theta > t]) \\ &= \text{Var}[\theta] - \left(F_\mu(t) \min_E \left\{ \frac{\int_{-\infty}^t (\theta - E)^2 dF_\mu(\theta)}{F_\mu(t)} \right\} + (1 - F_\mu(t)) \min_E \left\{ \frac{\int_t^\infty (\theta - E)^2 dF_\mu(\theta)}{1 - F_\mu(t)} \right\} \right) \\ &= \text{Var}[\theta] - \left(\min_E \left\{ \int_{-\infty}^t (\theta - E)^2 f_\mu(\theta) d\theta \right\} + \min_E \left\{ \int_t^\infty (\theta - E)^2 f_\mu(\theta) d\theta \right\} \right) \end{aligned}$$

By the envelope theorem,

$$\begin{aligned} c'(t) &= f_\mu(t)(t - \mathbb{E}_\mu[\theta | \theta > t])^2 - f_\mu(t)(t - \mathbb{E}_\mu[\theta | \theta < t])^2 \\ &= f_\mu(t)(\mathbb{E}_\mu[\theta - t | \theta > t]^2 - \mathbb{E}_\mu[t - \theta | \theta < t]^2). \end{aligned}$$

By Theorem 1.C.52 in Shaked and Shanthikumar (2007), $\mathbb{E}_\mu[\theta - t | \theta > t]$ is decreasing in t when f_μ is log-concave. By symmetry of f_μ around 0, $\mathbb{E}_\mu[t - \theta | \theta \leq t] = \mathbb{E}_\mu[\theta - (-t) | \theta > -t]$. Hence, for $t > 0$,

$$0 < \mathbb{E}_\mu[\theta - t | \theta > t] < \mathbb{E}_\mu[\theta | \theta > 0] < \mathbb{E}_\mu[\theta - (-t) | \theta > -t] = \mathbb{E}_\mu[t - \theta | \theta \leq t],$$

so $c'(t) < 0$ for $t > 0$.

For $\frac{x_a+x_b}{2} > 0$, any threshold $t < 0$ is dominated by its reflection $-t > 0$, which has the same information cost and strictly higher instrumental value. For $0 \leq t < \frac{x_a+x_b}{2}$, we have $b'(t) > 0$ and $c'(t) \leq 0$, so $t = \frac{x_a+x_b}{2}$ dominates every $t < \frac{x_a+x_b}{2}$. Finally, $b'(\frac{x_a+x_b}{2}) = 0$ and $c'(\frac{x_a+x_b}{2}) < 0$ give $\frac{d}{dt}[b(t) - \kappa c(t)]|_{t=\frac{x_a+x_b}{2}} > 0$, so every optimal t satisfies $t > \frac{x_a+x_b}{2}$. The optimal distribution over posteriors is unique. By log-concavity of the prior density, μ has connected support, so the optimal threshold t is unique, unless $t \in \{-\infty, \infty\}$ is optimal, in which case every threshold beyond the prior support on the corresponding side is also optimal.

Restrict t to $[\frac{x_a+x_b}{2}, \infty]$, where $t = \infty$ denotes always voting for party a . Since $c'(t) < 0$, the objective $b(t) - \kappa c(t)$ is supermodular in (κ, t) , so the optimal t weakly increases in κ . $\square$

B.0.2 Corollary 3

Proof. Denote $P_a := P_a(x_a, x_b)$.

Case (1): By ρ having mean zero, the vote share P_a satisfies

$$P_a \left(\frac{x_a + x_b}{2} - \frac{x_b - x_a}{2\kappa} \right) + (1 - P_a) \left(\frac{x_a + x_b}{2} + \frac{x_b - x_a}{2\kappa} \right) = 0 \Rightarrow P_a = \frac{1}{2} + \frac{\kappa}{2} \frac{x_a + x_b}{x_b - x_a}.$$

Case (2) is immediate.

Case (3): By case (3) of Proposition 3, $P_a = F_\mu(t)$. It remains to show that, if $\{\theta_a, \theta_b\}$ is not feasible, $0 \in [\theta_a, \theta_b]$, and $|x_a| > |x_b|$, then

$$P_a < \tilde{P}_a := \frac{1}{2} + \frac{\kappa}{2} \frac{x_a + x_b}{x_b - x_a}.$$

Since $x_a < x_b$ and $|x_a| > |x_b|$, we have $\frac{x_a+x_b}{2} < 0$, and hence $\tilde{P}_a < 1/2$. By $\theta_a < 0 < \theta_b$, $\tilde{P}_a = \theta_b/(\theta_b - \theta_a) \in (0, 1)$.

If the voter acquires no information, the posterior mean is 0, so all voters choose party b , which is closer to 0. Hence $P_a = 0 < \tilde{P}_a$. Suppose, therefore, that the voter acquires two posterior means. By Proposition 3, the optimal signal is a threshold signal, and an optimal threshold t induces posterior means satisfying

$$\theta_a \leq \mathbb{E}_\mu[\theta | \theta < t] < \frac{x_a+x_b}{2} < \mathbb{E}_\mu[\theta | \theta > t] \leq \theta_b. \quad (22)$$

Let t' be defined by $F_\mu(t') = \tilde{P}_a$. Since $\tilde{P}_a < 1/2$ and μ is symmetric, $t' < 0$. By Bayes plausibility,

$$0 = \tilde{P}_a \mathbb{E}_\mu[\theta | \theta < t'] + (1 - \tilde{P}_a) \mathbb{E}_\mu[\theta | \theta > t'] = \tilde{P}_a \theta_a + (1 - \tilde{P}_a) \theta_b. \quad (23)$$

Hence there exists $\alpha > 0$ such that

$$(\mathbb{E}_\mu[\theta | \theta < t'], \mathbb{E}_\mu[\theta | \theta > t']) = (\alpha \theta_a, \alpha \theta_b).$$

Since $\{\theta_a, \theta_b\}$ is not feasible, $\alpha < 1$: otherwise $\{\theta_a, \theta_b\}$ would be a mean-preserving contraction of the feasible threshold distribution induced by t' .

We show that no optimal threshold satisfies $t \geq t'$. If $\mathbb{E}_\mu[\theta | \theta < t'] \geq \frac{x_a+x_b}{2}$, this is immediate from (22), since $\mathbb{E}_\mu[\theta | \theta < t]$ is increasing in t . Hence suppose $\mathbb{E}_\mu[\theta | \theta < t'] < \frac{x_a+x_b}{2}$.

By Proposition 3, since $\frac{x_a+x_b}{2} < 0$, any informative optimal threshold satisfies $t < \frac{x_a+x_b}{2} < 0$. Hence, to rule out $t \geq t'$, it is enough to consider candidate thresholds $t \in [t', 0]$ satisfying (22).

Write

$$L(t) := \mathbb{E}_\mu[\theta | \theta < t], \quad R(t) := \mathbb{E}_\mu[\theta | \theta > t].$$

The objective of the voter problem is $F_\mu(t)v(L(t)) + (1 - F_\mu(t))v(R(t))$. We show that for any $t \in [t', 0]$ satisfying (22),

$$\begin{aligned} \frac{d}{dt} (F_\mu(t)v(L(t)) + (1 - F_\mu(t))v(R(t))) &= (F_\mu(t)v'(L(t)) + (1 - F_\mu(t))v'(R(t)))R'(t) \\ &\quad + F_\mu(t)v'(L(t))(L'(t) - R'(t)) \\ &\quad + f_\mu(t)(v(L(t)) - v(R(t))) < 0, \end{aligned}$$

hence t is suboptimal. We go through the three terms in turn.

The first term $(F_\mu(t)v'(L(t)) + (1 - F_\mu(t))v'(R(t)))R'(t)$ is weakly negative. By (22),

$$v'(L(t)) = -2\kappa(L(t) - \theta_a) \leq 0, \quad v'(R(t)) = 2\kappa(\theta_b - R(t)) \geq 0. \quad (24)$$

If $v'(R(t)) = 0$, we are done. If $v'(R(t)) > 0$, then for any $t \geq t'$ satisfying (22),

$$\frac{-v'(L(t))}{v'(R(t))} = \frac{L(t) - \theta_a}{\theta_b - R(t)} \geq \frac{L(t') - \theta_a}{\theta_b - R(t')} = \frac{\alpha\theta_a - \theta_a}{\theta_b - \alpha\theta_b} = \frac{-\theta_a}{\theta_b} = \frac{1 - F_\mu(t')}{F_\mu(t')}. \quad (25)$$

The inequality uses that both $L(t)$ and $R(t)$ are nondecreasing in t and the last equality uses (23).

By $F_\mu(t) \geq F_\mu(t')$ for $t \geq t'$ and (24), the previous display implies

$$F_\mu(t)v'(L(t)) + (1 - F_\mu(t))v'(R(t)) \leq 0. \quad (26)$$

The second term $F_\mu(t)v'(L(t))(L'(t) - R'(t))$ is weakly negative. By (24), $v'(L(t)) < 0$. By Jewitt's Lemma, $\Delta(t) = R(t) - L(t)$ is nonincreasing in t for $t < 0$. Hence $L'(t) - R'(t) \geq 0$.

The third term $f_\mu(t)(v(L(t)) - v(R(t)))$ is strictly negative. Since v is symmetric around $\frac{x_a+x_b}{2}$ and $v(\theta_a) = v(\theta_b)$, it is enough to compare the distances of $\alpha\theta_a$ and $\alpha\theta_b$ from $\frac{x_a+x_b}{2}$. Because $\theta_a < \frac{x_a+x_b}{2} < 0$ and $\alpha < 1$, $|\alpha\theta_b - \frac{x_a+x_b}{2}| < |\alpha\theta_b - \frac{x_a+x_b}{2}|$, so $v(\alpha\theta_a) < v(\alpha\theta_b)$. For $t \geq t'$, $L(t) \geq L(t') = \alpha\theta_a$ and $R(t) \geq R(t') = \alpha\theta_b$. Since v is decreasing on $[\theta_a, \frac{x_a+x_b}{2}]$ and increasing on $[\frac{x_a+x_b}{2}, \theta_b]$, (22) implies $v(L(t)) \leq v(\alpha\theta_a) < v(\alpha\theta_b) \leq v(R(t))$.

Thus the optimal threshold satisfies $t < t'$ and

$$P_a(x_a, x_b) = F_\mu(t) < F_\mu(t') = \tilde{P}_a = \frac{1}{2} + \frac{\kappa}{2} \frac{x_a + x_b}{x_b - x_a}. \quad \square$$

B.0.3 Corollary 4

Proof. We go through the three cases of Corollary 3 in order of increasing κ . We show in each case the vote share is biased toward the moderate party, and the bias weakly increases in κ . Dividing the objective in (14) by $x_b - x_a > 0$ does not change its maximizer and rewrites it as $|\theta - \frac{x_a+x_b}{2}| - \frac{\kappa}{x_b-x_a}\theta^2$, so the solution depends on $x_b - x_a$ and κ only through $\frac{\kappa}{x_b-x_a}$. Raising $x_b - x_a$ while holding $\frac{\kappa}{x_b-x_a}$ fixed lowers this ratio exactly as lowering κ does, so the comparative static in $x_b - x_a$ follows from the one in κ . Throughout, $\frac{x_a+x_b}{2} > 0$ because $x_a < x_b$ and $|x_a| < |x_b|$, so party a is the moderate party and a voter is closer to x_a than to x_b exactly when $\theta < \frac{x_a+x_b}{2}$. The share of voters whose true ideal point is closer to x_a than to x_b is therefore $F_\mu(\frac{x_a+x_b}{2})$.

For small κ we have $0 \in [\theta_a, \theta_b]$, and $\{\theta_a, \theta_b\}$ is not feasible because θ_a and θ_b diverge as $\kappa \rightarrow 0$. By Corollary 3 the solution is induced by a threshold signal with threshold t strictly more extreme than the party midpoint, so $\frac{x_a+x_b}{2} < t$ and $P_a(x_a, x_b) = F_\mu(t) \geq F_\mu(\frac{x_a+x_b}{2})$. As κ increases, t moves weakly farther from $\frac{x_a+x_b}{2}$ by Proposition 3, so the vote share $F_\mu(t)$ weakly increases.

While $0 \in [\theta_a, \theta_b]$, raising κ keeps $\{\theta_a, \theta_b\}$ feasible once it is feasible: raising κ moves θ_a, θ_b closer together which corresponds to a mean-preserving contraction. The set of mean-preserving contractions of μ is compact (Lemma 1), so if $\{\theta_a, \theta_b\}$ is feasible at any κ there is a smallest such κ_0 , and feasibility holds on $[\kappa_0, \bar{\kappa})$, where $\bar{\kappa}$ is the value at which 0 leaves the interval. At κ_0 the posterior means coincide with those induced by the threshold signal of the previous case, so the bias carries over by continuity. For $\kappa \in (\kappa_0, \bar{\kappa})$ the formula $P_a = \frac{1}{2} + \frac{\kappa}{2} \frac{x_a+x_b}{x_b-x_a}$ applies, and since $x_a + x_b = 2\frac{x_a+x_b}{2} > 0$ the vote share increases in κ .

For $\kappa \geq \bar{\kappa}$ we have $0 \notin [\theta_a, \theta_b]$, voters acquire no information, and all vote for the ex-ante more attractive party, which is party a . So $P_a = 1$, the maximal bias. $\square$

B.0.4 Lemma 4

Proof. Let ρ be supported on the line through the origin with direction $A(x_b - x_a)$. We first show that the objectives of (Pn) and (P1) agree up to a positive factor. Write $e = \frac{A(x_b - x_a)}{\|A(x_b - x_a)\|}$, so that $\theta \sim \rho$ can be represented as $\theta = \hat{\theta} e$ with $\hat{\theta} \sim \hat{\rho}$. Then $\theta^\top \theta = \hat{\theta}^2$ and

$$(x_b - x_a)^\top A(\theta - \frac{x_a + x_b}{2}) = \|A(x_b - x_a)\| \left(\hat{\theta} - \frac{\hat{x}_a + \hat{x}_b}{2} \right) = \frac{\|A(x_b - x_a)\|^2}{(x_b - x_a)^\top A(x_b - x_a)} (\hat{x}_b - \hat{x}_a) \left(\hat{\theta} - \frac{\hat{x}_a + \hat{x}_b}{2} \right),$$

where the last step uses $\hat{x}_b - \hat{x}_a = \frac{(x_b - x_a)^\top A(x_b - x_a)}{\|A(x_b - x_a)\|}$. Dividing the objective of (Pn) by the positive constant $\frac{\|A(x_b - x_a)\|^2}{(x_b - x_a)^\top A(x_b - x_a)}$ therefore yields the objective of (P1), with cost coefficient $\hat{\kappa}$ as in (17). Since the constant is positive, the two problems have the same maximizers.

For the constraint, we use that for ρ supported on the line through the origin with direction $A(x_b - x_a)$,

$$\rho \preceq_{\text{MPS}} \mu \iff \hat{\rho} \preceq_{\text{MPS}} \hat{\mu}.$$

The only if direction holds because the mean-preserving contraction order is preserved under scalar projection. For the if direction, let $\text{Proj}_{0, A(x_b - x_a)} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the orthogonal projection onto the line through the origin with direction $A(x_b - x_a)$, extended to distributions via the pushforward. Note this is different from the scalar projection $\text{proj}_{A(x_b - x_a)}$. Since ρ is supported on that line, $\hat{\rho} \preceq_{\text{MPS}} \hat{\mu}$ means $\rho \preceq_{\text{MPS}} \text{Proj}_{0, A(x_b - x_a)}(\mu)$. Because μ is spherical, $\text{Proj}_{0, A(x_b - x_a)}(\mu) \preceq_{\text{MPS}} \mu$, and transitivity gives $\rho \preceq_{\text{MPS}} \mu$.

Combining the two equivalences, among distributions ρ supported on the line through the origin with direction $A(x_b - x_a)$, ρ solves (Pn) if and only if $\hat{\rho}$ solves (P1). This establishes the if direction. For the only if direction, Theorem 1 implies that any solution to (Pn) is supported on that line, and the equivalence just shown then implies that $\hat{\rho}$ solves (P1). $\square$

B.0.5 Proposition 2

Proof. By Lemma 4, the optimal ρ projects to the optimal $\hat{\rho}$ of (P1), a one-dimensional voter learning problem with cost $\hat{\kappa}$, party platforms $\hat{x}_a, \hat{x}_b$, and prior $\hat{\mu}$. Moreover the vote shares from (Pn) equal the one-dimensional vote shares for (P1), and the share of voters whose true ideal point is A -closer to x_a equals $F_{\hat{\mu}}(\frac{\hat{x}_a + \hat{x}_b}{2})$.

The assumption $x_a^\top A x_a < x_b^\top A x_b$ gives $|\hat{x}_a| < |\hat{x}_b|$: with $d = x_b - x_a$,

$$|\hat{x}_a| < |\hat{x}_b| \iff (d^\top A x_b)^2 - (d^\top A x_a)^2 > 0 \iff (d^\top A(x_b + x_a))(d^\top A(x_b - x_a)) > 0.$$

The first factor equals $x_b^\top A x_b - x_a^\top A x_a > 0$ by assumption, and the second equals $d^\top A d > 0$ since $x_a \neq x_b$. Corollary 4 applied to (P1) then gives $P_a \geq F_{\hat{\mu}}(\frac{\hat{x}_a + \hat{x}_b}{2})$, which is the claimed bias, and shows that P_a increases in $\hat{\kappa}$ and decreases in $|\hat{x}_b - \hat{x}_a|$, holding $\frac{\hat{x}_a + \hat{x}_b}{2}$ fixed.

It remains to translate the comparative statics in Corollary 4 to the Proposition 2. Increasing κ is equivalent to increasing $\hat{\kappa}$ by (17), which weakly increases $P_a(x_a, x_b)$. Scaling up $x_b - x_a$ while holding $\frac{x_a + x_b}{2}$ fixed is equivalent to scaling up $\hat{x}_b - \hat{x}_a$ while holding $\frac{\hat{x}_a + \hat{x}_b}{2}$ and $\hat{\kappa}$ fixed, which weakly

decreases $P_a(x_a, x_b)$. □

B.0.6 Lemma 5

Proof. Write $\ell(\theta) := (x_b - x_a)^\top A(\theta - \frac{x_a + x_b}{2})$, so that $\mathcal{I} = \{\theta \in \mathbb{R}^n \mid \ell(\theta) = 0\}$ and the objective of (4) is $\mathbb{E}_\rho[|\ell(\theta)|] - \kappa \mathbb{E}_\rho[\theta^\top \theta]$.

Suppose ρ places mass on $\mathcal{I}$. By Theorem 1, ρ is supported on at most two points, one of which lies in $\mathcal{I}$ and contributes zero to $\mathbb{E}_\rho[|\ell(\theta)|]$. At most one support point therefore contributes, so the absolute value can be taken outside the expectation, and because ℓ is affine and ρ preserves the prior mean, $\mathbb{E}_\rho[|\ell(\theta)|] = |\mathbb{E}_\rho[\ell(\theta)]| = |\ell(0)|$. The value of ρ is thus $|\ell(0)| - \kappa \mathbb{E}_\rho[\theta^\top \theta]$, while the value of no information, δ_0 , is $|\ell(0)|$. Optimality of ρ hence requires $\mathbb{E}_\rho[\theta^\top \theta] = 0$, that is, $\rho = \delta_0$, and $0 \in \mathcal{I}$, that is, $x_a^\top A x_a = x_b^\top A x_b$.

It remains to rule out $\rho = \delta_0$ with $\ell(0) = 0$. Let $d := A(x_b - x_a)/\|A(x_b - x_a)\|$ and let ρ_ε place equal mass on εd and $-\varepsilon d$. Then $\ell(\pm \varepsilon d) = \pm \varepsilon \|A(x_b - x_a)\|$, so the value of ρ_ε is $\varepsilon \|A(x_b - x_a)\| - \kappa \varepsilon^2$, which is strictly positive for ε small and hence exceeds the value $|\ell(0)| = 0$ of δ_0 . For ε small enough $\rho_\varepsilon \preceq_{\text{MPS}} \mu$, so δ_0 is not optimal. □

C Online Appendix: Electoral Competition

C.1 One-Dimensional Policy Space

We first solve the electoral-competition game in the one-dimensional case $n = 1$, in which party platforms are scalars $x_a, x_b \in \mathbb{R}$ and the voter prior $\mu \in \Delta(\mathbb{R})$ has a log-concave density f_μ that is symmetric around 0. Recall that the definition (7) of the posterior means from the multidimensional case simplifies in one dimension to

$$\begin{aligned}\theta_a(x_a, x_b) &:= \frac{x_a + x_b}{2} - \frac{x_b - x_a}{2\kappa}, \\ \theta_b(x_a, x_b) &:= \frac{x_a + x_b}{2} + \frac{x_b - x_a}{2\kappa}.\end{aligned}\tag{27}$$

The following terminology is used in the following.

Definition 2. *We say*

- **feasibility holds at (x_a, x_b)** if $\{\theta_a(x_a, x_b), \theta_b(x_a, x_b)\}$ is feasible.
- **feasibility is slack at (x_a, x_b)** if feasibility holds on an open neighborhood of (x_a, x_b) .

We use slack feasibility in the best-response lemma below. Lemma 3 gives a useful sufficient condition for feasibility holding at (x_a, x_b) . The following strengthening of Lemma 3 gives a useful sufficient condition for feasibility being slack at (x_a, x_b) .

Lemma 6. *If $|x_b - x_a| < \kappa\Delta$ and $\theta_a(x_a, x_b) < 0 < \theta_b(x_a, x_b)$, then feasibility is slack at (x_a, x_b) .*

Proof. Let $|x_b - x_a| < \kappa\Delta$ and $\theta_a(x_a, x_b) < 0 < \theta_b(x_a, x_b)$. By Lemma 3 and $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$ being continuous in x_a and x_b , then feasibility holds on an open neighborhood of (x_a, x_b) . $\square$

Lemma 7. *Let $\kappa < 1$ and $x_a \leq x'_a \leq x'_b \leq x_b$, and suppose feasibility holds at (x_a, x_b) . If $\theta_a(x'_a, x'_b) \leq 0 \leq \theta_b(x'_a, x'_b)$, then feasibility holds at (x'_a, x'_b) .*

Proof. For $\kappa < 1$, the posterior mean $\theta_a(x_a, x_b) = \frac{x_a + x_b}{2} - \frac{x_b - x_a}{2\kappa}$ increases in x_a and decreases in x_b , and $\theta_b(x_a, x_b) = \frac{x_a + x_b}{2} + \frac{x_b - x_a}{2\kappa}$ decreases in x_a and increases in x_b . Therefore

$$\theta_a(x_a, x_b) \leq \theta_a(x'_a, x'_b) \leq \theta_b(x'_a, x'_b) \leq \theta_b(x_a, x_b).$$

By feasibility at (x_a, x_b) , the mean-zero distribution on $\{\theta_a(x_a, x_b), \theta_b(x_a, x_b)\}$ is a mean-preserving contraction of μ . By $\theta_a(x'_a, x'_b) \leq 0 \leq \theta_b(x'_a, x'_b)$ and the inclusion above, the mean-zero distribution on $\{\theta_a(x'_a, x'_b), \theta_b(x'_a, x'_b)\}$ is a mean-preserving contraction of the mean-zero distribution on $\{\theta_a(x_a, x_b), \theta_b(x_a, x_b)\}$, hence of μ . $\square$

C.1.1 Lemmas on Party Positions

The following lemmas describe properties of any equilibrium. Without loss, assume $x_a^* \leq x_b^*$.

Lemma 8 (Platform Proximity). *In any equilibrium, each party's platform is weakly closer to its own ideal policy than the opponent's platform, that is, $|x_j - x_j^*| \leq |x_{-j} - x_j^*|$ for $j \in \{a, b\}$.*

Proof. Recall that we restrict attention to equilibria where both parties obtain positive vote shares. Suppose $|x_j - x_j^*| > |x_{-j} - x_j^*|$ for some $j \in \{a, b\}$, so $u(x_j, x_j^*) < u(x_{-j}, x_j^*)$. Because party j wins with positive probability, its expected utility is strictly less than $u(x_{-j}, x_j^*)$. By deviating to the platform x_{-j} , policy x_{-j} is implemented with certainty and obtains party j obtains utility $u(x_{-j}, x_j^*)$. The deviation is profitable, so (x_a, x_b) is not an equilibrium. $\square$

Lemma 9 (Equilibrium Bounds). *If $x_a^* < x_b^*$, then in any equilibrium:*

- (1) $x_a \leq x_b$.
- (2) If $x_a^* \leq 0 \leq x_b^*$, then $2x_a^* \leq x_a \leq x_b^*$ and $x_a^* \leq x_b \leq 2x_b^*$.
- (3) If $0 < x_a^*$, then either $x_a = x_b$ or $x_a^* \leq x_a < x_b \leq 2x_b^* - x_a^*$. An analogous statement holds if $x_b^* < 0$.

In every case, $|x_b - x_a| \leq 2|x_b^ - x_a^*|$.*

Proof. (1) Suppose, for the sake of contradiction, that $x_a > x_b$. The inequality $|x_a - x_a^*| \leq |x_b - x_a^*|$ from Lemma 8 together with $x_a > x_b$ implies $\frac{x_a + x_b}{2} \leq p$. By the same argument with the roles of party a and b exchanged, $x_b^* \leq \frac{x_a + x_b}{2}$. Together, $x_b^* \leq x_a^*$, contradicting $x_a^* < x_b^*$.

(2) First we show $x_a \leq x_b^*$. Suppose $x_a > x_b^*$. Lemma 8 for party b gives $x_b \leq x_a$, so $x_a = x_b$ by (1). For small $\varepsilon > 0$, both posterior means $\theta_a(x_a - \varepsilon, x_a)$, $\theta_b(x_a - \varepsilon, x_a)$ are positive, so deviating to $x_a - \varepsilon$ gives party a all votes by Corollary 3 and a better policy by $x_a^* \leq 0$, a contradiction. By symmetry of the setup, an analogous argument gives $x_b \geq x_a^*$.

Next we show $2x_a^* \leq x_a$. Suppose $x_a < 2x_a^*$, so $x_a < x_a^* \leq 0$. We distinguish three cases.

If $x_b \leq 0$, then Lemma 8 and $x_a^* \leq x_b$ give $x_a \geq 2x_a^* - x_b \geq 2x_a^*$, a contradiction.

If $x_b > 0$ and $|x_a| \geq |x_b|$, then $P_a(x_a, x_b) \leq \frac{1}{2} \leq P_a(0, x_b)$ and $u(x_b, x_a^*) \leq u(x_a, x_a^*) < u(0, x_a^*)$ by Corollary 3, Lemma 8, and $x_a < 2x_a^*$. Together, $U_a(x_a, x_b) \leq \frac{1}{2}u(x_a, x_a^*) + \frac{1}{2}u(x_b, x_a^*) < U_a(0, x_b)$, so 0 is a profitable deviation.

If $x_b > 0$ and $|x_a| < |x_b|$, fix $\varepsilon > 0$ with $x_a + \varepsilon < x_a^*$, so $|x_a + \varepsilon| < |x_b|$. By $u(x_a + \varepsilon, x_a^*) > u(x_a, x_a^*) \geq u(x_b, x_a^*)$, where the second inequality holds by Lemma 8, the deviation to $x_a + \varepsilon$ is profitable if $P_a(x_a + \varepsilon, x_b) \geq P_a(x_a, x_b)$, which we show now. If feasibility holds at (x_a, x_b) , then $P_a(x_a, x_b) = \frac{1}{2} + \frac{\kappa}{2} \frac{x_b + x_a}{x_b - x_a} < 1$, which is increasing in x_a by $x_b > 0$, and Corollary 3 gives $P_a(x_a + \varepsilon, x_b) = 1$ or $P_a(x_a + \varepsilon, x_b) \geq \frac{1}{2} + \frac{\kappa}{2} \frac{x_b + (x_a + \varepsilon)}{x_b - (x_a + \varepsilon)}$. If feasibility does not hold at (x_a, x_b) , then $0 \in (\theta_a, \theta_b)$, since party a would otherwise obtain all votes. Both properties persist at $(x_a + \varepsilon, x_b)$ for small ε , because the posterior means are continuous in x_a and the set of feasible pairs is closed. By Proposition 3 and Corollary 3, a threshold signal is optimal at both position pairs, with thresholds $t, t_\varepsilon > \frac{x_a + x_b}{2} > x_a$ and vote shares $F_\mu(t), F_\mu(t_\varepsilon)$. The voter objective $b(t) - \kappa c(t)$ is supermodular in (x_a, t) on the region $t > x_a$, so $t_\varepsilon \geq t$ and $P_a(x_a + \varepsilon, x_b) \geq P_a(x_a, x_b)$. Both the vote share and the policy utility when winning increase when raising x_a , so $x_a < x_a^*$ is suboptimal.

Analogously, $x_b \leq 2x_b^*$.

(3) Let $0 < x_a^*$ and $x_a \neq x_b$, so $x_a < x_b$ by (1).

First we show $x_a^* \leq x_a$. Suppose $x_a < x_a^*$. If $x_b < x_a^*$, then $x_a \geq x_b$ by Lemma 8 contradicting $x_a < x_b$. If $x_b \geq x_a^*$ and $|x_a| \geq |x_b|$, then $x_a \leq -x_b$, so x_a is further from x_a^* than x_b contradicting Lemma 8. If $x_b \geq x_a^*$, $|x_a| \leq |x_b|$ and $x_a < x_a^*$, the third case in the proof of (2) applies verbatim and yields a profitable deviation via increasing x_a , a contradiction.

Next, we show $x_a \leq x_b^*$. If not, Lemma 8 for party b gives $x_b \leq x_a$, contradicting $x_a < x_b$.

Finally, by Lemma 8 for party b and $x_a^* \leq x_a \leq x_b^*$, we have $x_b - x_b^* \leq |x_a - x_b^*| = x_b^* - x_a \leq x_b^* - x_a^*$, so $x_b \leq 2x_b^* - x_a^*$.

The statement for $x_b^* < 0$ follows analogously. $\square$

Lemma 10. *If $x_a^* = x_b^*$, then $x_a = x_b$ in any equilibrium.*

Proof. Suppose $x_a \neq x_b$ and write $x^* := x_a^* = x_b^*$. Lemma 8 applied to both parties gives $|x_a - x^*| = |x_b - x^*| =: d$, so $d > 0$ and $x^* = \frac{x_a + x_b}{2}$. The implemented policy is at distance d from x^* regardless of the election outcome, so each party's expected utility equals $-d^2$. Let j be a party with $|x_j| = \min\{|x_a|, |x_b|\}$ and consider a deviation of party j to x^* . By $x_a \neq x_b$, $|x^*| < |x_{-j}|$, and the midpoint of x^* and x_{-j} lies strictly on the same side of 0 as x_{-j} . By Corollary 4, party j 's vote share P_j at the deviation weakly exceeds the share of voters whose true ideal point is closer to x^* than to x_{-j} , and that share is at least $\frac{1}{2}$ by the previous sentence. Party j 's expected utility from the deviation is at least $1/2 \cdot 0 + 1/2 \cdot (-d^2) > -d^2$, so (x_a, x_b) is not an equilibrium. $\square$

Lemma 11. *If $x_a^* \leq 0 \leq x_b^*$ and $\kappa \geq 1$, then the unique equilibrium is $(x_a, x_b) = (0, 0)$.*

Proof. Under $\kappa \geq 1$, the implemented policy is zero if one party chooses policy 0. This follows from Corollary 3 and the fact that $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$ are weakly between x_a and x_b for $\kappa \geq 1$. Therefore, $(x_a, x_b) = (0, 0)$ is an equilibrium.

By Lemma 9 (1) and Lemma 10, $x_a \leq x_b$ in any equilibrium.

Suppose (x_a, x_b) is an equilibrium. The average implemented policy is weakly positive or weakly negative. Suppose without loss that it is weakly positive. Then party a can obtain policy 0 for certain by choosing $x_a = 0$. This weakly improves the utility of party a because the policy outcome is weakly better in expectation and has no variance. Thus, if the policy outcome under (x_a, x_b) is not 0 for certain, then this is a profitable deviation. Finally, if the policy outcome is 0 for certain, then at least one party must choose position 0. The other party must also choose position 0, otherwise only one party would obtain a positive vote share. $\square$

C.1.2 Best Response Lemma

Next, we provide a lemma that partially characterizes parties' best responses. We need a few definitions. Define

$$\tilde{P}_a(x_a, x_b) := \frac{1}{2} + \frac{\kappa}{2} \frac{x_b + x_a}{x_b - x_a}, \quad (28)$$

$$\tilde{U}_a(x_a, x_b) := \tilde{P}_a(x_a, x_b)u(x_a, x_a^*) + (1 - \tilde{P}_a(x_a, x_b))u(x_b, x_a^*). \quad (29)$$

We call $\tilde{P}_a$ and $\tilde{U}_a$ the **pseudo vote share** and **pseudo utility**, respectively, because they are equal to the vote share and expected utility of party a only if feasibility holds at (x_a, x_b) , by Corollary 3.

Lemma 12 (Best Response). *Let $x_a^* < x_b^*$, $\kappa < 1$, and $x_b \in \mathbb{R}$. Define*

$$\hat{x}_a = x_a^* + \frac{\kappa}{1 - \kappa} x_b.$$

Necessity: *If x_a is a best response of party a to x_b and feasibility is slack at (x_a, x_b) , then $x_a = \hat{x}_a$.*

Sufficiency: *If feasibility holds at $(\hat{x}_a, x_b)$ and at $(-x_b, x_b)$, then $\hat{x}_a$ is a best response of party a to x_b and the unique best response if $x_b \neq (1 - \kappa)x_a^*$.*

Analogous statements hold when exchanging a and b (maintaining $x_a^ < x_b^*$).*

Proof. Necessity: If feasibility is slack at (x_a, x_b) , then $U_a(x_a, x_b)$ is locally given by the pseudo utility

$$\begin{aligned} \tilde{U}_a(x_a, x_b) &= \tilde{P}_a(x_a, x_b)u(x_a, x_a^*) + (1 - \tilde{P}_a(x_a, x_b))u(x_b, x_a^*) \\ &= (u(x_a, x_a^*) - u(x_b, x_a^*))\tilde{P}_a(x_a, x_b) + u(x_b, x_a^*) \\ &= 2\left(x_a^* - \frac{x_a + x_b}{2}\right)(x_a - x_b)\left(\frac{1}{2} + \frac{\kappa}{2} \frac{x_a + x_b}{x_b - x_a}\right) + u(x_b, x_a^*) \\ &= \left(\frac{x_a + x_b}{2} - x_a^*\right)(x_b(1 + \kappa) - x_a(1 - \kappa)) + u(x_b, x_a^*). \end{aligned} \quad (30)$$

Since the domain of x_a is unbounded, if x_a is a best response, then it satisfies the first-order condition $\frac{d\tilde{U}_a(x_a, x_b)}{dx_a} = 0$. The unique solution to the first-order condition is

$$\hat{x}_a = x_a^* + \frac{\kappa}{1 - \kappa} x_b.$$

Sufficiency: Inserting $\hat{x}_a$ into (30) gives

$$U_a(\hat{x}_a, x_b) = \frac{1 - \kappa}{2} \left(\frac{x_b}{1 - \kappa} - x_a^* \right)^2 + u(x_b, x_a^*) \geq u(x_b, x_a^*), \quad (31)$$

with equality if and only if $x_b = (1 - \kappa)x_a^*$. Take any $x_a \neq \hat{x}_a$.

If x_a is at least as far from x_a^* as x_b is, then $U_a(x_a, x_b) \leq u(x_b, x_a^*)$, which is less or equal to $U_a(\hat{x}_a, x_b)$ by (31). Assume for the rest of the proof that x_a is strictly closer to x_a^* than x_b is.

If feasibility holds at (x_a, x_b) , then $U_a(x_a, x_b) = \tilde{U}_a(x_a, x_b) < \tilde{U}_a(\hat{x}_a, x_b) = U_a(\hat{x}_a, x_b)$ by strict concavity, since $x_a \neq \hat{x}_a$ and $\hat{x}_a$ is the unique maximizer of the pseudo utility.

If feasibility does not hold at (x_a, x_b) , then by Proposition 3, either the origin does not lie between $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$, or the voter uses a threshold signal. We discuss both cases in turn.

If the origin does not lie between $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$, party a obtains either no votes or all votes. With no votes, $U_a(x_a, x_b) = u(x_b, x_a^*) \leq U_a(\hat{x}_a, x_b)$. With all votes, both posterior means lie on the same side of the origin with $\theta_a(x_a, x_b)$ closer to 0 than $\theta_b(x_a, x_b)$, so $\tilde{P}_a(x_a, x_b) \geq 1$. Together with x_a being closer to x_a^* than x_b ,

$$U_a(x_a, x_b) = u(x_a, x_a^*) \leq \tilde{U}_a(x_a, x_b) < \tilde{U}_a(\hat{x}_a, x_b) = U_a(\hat{x}_a, x_b).$$

If the voter uses a threshold signal, the origin lies between $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$, and we claim $|x_a| > |x_b|$. Suppose not, so $|x_a| \leq |x_b|$. Writing $x_b > 0$ and $x_a < x_b$ without loss gives $-x_b \leq x_a < x_b$. By feasibility at $(-x_b, x_b)$ and Lemma 7, feasibility holds at (x_a, x_b) , a contradiction. Thus $|x_a| > |x_b|$, and by Corollary 3(2), $P_a(x_a, x_b) < \tilde{P}_a(x_a, x_b)$. Since x_a is closer to x_a^* than x_b is,

$$U_a(x_a, x_b) \leq \tilde{U}_a(x_a, x_b) < \tilde{U}_a(\hat{x}_a, x_b) = U_a(\hat{x}_a, x_b).$$

In every case $U_a(x_a, x_b) \leq U_a(\hat{x}_a, x_b)$. Hence $\hat{x}_a$ is a best response.

Equality can hold only through $U_a(x_a, x_b) \leq u(x_b, x_a^*)$, which arises when x_a is at least as far from x_a^* as x_b is and in the no-votes subcase. In both, (31) makes the comparison with $U_a(\hat{x}_a, x_b)$ strict unless $x_b = (1 - \kappa)x_a^*$. In the remaining cases the strict concavity of $\tilde{U}_a(\cdot, x_b)$ already gives $U_a(x_a, x_b) < U_a(\hat{x}_a, x_b)$. Hence $\hat{x}_a$ is the unique best response whenever $x_b \neq (1 - \kappa)x_a^*$.

By symmetry, in particular of the prior around 0, analogous statements hold when exchanging a and b . $\square$

C.1.3 Equilibrium

We first describe equilibrium in a symmetric setup and then in an asymmetric setup. Define $\underline{\kappa}$ by

$$\frac{1 - \underline{\kappa}}{\underline{\kappa}} |x_a^*| = \mathbb{E}_\mu[|\theta|].$$

The parameter $\underline{\kappa}$ is the smallest information cost parameter for which feasibility holds at the equilibrium positions (x_a, x_b) according to the following proposition.

Proposition 4. *Let $x_a^* < 0 < x_b^* = -x_a^*$. The following constitutes an equilibrium.*

	Party platforms	Revealed voter ideology
$\kappa \geq 1$:	Downsian equilibrium $(x_a, x_b) = (0, 0)$	$\rho = \delta(0)$
$\kappa \in (\underline{\kappa}, 1)$:	Polarizing equilibrium $(x_a, x_b) = (1 - \kappa)(x_a^*, x_b^*)$	$\rho = \frac{1}{2}\delta\left(\frac{x_a}{\kappa}\right) + \frac{1}{2}\delta\left(\frac{x_b}{\kappa}\right)$

The equilibrium platforms are unique if $|x_b^* - x_a^*| < \kappa \mathbb{E}_\mu[|\theta|]$ and $\kappa \neq 1/2$.

Proof. Lemma 11 shows that the unique equilibrium is $(x_a, x_b) = (0, 0)$ if $\kappa \geq 1$. Let $\kappa < 1$.

Equilibrium verification: The positions $(x_a, x_b) = (1 - \kappa)(x_a^*, x_b^*)$ solve $x_a = x_a^* + \frac{\kappa}{1 - \kappa}x_b$ and $x_b = x_b^* + \frac{\kappa}{1 - \kappa}x_a$. By $\kappa > \underline{\kappa}$, feasibility holds at (x_a, x_b) . By $\kappa < 1$, $|x_a| = |x_b|$, and the sufficiency part of Lemma 12, this constitutes mutual best responses and hence an equilibrium.

Uniqueness: By Lemma 9 and $|x_b^* - x_a^*| < \kappa \mathbb{E}_\mu[|\theta|]$, in any equilibrium (x_a, x_b) ,

$$|x_b - x_a| \leq 2|x_b^* - x_a^*| < 2\kappa \mathbb{E}_\mu[|\theta|] = \kappa \Delta.$$

By Lemma 6, either $0 \notin (\theta_a(x_a, x_b), \theta_b(x_a, x_b))$ or feasibility is slack at (x_a, x_b) . In the former case, one party obtains all votes or $x_a = x_b$. Let $x_a = x_b \geq 0$, the other case being analogous. By Corollary 3, party a obtains positive votes by deviating to $x_a - \varepsilon$ for ε small enough, so $x_a - \varepsilon$ is a profitable deviation. Finally, let feasibility be slack at (x_a, x_b) . By $\kappa \in (0, 1)$ and the necessity part of Lemma 12, then $x_a = x_a^* + \frac{\kappa}{1 - \kappa}x_b$ and $x_b = x_b^* + \frac{\kappa}{1 - \kappa}x_a$. Solving, we obtain

$$(x_a, x_b) = (1 - \kappa)(x_a^*, x_b^*) \tag{32}$$

if $\kappa \neq 1/2$.¹⁴ □

Proposition 5. Suppose $x_a^* < 0 < x_b^*$ and $|x_a^*| < |x_b^*|$. The positions

$$\begin{aligned} x_a(\kappa) &= \frac{1 - \kappa}{1 - 2\kappa} \left((1 - \kappa)x_a^* + \kappa x_b^* \right) \\ x_b(\kappa) &= \frac{1 - \kappa}{1 - 2\kappa} \left((1 - \kappa)x_b^* + \kappa x_a^* \right) \end{aligned}$$

constitute an equilibrium for

$$\kappa \in \left(0, \frac{x_b^* - x_a^*}{3x_b^* - x_a^*} \right) \cup \left(\frac{x_b^* - x_a^*}{x_b^* - 3x_a^*}, 1 \right) \quad \text{and} \quad \max\{|x_a(\kappa)|, |x_b(\kappa)|\} < \kappa \mathbb{E}_\mu[|\theta|]. \tag{33}$$

These are the unique equilibrium platforms for κ satisfying (33) if additionally $|x_b^* - x_a^*| < \kappa \mathbb{E}_\mu[|\theta|]$.

If $\kappa \geq 1$, the unique equilibrium platforms are $(x_a, x_b) = (0, 0)$.

Proof. If $\kappa \geq 1$, the unique equilibrium is $(x_a, x_b) = (0, 0)$ by Lemma 11. Assume $\kappa < 1$ henceforth. Imposing the best-response conditions from Lemma 12,

$$\begin{aligned} x_a &= x_a^* + \frac{\kappa}{1 - \kappa}x_b, \\ x_b &= x_b^* + \frac{\kappa}{1 - \kappa}x_a, \end{aligned} \tag{34}$$

and solving yields the unique solution $(\tilde{x}_a, \tilde{x}_b) = (x_a(\kappa), x_b(\kappa))$ for $\kappa \neq 1/2$. We first verify that $(\tilde{x}_a, \tilde{x}_b)$ constitutes an equilibrium under (33), then establish uniqueness.

¹⁴If $\kappa = 1/2$, the best response function has slope 1 and all (x_a, x_b) with $x_b = x_a + \frac{1}{2}(x_b^* - x_a^*)$ solve the equations $x_a = x_a^* + \frac{\kappa}{1 - \kappa}x_b$ and $x_b = x_b^* + \frac{\kappa}{1 - \kappa}x_a$.

Equilibrium verification: By the sufficiency part of Lemma 12, $(\tilde{x}_a, \tilde{x}_b)$ constitutes mutual best responses whenever feasibility holds at $(\tilde{x}_a, \tilde{x}_b)$, $(-\tilde{x}_b, \tilde{x}_b)$, and $(\tilde{x}_a, -\tilde{x}_a)$. Take $|\tilde{x}_b| \geq |\tilde{x}_a|$, the other case being analogous.

Since $\max\{|\tilde{x}_a|, |\tilde{x}_b|\} < \kappa \mathbb{E}_\mu[|\theta|]$, we have $|\tilde{x}_b - (-\tilde{x}_b)| = 2|\tilde{x}_b| < 2\kappa \mathbb{E}_\mu[|\theta|] = \kappa \Delta$. Together with $\theta_a(-|\tilde{x}_b|, |\tilde{x}_b|) \leq 0 \leq \theta_b(-|\tilde{x}_b|, |\tilde{x}_b|)$ and Lemma 3, feasibility holds at $(-\tilde{x}_b, \tilde{x}_b)$. By $|\tilde{x}_a| \leq |\tilde{x}_b|$, the platforms of both $(\tilde{x}_a, \tilde{x}_b)$ and $(\tilde{x}_a, -\tilde{x}_a)$ lie in $[-|\tilde{x}_b|, |\tilde{x}_b|]$, so by Lemma 7 feasibility holds at either pair once the origin lies between its posterior means θ_a, θ_b . The posterior means at $(\tilde{x}_a, -\tilde{x}_a)$ equal $\pm|\tilde{x}_a|/\kappa$, so the origin lies between them and feasibility holds at $(\tilde{x}_a, -\tilde{x}_a)$. For $(\tilde{x}_a, \tilde{x}_b)$ the same conclusion follows once $\theta_a(\tilde{x}_a, \tilde{x}_b) \leq 0 \leq \theta_b(\tilde{x}_a, \tilde{x}_b)$, so establishing that inequality suffices to conclude that feasibility holds at $(\tilde{x}_a, \tilde{x}_b)$.

Under $\kappa > 1/2$, $\tilde{x}_a$ is the more extreme platform by $|x_a^*| < |x_b^*|$, so $\theta_a(\tilde{x}_a, \tilde{x}_b) < 0$ automatically, and $\theta_a(\tilde{x}_a, \tilde{x}_b) \leq 0 \leq \theta_b(\tilde{x}_a, \tilde{x}_b)$ holds if

$$\theta_b(\tilde{x}_a, \tilde{x}_b) = \frac{\tilde{x}_a + \tilde{x}_b}{2} + \frac{\tilde{x}_b - \tilde{x}_a}{2\kappa} > 0 \iff \kappa > \frac{x_b^* - x_a^*}{x_b^* - 3x_a^*} > \frac{1}{2}.$$

Under $\kappa < 1/2$, $\tilde{x}_b$ is the more extreme platform by $|x_a^*| < |x_b^*|$, so $\theta_b(\tilde{x}_a, \tilde{x}_b) > 0$ automatically, and $\theta_a(\tilde{x}_a, \tilde{x}_b) \leq 0 \leq \theta_b(\tilde{x}_a, \tilde{x}_b)$ holds if

$$\theta_a(\tilde{x}_a, \tilde{x}_b) = \frac{\tilde{x}_a + \tilde{x}_b}{2} - \frac{\tilde{x}_b - \tilde{x}_a}{2\kappa} < 0 \iff \kappa < \frac{x_b^* - x_a^*}{3x_b^* - x_a^*} < \frac{1}{2}.$$

The bounds hold by (33).

Uniqueness: The argument follows the proof of Proposition 4, using that $(\tilde{x}_a, \tilde{x}_b)$ is the unique solution to (34). $\square$

C.2 Multidimensional Policy Space

We prove Theorems 2 and 3 in parallel through several steps, building on our results for a one-dimensional policy space. In Section C.2.1 we show how the party objective relates to the objective under a one-dimensional policy space. In Section C.2.2 we combine this reduction with the one-dimensional best response Lemma 12 to prove a best response lemma for the multidimensional policy space. In Section C.2.3 we verify that our equilibrium candidate is an equilibrium, using the sufficiency part of the best response lemma. In Section C.2.4 we prove uniqueness. Finally, in Section C.2.5 we treat the case $\kappa \geq 1$.

We set $A = I_n$, the assumption maintained in the electoral competition section. Then, in the reduction of the voter learning problem to one dimension (P1), $\hat{\kappa} = \kappa$, and $\text{proj}_{A(x_b - x_a)} = \text{proj}_{x_b - x_a}$ is the scalar projection onto the line through the origin with direction $x_b - x_a$. We generalize Definitions 1 and 2 related to feasibility verbatim to the multidimensional policy space using the definition of $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$ from (7).

C.2.1 Reduction to One Dimension

We now show that confining both parties to a common line reduces the party game to a one-dimensional game. Define $\hat{x}_a, \hat{x}_b$ as in (16) with $A = I_n$ and

$$\hat{x}_a^* = \text{proj}_{x_b - x_a}(x_a^*), \quad \hat{x}_b^* = \text{proj}_{x_b - x_a}(x_b^*).$$

Define $\hat{P}_a(x_a, x_b)$ as the vote share for party a under optimal learning in the one-dimensional model with party platforms $\hat{x}_a, \hat{x}_b$ and voter prior $\hat{\mu} = \text{proj}_{x_b - x_a}(\mu)$. Party a 's objective under a one-dimensional policy space is

$$\hat{U}_a(\hat{x}_a, \hat{x}_b) = \hat{P}_a(\hat{x}_a, \hat{x}_b) \hat{U}(\hat{x}_a, \hat{x}_a^*) + (1 - \hat{P}_a(\hat{x}_a, \hat{x}_b)) \hat{U}(\hat{x}_b, \hat{x}_a^*),$$

where $\hat{U}(x, y) = -(x - y)^2$. Define $\hat{U}_b$ analogously. Let D_a^2 be the squared Euclidean distance of x_a^* from the line through x_a and x_b . The next lemma shows that the party objective coincides with the one-dimensional objective up to the additive constant D_a^2 , which depends on x_a, x_b only through the line they span. The result follows from the one-dimensional reduction of voter learning and parties' quadratic preferences.

Lemma 13. *Let $x_a \neq x_b$. Party a 's utility satisfies*

$$U_a(x_a, x_b) = \hat{U}_a(\hat{x}_a, \hat{x}_b) - D_a^2 = \hat{P}_a(\hat{x}_a, \hat{x}_b) 2(\hat{x}_a - \hat{x}_b) \left(\hat{x}_a^* - \frac{\hat{x}_a + \hat{x}_b}{2} \right) + u(x_b, x_a^*).$$

The analogous identity holds after exchanging a and b .

Proof. By Lemma 4, $P_a(x_a, x_b) = \hat{P}_a(\hat{x}_a, \hat{x}_b)$. Hence

$$\begin{aligned} U_a(x_a, x_b) &= P_a(x_a, x_b) (u(x_a, x_a^*) - u(x_b, x_a^*)) + u(x_b, x_a^*) \\ &= \hat{P}_a(\hat{x}_a, \hat{x}_b) 2(x_a - x_b)^\top \left(x_a^* - \frac{x_a + x_b}{2} \right) + u(x_b, x_a^*) \\ &= \hat{P}_a(\hat{x}_a, \hat{x}_b) 2(\hat{x}_a - \hat{x}_b) \left(\hat{x}_a^* - \frac{\hat{x}_a + \hat{x}_b}{2} \right) + \hat{U}(\hat{x}_b, \hat{x}_a^*) - D_a^2 \\ &= \hat{U}_a(\hat{x}_a, \hat{x}_b) - D_a^2, \end{aligned}$$

where the third line uses Pythagoras. □

In particular, any equilibrium (x_a, x_b) remains an equilibrium when both parties are restricted to the line through x_a and x_b , where it is an equilibrium of the corresponding one-dimensional game.

C.2.2 Multidimensional Best Response Lemma

The following lemma on best responses exploits the reduction to one dimension from Lemma 13. Following definition (6), $\text{Proj}_{(1-\kappa)x_a^*, x_b}$ denotes the orthogonal projection of $\mathbb{R}^n$ onto the line through $(1 - \kappa)x_a^*$ and x_b .

As under one dimension, we define the pseudo vote-share and pseudo utility as the vote share and utility when feasibility holds at (x_a, x_b) :

$$\begin{aligned}\tilde{P}_a(x_a, x_b) &:= \frac{1}{2} + \frac{\kappa \hat{x}_a + \hat{x}_b}{2 \hat{x}_b - \hat{x}_a}, \\ \tilde{U}_a(x_a, x_b) &:= \tilde{P}_a(x_a, x_b)u(x_a, x_a^*) + (1 - \tilde{P}_a(x_a, x_b))u(x_b, x_a^*).\end{aligned}$$

Lemma 14 (Multidimensional Best Response Lemma). *Let $\kappa < 1$, $x_b \in \mathbb{R}^n$. Define*

$$\tilde{x}_a := \text{Proj}_{(1-\kappa)x_a^*, x_b} \left(x_a^* + \frac{\kappa}{1-\kappa} x_b \right).$$

Necessity: *If x_a is the best response of party a to x_b and feasibility is slack at (x_a, x_b) , then $x_a = \tilde{x}_a$.*

Sufficiency: *If feasibility holds at $(\tilde{x}_a, x_b)$ and at $(-x_b, x_b)$, then $\tilde{x}_a$ is the best response to x_b .*

Proof. Necessity: Suppose x_a is a best response to x_b and feasibility is slack at (x_a, x_b) . By Lemma 13,

$$U_a(x_a, x_b) = \hat{P}_a(\hat{x}_a, \hat{x}_b)2(\hat{x}_a - \hat{x}_b) \left(\hat{x}_a^* - \frac{\hat{x}_a + \hat{x}_b}{2} \right) + u(x_b, x_a^*),$$

where $\hat{x}_a, \hat{x}_b, \hat{x}_a^*$ are the scalar projections of x_a, x_b, x_a^* on the line through x_a and x_b . If feasibility is slack at (x_a, x_b) , then $U_a(x_a, x_b)$ is locally given by the pseudo utility

$$\begin{aligned}\tilde{U}_a(x_a, x_b) &= \tilde{P}_a(x_a, x_b)2(\hat{x}_a - \hat{x}_b) \left(\hat{x}_a^* - \frac{\hat{x}_a + \hat{x}_b}{2} \right) + u(x_b, x_a^*) \\ &= \left(\frac{1}{2} + \frac{\kappa \hat{x}_a + \hat{x}_b}{2 \hat{x}_b - \hat{x}_a} \right) 2(\hat{x}_a - \hat{x}_b) \left(\hat{x}_a^* - \frac{\hat{x}_a + \hat{x}_b}{2} \right) + u(x_b, x_a^*) \\ &= ((1-\kappa)\hat{x}_a - (1+\kappa)\hat{x}_b) \left(\hat{x}_a^* - \frac{\hat{x}_a + \hat{x}_b}{2} \right) + u(x_b, x_a^*).\end{aligned}\tag{35}$$

If x_a is a best response, then x_a is a local maximum of the pseudo utility $\tilde{U}_a(x_a, x_b)$. Then, the projection $\hat{x}_a = \text{proj}(x_a)$ must satisfy the first-order condition of (35). By Lemma 12 the unique $\hat{x}_a$ that satisfies the first-order condition is $\hat{x}_a = \hat{x}_a^* + \frac{\kappa}{1-\kappa} \hat{x}_b$. Inserting $\hat{x}_a$ into the pseudo utility, we obtain

$$\begin{aligned}U_a(x_a, x_b) &= \left((1-\kappa) \left(\hat{x}_a^* + \frac{\kappa}{1-\kappa} \hat{x}_b \right) - (1+\kappa)\hat{x}_b \right) \left(\hat{x}_a^* - \frac{\hat{x}_a^* + \frac{\kappa}{1-\kappa} \hat{x}_b + \hat{x}_b}{2} \right) + u(x_b, x_a^*) \\ &= \left((1-\kappa)\hat{x}_a^* - \hat{x}_b \right) \frac{1}{2} \left(\hat{x}_a^* - \frac{\hat{x}_b}{1-\kappa} \right) + u(x_b, x_a^*) \\ &= \frac{1}{2(1-\kappa)} (\hat{x}_b - (1-\kappa)\hat{x}_a^*)^2 + u(x_b, x_a^*) \\ &= \frac{1}{2(1-\kappa)} \left(\frac{(x_b - x_a)^\top (x_b - (1-\kappa)x_a^*)}{\|x_b - x_a\|} \right)^2 + u(x_b, x_a^*).\end{aligned}\tag{36}$$

If x_a is a local maximum of the pseudo utility, then the direction $\frac{x_a - x_b}{\|x_a - x_b\|}$ of x_a from x_b must be a local optimum of (36), given that the projection $\hat{x}_a$ on this direction is optimally chosen. The only local optimum is when $x_b - x_a$ is parallel to $x_b - (1-\kappa)x_a^*$, that is, when x_a is on the line through

$(1 - \kappa)x_a^*$ and x_b .

Together, we have shown that x_a is on the line through $(1 - \kappa)x_a^*$ and x_b and the scalar projection of x_a on this line is $\hat{x}_a^* + \frac{\kappa}{1 - \kappa}\hat{x}_b$. The unique point that satisfies these two conditions is $\tilde{x}_a$.

Sufficiency: By (36), if feasibility holds at $(\tilde{x}_a, x_b)$, then

$$U_a(\tilde{x}_a, x_b) = \frac{1}{2(1 - \kappa)} \|x_b - (1 - \kappa)x_a^*\|^2 + u(x_b, x_a^*). \quad (37)$$

For any $d \in \mathbb{R}^n$ with $\|d\| = 1$, define

$$B(d) := \frac{1}{2(1 - \kappa)} (d^\top (x_b - (1 - \kappa)x_a^*))^2 + u(x_b, x_a^*),$$

which is bounded above by $U_a(\tilde{x}_a, x_b)$ by the Cauchy-Schwarz inequality. Given that $U_a(x_b, x_b) \leq U_a(\tilde{x}_a, x_b)$ by (37), the following lemma concludes the proof.

Lemma 15. *For any $x_a \in \mathbb{R}^n$ with $x_a \neq x_b$, the utility $U_a(x_a, x_b)$ is bounded above by $B\left(\frac{x_b - x_a}{\|x_b - x_a\|}\right)$.*

Proof. Write $d(x_a) = \frac{x_b - x_a}{\|x_b - x_a\|}$. By (36), $B(d(x_a))$ is the pseudo utility $\tilde{U}_a(x_a, x_b)$ maximized over the scalar projection $\hat{x}_a$ in direction $d(x_a)$, so $\tilde{U}_a(x_a, x_b) \leq B(d(x_a))$.

If $u(x_a, x_a^*) \leq u(x_b, x_a^*)$, then $U_a(x_a, x_b) \leq u(x_b, x_a^*) \leq B(d(x_a))$. Assume for the rest of the proof that $u(x_a, x_a^*) > u(x_b, x_a^*)$.

If feasibility holds at (x_a, x_b) , then the vote share equals the pseudo vote share, so $U_a(x_a, x_b)$ equals the pseudo utility, which is bounded from above by $B(d(x_a))$.

If feasibility does not hold at (x_a, x_b) , then by Proposition 3 either the origin does not lie between the posterior means $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$, or voters acquire a threshold signal. If the origin does not lie between $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$, then one party receives all votes. If party a receives all votes, then $P_a(x_a, x_b) = 1 \leq \tilde{P}_a(x_a, x_b)$, so $U_a(x_a, x_b)$ is bounded above by the pseudo utility. If party b receives all votes, then $U_a(x_a, x_b) = u(x_b, x_a^*) \leq B(d(x_a))$. Finally, suppose voters acquire a threshold signal, so the origin lies between $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$. We claim $\|x_a\| > \|x_b\|$. Suppose instead $\|x_a\| \leq \|x_b\|$. By the Cauchy-Schwarz inequality the scalar projections $\hat{x}_a$ and $\hat{x}_b$ onto the line through x_a and x_b satisfy $|\hat{x}_b| \leq \|x_b\|$ and $|\hat{x}_a| \leq \|x_a\| \leq \|x_b\|$, so $[\hat{x}_a, \hat{x}_b] \subseteq [-\|x_b\|, \|x_b\|]$. Because the density of μ is spherical, its projection onto every line through the origin is the same one-dimensional law, in which feasibility holds at $(-\|x_b\|, \|x_b\|)$ by feasibility at $(-x_b, x_b)$. Lemma 7 then gives feasibility at (x_a, x_b) , contradicting that feasibility does not hold there. Hence $\|x_a\| > \|x_b\|$, so $|\hat{x}_a| > |\hat{x}_b|$, and by Corollary 3(3), $P_a(x_a, x_b) < \tilde{P}_a(x_a, x_b)$. Together with $u(x_a, x_a^*) > u(x_b, x_a^*)$, $U_a(x_a, x_b)$ is bounded from above by the pseudo utility, which is bounded from above by $B(d(x_a))$. $\square$

This concludes the proof of sufficiency. $\square$

C.2.3 Equilibrium Verification

We use the best response lemma to verify that the polarizing platform pairs of Theorems 2 and 3 constitute equilibria.

Consider the symmetric setup with $\kappa < 1$, $\|x_a^*\| = \|x_b^*\|$, and $\frac{1-\kappa}{\kappa}\|x_a^*\| \leq \mathbb{E}_\mu[|\theta_1|]$. Set $x_b = (1-\kappa)x_b^*$. By $\|x_a^*\| = \|x_b^*\|$,

$$\text{Proj}_{(1-\kappa)x_a^*, x_b} \left(x_a^* + \frac{\kappa}{1-\kappa} x_b \right) = \text{Proj}_{(1-\kappa)x_a^*, (1-\kappa)x_b^*} (x_a^* + \kappa x_b^*) = (1-\kappa)x_a^*,$$

so the point $\tilde{x}_a$ of Lemma 14 equals $(1-\kappa)x_a^*$. By $\frac{1-\kappa}{\kappa}\|x_a^*\| = \frac{1-\kappa}{\kappa}\|x_b^*\| \leq \mathbb{E}_\mu[|\theta_1|]$ and μ being spherical, the posterior means θ_a, θ_b under $(\tilde{x}_a, x_b)$ and under $(-x_b, x_b)$ are less far apart than those induced by the threshold signal informing voters whether $(x_b - x_a)^\top \theta \gtrless 0$. Because the induced posterior means are also symmetric around 0, feasibility holds at $(\tilde{x}_a, x_b)$ and $(-x_b, x_b)$. By the sufficiency part of Lemma 14, $x_a = (1-\kappa)x_a^*$ is a best response to $x_b = (1-\kappa)x_b^*$. By symmetry, x_b is a best response to x_a , so $(1-\kappa)(x_a^*, x_b^*)$ is an equilibrium.

For the asymmetric setup, the positions $(x_a(\kappa), x_b(\kappa))$ satisfy

$$x_j = \text{Proj}_{(1-\kappa)x_j^*, x_{-j}} \left(x_j^* + \frac{\kappa}{1-\kappa} x_{-j} \right)$$

for $j \in \{a, b\}$ by Proposition 5. If $\max\{\|x_a(\kappa)\|, \|x_b(\kappa)\|\} \leq \kappa \mathbb{E}_\mu[|\theta_1|]$, then feasibility holds at $(x_a(\kappa), x_b(\kappa))$, $(-x_b(\kappa), x_b(\kappa))$, and $(x_a(\kappa), -x_a(\kappa))$ by Lemma 3 and Lemma 4, analogous to the argument in the equilibrium verification for Proposition 5. By the sufficiency part of Lemma 14 applied to each party, $(x_a(\kappa), x_b(\kappa))$ is an equilibrium.

Finally, we verify the formulas for the revealed ideal points and the vote share.

Derivation of θ_a, θ_b, P_a : Write $d = x_b^* - x_a^*$ and let $\Pi := \text{Proj}_{(1-\kappa)x_a^*, (1-\kappa)x_b^*}$ denote the orthogonal projection onto the line L through $(1-\kappa)x_a^*$ and $(1-\kappa)x_b^*$, which has direction d . The map Π is affine, so the difference of the party platforms in (8) equals the projection onto L of the difference of the arguments, and that difference is $\frac{1-\kappa}{1-2\kappa}(1-2\kappa)d = (1-\kappa)d$, which lies along L . Hence

$$x_b(\kappa) - x_a(\kappa) = (1-\kappa)(x_b^* - x_a^*). \quad (38)$$

Because Π is affine and the two arguments average to $\frac{1-\kappa}{1-2\kappa} \frac{x_a^* + x_b^*}{2}$,

$$\frac{x_a(\kappa) + x_b(\kappa)}{2} = \Pi \left(\frac{1-\kappa}{1-2\kappa} \frac{x_a^* + x_b^*}{2} \right). \quad (39)$$

Orthogonal projection onto a line leaves the component along that line unchanged, so for every y and w , $d^\top \text{Proj}_{y, y+d}(w) = d^\top w$. Since $\text{Proj}_{0,d}(u) = \frac{d^\top u}{\|d\|^2} d$, this yields the parallel-projection identity

$$\text{Proj}_{0,d}(\text{Proj}_{y, y+d}(w)) = \text{Proj}_{0,d}(w). \quad (40)$$

By (38), $x_b(\kappa) - x_a(\kappa)$ is a positive multiple of d , so $\text{Proj}_{0, x_b(\kappa) - x_a(\kappa)} = \text{Proj}_{0,d}$ and $\frac{x_b(\kappa) - x_a(\kappa)}{2\kappa} =$

$\frac{1-\kappa}{2\kappa}d$. Applying (40) to (39) with $y = (1-\kappa)x_a^*$ and $w = \frac{1-\kappa}{1-2\kappa} \frac{x_a^*+x_b^*}{2}$,

$$\text{Proj}_{0, x_b(\kappa)-x_a(\kappa)} \left(\frac{x_a(\kappa)+x_b(\kappa)}{2} \right) = \text{Proj}_{0, x_b^*-x_a^*} \left(\frac{1-\kappa}{1-2\kappa} \frac{x_a^*+x_b^*}{2} \right). \quad (41)$$

Substituting (41) and $\frac{x_b(\kappa)-x_a(\kappa)}{2\kappa} = \frac{1-\kappa}{2\kappa}(x_b^*-x_a^*)$ into (7) gives, for $j \in \{a, b\}$,

$$\theta_j(\kappa) = \text{Proj}_{0, x_b^*-x_a^*} \left(\frac{1-\kappa}{1-2\kappa} \frac{x_a^*+x_b^*}{2} \right) + \frac{1-\kappa}{2\kappa}(x_j^*-x_{-j}^*).$$

For the vote share, $\|x_b\|^2 - \|x_a\|^2 = (x_b - x_a)^\top (x_a + x_b)$. Using (38), then $d^\top \text{Proj}_{y, y+d}(w) = d^\top w$ on (39),

$$(x_b^* - x_a^*)^\top \frac{x_a(\kappa)+x_b(\kappa)}{2} = \frac{1-\kappa}{1-2\kappa} (x_b^* - x_a^*)^\top \frac{x_a^*+x_b^*}{2} = \frac{1-\kappa}{1-2\kappa} \frac{\|x_b^*\|^2 - \|x_a^*\|^2}{2},$$

so that

$$\|x_b(\kappa)\|^2 - \|x_a(\kappa)\|^2 = \frac{(1-\kappa)^2}{1-2\kappa} (\|x_b^*\|^2 - \|x_a^*\|^2).$$

With $\|x_b(\kappa) - x_a(\kappa)\|^2 = (1-\kappa)^2 \|x_b^* - x_a^*\|^2$ from (38), the vote-share expression $P_a = \frac{1}{2} + \frac{\kappa}{2} \frac{\|x_b\|^2 - \|x_a\|^2}{\|x_b - x_a\|^2}$ becomes

$$P_a(\kappa) = \frac{1}{2} + \frac{\kappa}{2(1-2\kappa)} \frac{\|x_b^*\|^2 - \|x_a^*\|^2}{\|x_b^* - x_a^*\|^2}.$$

C.2.4 Uniqueness

Let $\kappa < 1$. We prove uniqueness in two steps, under $\|x_b^* - x_a^*\| < \kappa \mathbb{E}_\mu[|\theta_1|]$. First, we show feasibility is slack in an equilibrium. Second, we apply the necessity part of the multidimensional best response lemma to reduce the game to one dimension and invoke the one-dimensional uniqueness results. Let (x_a, x_b) be an equilibrium.

First step: feasibility is slack. Denote scalar projections onto the direction $x_b - x_a$ by a hat, such as $\hat{x}_a = \text{proj}_{x_b - x_a}(x_a)$ and $\hat{x}_a^* = \text{proj}_{x_b - x_a}(x_a^*)$. Restricting both parties to positions on the line through x_a and x_b leaves (x_a, x_b) an equilibrium, so by Lemma 13 the pair $(\hat{x}_a, \hat{x}_b)$ is an equilibrium of the induced one-dimensional game with voter prior $\hat{\mu}$. Because μ is spherical, its scalar projection onto every line through the origin is the same one-dimensional distribution $\hat{\mu}$, with

$$\Delta := \mathbb{E}_{\hat{\theta} \sim \hat{\mu}}[\hat{\theta} \mid \hat{\theta} > 0] - \mathbb{E}_{\hat{\theta} \sim \hat{\mu}}[\hat{\theta} \mid \hat{\theta} < 0] = 2\mathbb{E}_\mu[|\theta_1|].$$

By Lemmas 9 and 11, $|\hat{x}_b - \hat{x}_a| \leq 2|\hat{x}_b^* - \hat{x}_a^*| \leq 2\|x_b^* - x_a^*\|$. Together with $\|x_b^* - x_a^*\| < \kappa \mathbb{E}_\mu[|\theta_1|]$,

$$|\hat{x}_b - \hat{x}_a| < 2\kappa \mathbb{E}_\mu[|\theta_1|] = \kappa \Delta.$$

If $0 \in (\theta_a(\hat{x}_a, \hat{x}_b), \theta_b(\hat{x}_a, \hat{x}_b))$, then by Lemma 6 feasibility is slack at $(\hat{x}_a, \hat{x}_b)$ and therefore at (x_a, x_b) . Otherwise, if $x_a \neq x_b$, one party receives all votes by Corollary 3, which violates our equilibrium definition. We now rule out $x_a = x_b$.

First, $(x_a, x_b) = (0, 0)$ is not an equilibrium for $\kappa < 1$. By $x_a^* \neq x_b^*$, at least one ideal policy is nonzero, say $x_a^* \neq 0$. Moving toward x_a^* gives party a a positive vote share for $\kappa < 1$ and a policy closer to x_a^* when winning, a profitable deviation.

Second, we rule out $(x_a, x_b) = (x, x)$ with $x \neq 0$.

Lemma 16. *For any κ , if $x \neq 0$ does not lie on both the line through 0 and x_a^* and the line through 0 and x_b^* , then (x, x) is not an equilibrium.*

Proof. Let $x_a = x_b = x \neq 0$. Given that x does not lie on the line through 0 and x_a^* , every ε -neighborhood of x contains a point x' closer to both 0 and x_a^* , obtained by moving x in the direction

$$\frac{1}{2} \frac{x_a^* - x}{\|x_a^* - x\|} + \frac{1}{2} \frac{0 - x}{\|0 - x\|}.$$

For small $\varepsilon > 0$, adopting x' gives party a all votes and a higher policy utility when winning, a profitable deviation. The argument for party b and the line through 0 and x_b^* is analogous. $\square$

If 0, x_a^* , and x_b^* are not jointly on a line, the two lines meet only at 0, so every $x \neq 0$ satisfies the hypothesis of Lemma 16 and (x, x) is not an equilibrium. If instead 0, x_a^* , and x_b^* are jointly on a line and $x \neq 0$ lies on this line, then 0 lies strictly between the projections of x_a^* and x_b^* : for Theorem 2 by $\|x_a^*\| = \|x_b^*\|$ and $x_a^* \neq x_b^*$, and for Theorem 3 by assumption. As in the proof of Lemma 16, at least one party then profits from moving slightly toward the origin, obtaining all votes and a policy closer to its ideal point. Hence $x_a = x_b$ does not occur in equilibrium, and feasibility is slack.

Second step: reduction to one dimension. By the first step, feasibility is slack in any equilibrium (x_a, x_b) . The necessity part of the multidimensional best response lemma then places x_a and x_b on the line through $(1 - \kappa)x_a^*$ and $(1 - \kappa)x_b^*$. On that line the induced game is the one-dimensional game by Lemma 13, so the one-dimensional uniqueness results in Propositions 4 and 5 apply.

C.2.5 Equilibrium if $\kappa \geq 1$

Define $\tilde{x}_a^* := \text{proj}_{x_b^* - x_a^*}(x_a^*)$ and $\tilde{x}_b^* := \text{proj}_{x_b^* - x_a^*}(x_b^*)$. The ordering $\tilde{x}_a^* < 0 < \tilde{x}_b^*$ is assumed in Theorem 3 and implied by $\|x_a^*\| = \|x_b^*\|$ and $x_a^* \neq x_b^*$ in Theorem 2. We establish if $\kappa \geq 1$ and $\tilde{x}_a^* < 0 < \tilde{x}_b^*$, the unique equilibrium is $(0, 0)$.

First, $(0, 0)$ is an equilibrium. The reason is that under $\kappa \geq 1$, the implemented policy is 0 if one party chooses position 0. This follows from Corollary 3 and the fact that $\theta_a(x_a, x_b)$ and $\theta_b(x_a, x_b)$ are weakly between x_a and x_b for $\kappa \geq 1$. Hence, no party has a profitable deviation.

Second, we show uniqueness. Suppose, $(x_a, x_b) \neq (0, 0)$ is an equilibrium.

Consider first $x_a = x_b = x \neq 0$. If the origin, x_a^* , and x_b^* are not on a joint line, then x fails to lie on at least one of the lines through the origin and x_a^* and through the origin and x_b^* , so (x, x) is not an equilibrium by Lemma 16. If instead the origin, x_a^* , and x_b^* lie on a joint line and x lies on this line, then $\tilde{x}_a^* < 0 < \tilde{x}_b^*$. The party whose ideal point lies on the opposite side of the origin from x then profits from moving slightly toward the origin, obtaining all votes and a policy closer to its ideal point. So $x_a = x_b \neq 0$ does not occur in equilibrium.

Consider now $x_a \neq x_b$ and define $\hat{x}_a^* := \text{proj}_{x_b - x_a}(x_a^*)$ and $\hat{x}_b^* := \text{proj}_{x_b - x_a}(x_b^*)$. By Lemma 13, $(\hat{x}_a, \hat{x}_b) = (\text{proj}_{x_b - x_a}(x_a), \text{proj}_{x_b - x_a}(x_b))$ is an equilibrium of the induced one-dimensional game with ideal points $\hat{x}_a^*, \hat{x}_b^*$. If $\hat{x}_a^* \leq 0 \leq \hat{x}_b^*$, then $(x_a, x_b) = (0, 0)$ by Lemma 11. Otherwise, the origin does not lie between $\hat{x}_a^*$ and $\hat{x}_b^*$. Without loss, suppose, $0 < \hat{x}_a^*$ and $0 < \hat{x}_b^*$. By Lemma 9 (3) and $x_a \neq x_b$,

$$\hat{x}_a^* \leq \hat{x}_a < \hat{x}_b \leq 2\hat{x}_b^* - \hat{x}_a^*.$$

Thus $0 < \hat{x}_a < \hat{x}_b$, so for $\kappa \geq 1$ the posterior means θ_a, θ_b lie weakly between $\hat{x}_a$ and $\hat{x}_b$, so party a obtains all the votes, contradicting our equilibrium definition.

C.2.6 Corollary 2

Proof. Denote the equilibrium positions by $x_j(\kappa) := (1 - \kappa)x_j^*$ for $j \in \{a, b\}$. For any $\kappa \in (\underline{\kappa}, 1]$, $P_a(x_a(\kappa), x_b(\kappa)) = 1/2$ and by μ being spherical $\mathbb{E}_\mu[u(x_a(\kappa), \theta)] = \mathbb{E}_\mu[u(x_b(\kappa), \theta)]$. By a bias-variance decomposition, aggregate policy utility is

$$-\mathbb{E}_\mu[\|\theta - (1 - \kappa)x_a^*\|^2] = -\text{Var}[\theta] - (1 - \kappa)^2\|x_a^*\|^2,$$

which is increasing in κ and strictly increasing in κ if $x_a^* \neq 0$. By $\|x_a^*\| = \|x_b^*\|$ and $x_a^* \neq x_b^*$, we have $x_a^* \neq 0$.

The total information cost is

$$\begin{aligned} \kappa c(\rho) &= \kappa \mathbb{E}_{\theta \sim \rho}[\|\theta\|^2] = \kappa \left\| \text{Proj}_{0, x_b^* - x_a^*} \left(\frac{x_a(\kappa)}{\kappa} \right) \right\|^2 \\ &= \kappa \left\| \text{Proj}_{0, x_b^* - x_a^*} \left(\frac{(1 - \kappa)x_a^*}{\kappa} \right) \right\|^2 = \frac{(1 - \kappa)^2}{\kappa} \left\| \text{Proj}_{0, x_b^* - x_a^*}(x_a^*) \right\|^2. \end{aligned}$$

Thus the information cost is decreasing in κ for $\kappa < 1$. By $\|x_a^*\| = \|x_b^*\|$ and $x_a^* \neq x_b^*$, $\text{Proj}_{0, x_b^* - x_a^*}(x_a^*) \neq 0$, so the information cost is strictly decreasing in κ . Aggregate policy utility strictly increases in κ and the information cost strictly decreases in κ , so aggregate voter welfare strictly increases in κ on $(\underline{\kappa}, 1]$. $\square$

D Online Appendix: Further Results and Proofs

D.1 Imperfect Observation of Party Platforms

Our results on voter learning are under some conditions robust to voters observing a *noisy* signal about party platforms before learning about ideal points.

Suppose that platforms are stochastic and independent of each other and of voters' ideal points. This stochasticity may stem from parties making random errors when choosing their platforms (Matějka and Tabellini 2021) or from parties having stochastic and private ideal points. Further, assume that, after a common signal about platforms, voters acquire a signal about their ideal points. Because both signals preserve the independence of platforms and ideal points, the expected policy utility from voting for unknown platform x under unknown ideal point θ can be written as

$$\mathbb{E}[u(x, \theta)] = u(\mathbb{E}[x], \mathbb{E}[\theta]) - \mathbb{E}[(x - \mathbb{E}[x])^\top A(x - \mathbb{E}[x])] - \mathbb{E}[(\theta - \mathbb{E}[\theta])^\top A(\theta - \mathbb{E}[\theta])].$$

Up to the constant $\mathbb{E}[(x - \mathbb{E}[x])^\top A(x - \mathbb{E}[x])]$, the agent's utility is as under known platforms, (3), except for replacing the platform x with the expected platform $\mathbb{E}[x]$. Thus, our results from section 3 remain to hold after replacing party platforms with their expectation.

If voters observe heterogeneous signals about platforms, this creates heterogeneity in their learning strategies, such as the direction in which they learn, because they perceive different $\mathbb{E}[x]$. However, as long as their signals about platforms are similar enough, our results carry over qualitatively.

D.2 Proof of Theorem 1 under General Properties of Information Cost

Appendix A.2 proves Theorem 1 for the posterior-variance cost. The proof shows uses only three properties of the information cost, which we state and use here to prove the theorem for any information cost satisfying them.

Let $\Gamma(\mu)$ denote the set of Bayes-consistent distributions $\tau \in \Delta(\Delta(\mathbb{R}^n))$ over posteriors, that is, those satisfying $\int_{\Delta(\mathbb{R}^n)} \pi d\tau(\pi) = \mu$, and let $c : \Gamma(\mu) \rightarrow \mathbb{R} \cup \{\infty\}$ denote an information cost, normalized so that no information has zero cost, $c(\delta_\mu) = 0$. For $\tau \in \Gamma(\mu)$, write $\rho_\tau \in \Delta(\mathbb{R}^n)$ for the induced distribution over posterior means, that is, the distribution of $\mathbb{E}_\pi[\theta]$ when $\pi \sim \tau$. Say τ' is a mean-preserving contraction of τ , written $\tau' \preceq_{\text{MPS}} \tau$, if there exist $\Delta(\mathbb{R}^n)$ -valued random variables $\Pi' \sim \tau'$ and $\Pi \sim \tau$ with $\mathbb{E}[\Pi | \Pi'] = \Pi'$; this corresponds to a garbling of the underlying signal structure. The contraction is strict if in addition $\tau' \neq \tau$. Because the posterior mean is linear in the posterior, $\tau' \preceq_{\text{MPS}} \tau$ implies $\rho_{\tau'} \preceq_{\text{MPS}} \rho_\tau$.

Property 1 (Strict Blackwell monotonicity). *Let $\tau, \tau' \in \Gamma(\mu)$ with $c(\tau) < \infty$. If τ' is a strict mean-preserving contraction of τ and $\rho_{\tau'}$ is a strict mean-preserving contraction of ρ_τ , then $c(\tau') < c(\tau)$.*

This property weakens standard strict Blackwell monotonicity by adding one requirement: the contraction must also be strict at the level of posterior means, not merely at the level of posteriors.

It holds for our posterior-variance cost which depends on τ only through ρ_τ : for $X \sim \rho_{\tau'}$, $Y \sim \rho_\tau$ with $\mathbb{E}[Y | X] = X$, $c(\tau) - c(\tau') = \mathbb{E}[(Y - X)^\top (Y - X)] > 0$.

Property 2 (Mixing). *If $c(\tau) = c(\tau') < \infty$, then $c(\frac{1}{2}\tau + \frac{1}{2}\tau') \leq c(\tau)$.*

This property says that randomizing between two equally costly signal structures does not raise the cost. It holds for any posterior-separable cost, since these are linear in τ . The property also holds for monotone transformations of posterior-separable costs, such as costs that are convex in mutual information.

For the last property, recall that a matrix $Q \in \mathbb{R}^{n \times n}$ is orthogonal if $Q^\top Q = I_n$. We extend Q to posteriors and to distributions over posteriors via the pushforward. Because the prior μ is spherical, $Q(\mu) = \mu$ for every orthogonal Q , so Q maps $\Gamma(\mu)$ to itself and the following property is well-defined.

Property 3 (Reflection invariance). *For every orthogonal matrix $Q \in \mathbb{R}^{n \times n}$, $c(Q(\tau)) = c(\tau)$.*

The property says that learning is equally costly in every direction. Every orthogonal matrix is a composition of reflections across hyperplanes by the Cartan–Dieudonné theorem, so the property is equivalent to invariance under reflections only, which explains its name. It holds for our posterior-variance cost because orthogonal transformations preserve $\theta^\top \theta$. The property holds for mutual information, which is preserved under any invertible transformation of the state space, and for distance-based information costs based on the Euclidean distance.

The proof of Theorem 1 under these properties follows the same four steps as the proof for the posterior-variance cost (Appendix A.2), now applied to the distribution over posteriors τ rather than to ρ directly. Fix $\tau \in \Gamma(\mu)$ with $c(\tau) < \infty$ and write $\rho = \rho_\tau$. Suppose ρ does not place mass 1 on $L = \text{span}(v)$, $v = A(x_b - x_a)$, or induces more than two posterior means. First, reflect τ across L to obtain $\text{ref}(\tau)$. Second, randomize equally between τ and $\text{ref}(\tau)$ to obtain τ^m . Third, garble τ^m by pooling posteriors that induce the same projection onto L of their posterior mean. This yields τ^p , a mean-preserving contraction of τ^m that induces posterior means on L . Fourth, garble τ^p further by pooling posteriors according to whether the induced posterior mean lies in $R_a = \{\theta | v^\top (\theta - \frac{x_a + x_b}{2}) \leq 0\}$ or $R_b = \{\theta | v^\top (\theta - \frac{x_a + x_b}{2}) > 0\}$. This yields τ^* , a mean-preserving contraction of τ^p that induces a ρ^* supported on L with at most two points. As in the proof for the posterior-variance cost, the instrumental value is preserved throughout. Since ρ is not supported on L with at most two points, at least one of the two garblings in Steps 3 and 4 is strict, so τ^* is a strict mean-preserving contraction of τ^m with ρ_{τ^*} a strict mean-preserving contraction of ρ_{τ^m} . Property 1 then gives $c(\tau^*) < c(\tau^m) \leq c(\tau)$, where the last inequality is by Property 2 and 3. Hence τ^* is strictly better than τ , so any solution to the voter problem induces at most two posterior means supported on L .

D.3 Heterogeneous Prior Expectations

Throughout the paper voters share the prior μ , whose mean is normalized to 0. Suppose instead that voters differ in their prior expectation $m \in \mathbb{R}^n$. A voter with expectation m holds the prior

μ_m obtained by shifting μ to have mean m . We set $A = I_n$ to match the formulas of the main text.

The substitution $\eta = \theta - m$ turns the voter's problem into the problem of a voter with expectation 0 who faces the platforms $x_a - m$ and $x_b - m$. Shifting both platforms leaves their difference unchanged and moves their midpoint by $-m$. Applying (7) to the shifted problem and adding m back, the two candidate revealed ideal points of a voter with expectation m are

$$\begin{aligned}\theta_a(m) &= \theta_a(x_a, x_b) + (m - \text{Proj}_{0, x_b - x_a}(m)) \\ \theta_b(m) &= \theta_b(x_a, x_b) + (m - \text{Proj}_{0, x_b - x_a}(m)),\end{aligned}\tag{42}$$

where $m - \text{Proj}_{0, x_b - x_a}(m)$ is the component of the prior expectation orthogonal to $x_b - x_a$. By Proposition 3 and Lemma 4, the two points in (42) are the voter's revealed ideal points whenever they are feasible under μ_m , which requires m to lie between $\theta_a(m)$ and $\theta_b(m)$ and requires the prior to be dispersed enough along $x_b - x_a$. If the prior expectation m falls outside the segment between $\theta_a(m)$ and $\theta_b(m)$, the voter is already sure enough which party she prefers and does not learn. If it falls inside but the prior is not dispersed enough, she learns through a threshold signal and her two revealed ideal points lie strictly between $\theta_a(m)$ and $\theta_b(m)$.

Three implications are worth drawing out. First, $\theta_a(m)$ and $\theta_b(m)$ do not change when m moves along $x_b - x_a$, since the displacement in (42) depends on m only through its component orthogonal to $x_b - x_a$. Heterogeneous expectations along the direction $x_b - x_a$ therefore do not change the two ideological poles $\theta_a(m)$ and $\theta_b(m)$, as long as $\{\theta_a(m), \theta_b(m)\}$ is feasible for every voter. Second, heterogeneity orthogonal to $x_b - x_a$ displaces both revealed ideal points by the same amount. The cloud of prior expectations is thus mapped onto two segments orthogonal to $x_b - x_a$, as in Figure 4 (b). Third, the distance between the two poles is $\|x_b - x_a\|/\kappa$, while the dispersion orthogonal to $x_b - x_a$ is the dispersion of prior expectations and responds to neither κ nor the platforms.

The comparative statics of Proposition 1 therefore extend to heterogeneous prior expectations. Cheaper information and more distant platforms move the two poles apart and leave the orthogonal dispersion untouched. These results do not rely on $A = I_n$. For general priorities the direction of learning is $A(x_b - x_a)$, and the qualitative conclusions are unchanged at the cost of more involved formulas.

D.4 Corollary 1

Proof. Define

$$V := \text{span}(\{A(x_j - x_1)\}_{j=2, \dots, k}),$$

the subspace of dimension at most $k-1$ through the prior mean from the statement of the corollary. As in the proof of Theorem 1, write

$$c(\rho) = \mathbb{E}_{\theta \sim \rho}[\theta^\top \theta]$$

for the cost of $\rho \in \Delta(\mathbb{R}^n)$, and say ρ is feasible if $\rho \preceq_{\text{MPS}} \mu$.

We first show the instrumental value of information depends only on the projection of the

posterior mean onto V . Analogously to the two-party objective (11),

$$\begin{aligned} & \mathbb{E}_\tau \left[\max_j \mathbb{E}_\pi [- (\theta - x_j)^\top A (\theta - x_j)] \right] \\ &= \mathbb{E}_\tau \left[\max_j \{ -x_j^\top A x_j + 2x_j^\top A \mathbb{E}_\pi[\theta] \} \right] - \mathbb{E}_\mu [\theta^\top A \theta] \\ &= \mathbb{E}_\tau \left[\max_j \{ -x_j^\top A x_j + 2(x_j - x_1)^\top A \mathbb{E}_\pi[\theta] \} \right] + C, \end{aligned}$$

where we use the law of iterated expectations and the second inequality adds and subtracts $2x_1^\top A \mathbb{E}_\pi[\theta]$ so that

$$C = 2x_1^\top A \mathbb{E}_\mu[\theta] - \mathbb{E}_\mu[\theta^\top A \theta]$$

does not depend on τ and can be ignored. Each term $(x_j - x_1)^\top A \mathbb{E}_\pi[\theta]$ depends on $\mathbb{E}_\pi[\theta]$ only through its projection onto V , so only this projection is payoff-relevant.

The four steps of the proof of Theorem 1 now apply with the following modifications. Step 1 reflects ρ across V instead of L : ref is replaced by the orthogonal reflection across V ,

$$\text{ref}_k(\theta) := 2 \text{Proj}_V(\theta) - \theta,$$

where Proj_V denotes the orthogonal projection onto V . The reflection of ρ is feasible because ref_k is orthogonal and μ is spherical, it preserves the instrumental value because it fixes the projection onto V , and it preserves the cost because it preserves $\theta^\top \theta$. Step 2 mixes ρ with $\text{ref}_k(\rho)$, yielding a feasible distribution symmetric about V with the same value and cost. Step 3 projects this mixture orthogonally onto V , which by the symmetry of the mixture is a mean-preserving contraction, preserving the instrumental value while weakly reducing the cost. Step 4 garbles according to the action recommendation, partitioning the state space into the k cells on which each platform is optimal and replacing the distribution by its conditional means on these cells. This garbling preserves the instrumental value, weakly reduces the cost, and produces a distribution supported on at most k points in V . As in the proof of Theorem 1, Step 3 or 4 strictly reduce the information cost unless ρ is supported on at most k points, contained in V .

Hence any solution is supported on at most k revealed ideal points, contained in V . $\square$

D.5 Theorem 1 and Evidence on Issue Alignment

Section 1.1.1 reports that voters' survey responses are well explained by a one-dimensional spatial model with quadratic utility, and that adding dimensions raises explanatory power only marginally (Jessee 2012; Shor and Rogowski 2018, among others). Upon closer examination, it is not clear whether this finding confirms the prediction of Theorem 1. First, Theorem 1 may be proving too much, because ideal points on any one-dimensional *curve*, and not only on a line, might already suffice for a one-dimensional spatial model to explain survey responses. Second, Theorem 1 may be proving too little, because it delivers one-dimensional ideal points within a multidimensional policy space, whereas in a one-dimensional spatial model the *policies* live in a one-dimensional space as well.

In the following, we show neither is the case and Theorem 1 predicts exactly the property of voter ideology supported by the evidence, namely the property that makes voters' survey responses explainable by a one-dimensional quadratic spatial model.

We first state the model that the studies discussed in Section 1.1.1 estimate. Let $y_{ij} \in \{0, 1\}$ denote the response of individual $i \in I$ to question $j \in J$. Under the *one-dimensional spatial model*, the likelihood is $\Pr(y_{ij} = 1) = \Phi(u(x_{j1}, \theta_i) - u(x_{j2}, \theta_i))$, where Φ is the logistic or the normal cumulative distribution function and utility is quadratic, $u(x, \theta) = -(x - \theta)^2$. The to-be-estimated parameters are the ideal points $\theta_i \in \mathbb{R}$ of each individual and the positions $x_{j1}, x_{j2} \in \mathbb{R}$ of the policies corresponding to the two responses of each question, for instance a minimum wage raise and the status quo. The model is a standard logit or probit discrete choice model, in which an individual responds more likely with the policy closer to her ideal point. A *multidimensional spatial model* is defined identically (with the same stochastic choice function Φ), except that the parameters $(\theta_i, x_{j1}, x_{j2})_{i \in I, j \in J}$ are elements of $\mathbb{R}^n$ and utility is the multidimensional analogue of quadratic utility, $u(x, \theta) = -(x - \theta)^\top (x - \theta)$.¹⁵

It turns out that the property of a multidimensional spatial model supported by the evidence is that respondents' *ideal points are on a line when projected onto the space spanned by the survey questions*. Formally, this property, to which we refer as the *line property*, states that there are $(\lambda_i)_{i \in I}$ in $\mathbb{R}$, $\Delta\theta \in \mathbb{R}^n$ and $(\theta_i^\perp)_{i \in I}$ in $\mathbb{R}^n$ such that

$$\begin{aligned} \forall i: \theta_i &= \theta_1 + \lambda_i \Delta\theta + \theta_i^\perp, \\ \forall i, j: (x_{j1} - x_{j2})^\top \theta_i^\perp &= 0. \end{aligned}$$

That is, each ideal point θ_i is on the line through θ_1 with direction $\Delta\theta$, modulo a component $\theta_i^\perp$ that is orthogonal to the policy difference $x_{j1} - x_{j2}$ of each question j . We say that a multidimensional spatial model with (n -dimensional) parameters $(\theta_i, x_{j1}, x_{j2})_{i \in I, j \in J}$ is *observationally equivalent* to a one-dimensional spatial model with (one-dimensional) parameters $(\hat{\theta}_i, \hat{x}_{j1}, \hat{x}_{j2})_{i \in I, j \in J}$ if the two models predict the same likelihood $\Pr(y_{ij} = 1)$ for all i and j .

Proposition 6. *A multidimensional spatial model is observationally equivalent to some one-dimensional spatial model if and only if the multidimensional spatial model satisfies the line property.*

Proposition 6 identifies ideal points being on a line (when projected onto the space spanned by the survey questions) as the property of ideal points that makes survey behavior explainable by a one-dimensional spatial model. The parenthesized caveat holds because, naturally, survey responses are not affected by policy dimensions that are orthogonal to all survey questions. Because voter surveys try to cover most relevant policy dimensions, we take the caveat to be of limited importance. The studies discussed in Section 1.1.1 find that a one-dimensional spatial model already explains voters' responses well and that further dimensions add little, so by Proposition 6 their finding is evidence that the ideal points they estimate are on a line. These estimated ideal points are the

¹⁵The result remains the same if we assume a general quadratic form $u(x, \theta) = -(x - \theta)^\top A(x - \theta)$, because we can switch to a basis in which A becomes the identity matrix.

revealed ideal points of Remark 1, which Theorem 1 predicts to be on a line, so the evidence confirms the prediction of Theorem 1.

The high-level intuition for Proposition 6 is as follows. For the “only if”-direction, suppose voter ideal points were not on a line but, say, on a U-shaped curve. Then, two extreme individuals at the opposite ends of the U would respond alike on a question whose policy difference separates the ends from the bottom of the U. Such non-monotonic behavior is ruled out by a one-dimensional ideological spectrum. For the “if”-direction, for any survey question, one can find suitable projections of its two policies onto the voter line that do not change predicted behavior and make the model one-dimensional.

It is known (e.g., Ladha 1991) that the one-dimensional spatial model with quadratic utility is equivalent to the one-dimensional item-response theory (IRT) model, which the empirical papers discussed in Section 1.1.1 estimate (Jessee 2009; Jessee 2012; Tausanovitch and Warshaw 2012; Shor and Rogowski 2018; Fowler, Hill, Lewis, Tausanovitch, Vavreck, and Warshaw 2023). Under a one-dimensional IRT model, the likelihood is

$$\Pr(y_{ij} = 1) = \Phi(\alpha_j + \beta_j p_i),$$

where $\alpha_j, \beta_j \in \mathbb{R}$ are question-specific parameters and $p_i \in \mathbb{R}$ are individual-specific parameters.

In our proof, we show (Part I) that if a multidimensional spatial model has the line property, then there is a one-dimensional IRT model that is observationally equivalent, and (Part II) that if a one-dimensional IRT model is observationally equivalent to a given multidimensional spatial model, then the spatial model satisfies the line property. Because one-dimensional IRT models and one-dimensional spatial models with quadratic utility are equivalent, this establishes our result.

Proof. First, in the multidimensional spatial model, we have

$$u(x_{j1}, \theta_i) - u(x_{j2}, \theta_i) = 2(\Delta x_j)^\top (\theta_i - \bar{x}_j),$$

where $\Delta x_j := x_{j1} - x_{j2}$ and $\bar{x}_j := \frac{x_{j1} + x_{j2}}{2}$. We suppose that no response item belongs to two different questions, which seems to be satisfied in practice, so that x_{j1} and x_{j2} are unrestricted across questions.¹⁶ Hence, Δx_j and $\bar{x}_j$ are unrestricted as well, and we can reparametrize the model through $(\Delta x_j, \bar{x}_j, \theta_i)_{i \in I, j \in J}$ instead of $(x_{j1}, x_{j2}, \theta_i)_{i \in I, j \in J}$.

The multidimensional spatial model with parameters $(\Delta x_j, \bar{x}_j, \theta_i)_{i \in I, j \in J}$ is observationally equivalent to the one-dimensional IRT model with parameters $(\alpha_j, \beta_j, p_i)_{i \in I, j \in J}$ if and only if

$$\forall i, j: 2(\Delta x_j)^\top (\theta_i - \bar{x}_j) = \alpha_j + \beta_j p_i. \quad (43)$$

¹⁶This condition is sufficient but not necessary for our proof.

Part I: Suppose the multidimensional spatial model satisfies the line property and define

$$\begin{aligned} p_i &:= \lambda_i \\ \beta_j &:= 2(\Delta x_j)^\top \Delta \theta \\ \alpha_j &:= 2(\Delta x_j)^\top (\theta_1 - \bar{x}_j). \end{aligned}$$

Then, (43) holds by

$$2(\Delta x_j)^\top (\theta_i - \bar{x}_j) = 2(\Delta x_j)^\top (\theta_1 - \bar{x}_j + \lambda_i \Delta \theta + \theta_i^\perp) = \alpha_j + \beta_j \lambda_i,$$

so the IRT model with parameters $(\alpha_j, \beta_j, p_i)_{i \in I, j \in J}$ is observationally equivalent.

Part II: If a one-dimensional IRT model is observationally equivalent to the multidimensional spatial model, then (43) holds, which implies

$$2(\Delta x_j)^\top (\theta_i - \theta_1) = \beta_j (p_i - p_1)$$

for all i and j . Let S denote the space spanned by $(\Delta x_j)_{j \in J}$. If $p_i = p_1$ for all i , then the left-hand side vanishes for all i and j , so the line property holds with $\lambda_i = 0$ and any $\Delta \theta$. Otherwise, relabel individuals so that $p_2 \neq p_1$, and define $\Delta \theta$ as the orthogonal projection of $\theta_2 - \theta_1$ onto S and $\lambda_i := \frac{p_i - p_1}{p_2 - p_1}$. Then

$$2(\Delta x_j)^\top (\theta_i - \theta_1) = \lambda_i \beta_j (p_2 - p_1) = 2\lambda_i (\Delta x_j)^\top (\theta_2 - \theta_1) = 2\lambda_i (\Delta x_j)^\top \Delta \theta,$$

where the last equality holds because $\Delta x_j \in S$. Hence, $\theta_i^\perp := \theta_i - \theta_1 - \lambda_i \Delta \theta$ satisfies $(\Delta x_j)^\top \theta_i^\perp = 0$ for every j , which is the line property. $\square$